Localizing without a full place code: novel-view behavior and heading invariance in vision-language models

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Abstract

Hippocampal place cells fire at a location regardless of which way the animal faces. We ask whether vision-language models (VLMs) represent where they are in the same way. Standard place-cell criteria cannot tell: units that pass them appear in raw pixels and random networks. We instead combine three tests: novel-view behavior, heading invariance and causal intervention. Models watch a walk through a synthetic room, then judge which of two frames was taken closer to a new view facing a direction never seen from that spot. Open VLMs tend to pick the frame that looks similar, not the one taken nearby. A Qwen3-VL-8B fine-tuned on walks solves the task in unseen procedural rooms (97%, against at most 68% for 149 retrieval rules). Its answers depend on the walk's order and on where the walk frames were taken, yet its MLP units are only partly heading-invariant and their population code for views off the walk is partial. In VGGT, a 3D pose model, a 10-dimensional subspace of the camera token carries pose causally: copying it between frames moves the predicted position 86% of the way to the donor. Passing place-cell criteria, decoding position and localizing well are each insufficient evidence of a place code, and a model can learn to localize while the units we record show no full place code.

1 Introduction

In the rodent hippocampus, most place cells in open arenas fire at the same location regardless of heading (Navratilova et al., 2012), and grid cells tile the environment with periodic fields (Hafting et al., 2005). Units that pass the same statistical criteria appear in path-integrating networks (Banino et al., 2018; Cueva and Wei, 2018) and transformers (Whittington et al., 2022). Linear spatial readouts emerge in language models (Gurnee and Tegmark, 2024), and in one case non-spatial variables tile activation space in place-cell-like patterns validated by causal interventions (Gurnee et al., 2026). However, random-weight networks also contain place-, border- and head-direction-like units (Luo et al., 2025), and grid-cell emergence depends on the readout, the training targets and the hyperparameters (Schaeffer et al., 2022). These criteria alone cannot tell a genuine spatial code from an image-driven artifact. A stronger test is needed, especially for vision-language models (VLMs).

This paper asks **whether a model that localizes novel views has a place code.** We answer the question along three legs, each tighter than the last:

1. **Behavior:** which models localize novel views, and whether the fine-tuned model does more than the tested retrieval rules.
2. **Representation:** whether there is a heading-invariant place code, in individual units or in the population.

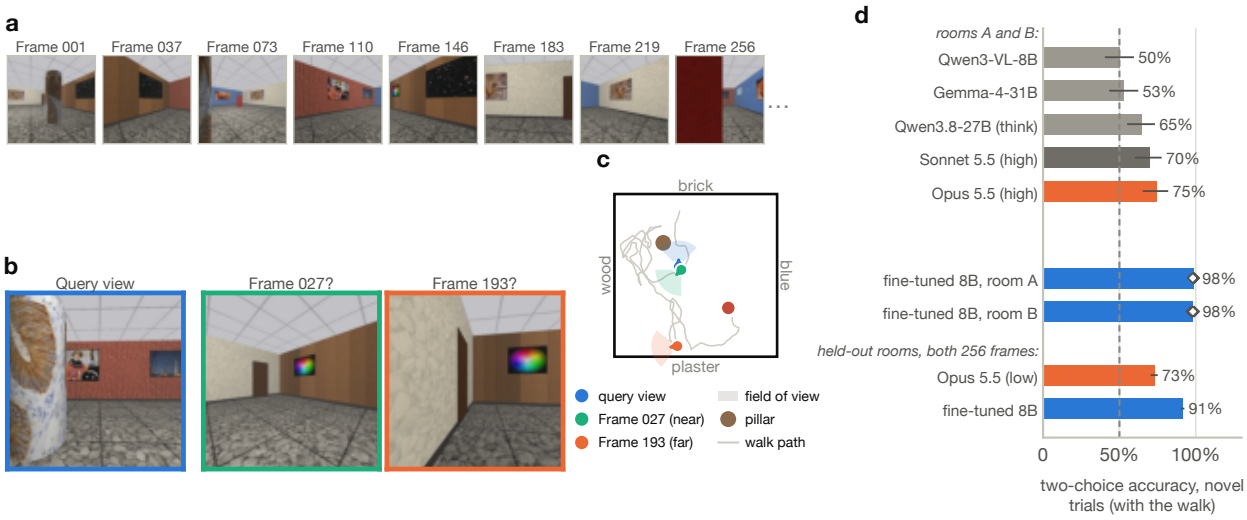


Figure 1. The fine-tuned model localizes novel views that defeat open VLMs. ($n = 102$ novel in room A, 106 in room B; the fine-tuned model uses 64-frame walks, others 256; Claude at high effort. Fine-tuned model: bar = mean of its five runs, diamonds = single runs. The bottom pair in (d) is the matched comparison: the ten held-out rooms ($n = 1{,}299$ novel), both models given the same 256-frame walk, Opus at low effort, whiskers = 95% intervals. Bars in the top group differ in rooms, walk lengths and effort.) (a) Walk frames from a foraging path. (b) Query view (blue) and two candidates: the near frame (green, 0.25 m away) faces 133° from the query; the far frame (orange, 4.9 m away) visually resembles the query. (c) Top-down map showing the walk path (gray), query position (blue dot) and candidates (green, orange); wedges show the field of view. (d) Lure-free (neither option shares a landmark with the query) novel-trial accuracy with the walk; dashed line is 50% chance.

3. **Causality:** whether a model that localizes uses a position code.

A place cell fires whenever the view is from the same place, regardless of heading (Rolls, 1999; Franzius et al., 2007; Vijayabaskaran and Cheng, 2022). Novel-view localization tests the behavioral counterpart: the task removes the simplest view-matching shortcuts (Section 2). A causal intervention is the strongest test: decodability does not imply use (Hewitt and Liang, 2019; Ravichander et al., 2021), so we check whether the output follows when part of the model's state is copied or erased. A **full place code** has place units that nearly all keep their field across headings and a population whose position carries across headings. Path-integrating RNNs show that the tests detect such a code when it is present. The fine-tuned model falls short on the strict unit test and, for the population, on teleport views. We apply these three tests, as far as each model allows, to open VLMs (Qwen3-VL (Bai et al., 2025), Qwen3.6 (Qwen Team, 2026a), Qwen3.8 (Qwen Team, 2026b), Gemma 4 (Gemma Team, 2026)), a fine-tuned 8B model, the frontier model Claude Opus 5.5 (Anthropic, 2026b) and the 3D pose model VGGT (Wang et al., 2025). Table 1 summarizes the answer model by model. In short: the fine-tuned model localizes novel views but lacks a full place code, open VLMs match views, and VGGT carries a causal position code with intermediate heading invariance.

The task is a two-choice question in a 10 m synthetically ray-cast room. Figure 1a–c shows an example: the model sees a sequence of walk frames (each labeled with its frame number), then a query image taken close to a visited position but facing a random direction. It must decide whether the query view was taken closer to Frame *a* or Frame *b*, where one option is within 1 m and the other more than 3 m away. Chance is 50%. In the example, the near frame is only 0.25 m from the query but faces 133° away from it, whereas the far frame is 4.9 m away, but visually resembles the query.

Open VLMs pick the frame that visually resembles the query, not the frame taken near it. On lure trials, where only the far frame shares a landmark with the query, open models mostly choose the far frame. The

base 8B stays at chance in the held-out rooms (52%; Sections 3.1–3.2). If these models are merely matching views, units that pass place-cell criteria should reflect the view, not the position.

We fine-tune Qwen3-VL-8B with LoRA on walks through procedural rooms and report the mean of five training runs. The fine-tuned model reaches 97.1% on novel trials in 10 held-out rooms drawn from its training generator. No tested retrieval rule comes within 15 points of it, the margin we require. The retrieval baseline, the highest-scoring of those rules, reaches 67.7% (Section 3.4). Every run depends on the walk's chronological order: shuffling the frames costs 28.0 points on average in the held-out rooms. In both runs we tested (two of the five), changing which walk frames the fine-tuned model sees shows that where those frames were taken matters more than what they show. Moving an option's neighbors toward the new view pulls the decision, but only partly (Section 5).

Claude Opus 5.5 also gains from the walk but does not clearly beat the retrieval baseline in the held-out rooms. The fine-tuned model leads Opus there and on all trials in the photoreal scenes (Sections 3.2–3.6).

Place-cell criteria are easy to satisfy: raw pixels produce place units. Heading invariance is a harder test. On single frames, few VLM place units pass the strict test (a field kept in all six pairs of 90° heading sectors). The share rises with the walk, but a moving average of single-frame activations reproduces it, so the rise is not evidence of an internal map (Section 4).

Despite solving the task, the fine-tuned model has no full place code in the units we record. Only 8%–17% of its place units keep their field when the camera turns, fewer than a moving average of single frames gives. At the population level, position carries across headings for walk frames but only partly for teleport views (Section 4).

VGGT, a 3D pose model, serves as a positive control for the causal tests: position must be computed somewhere. Copying a ten-dimensional subspace of its camera token, found by ordinary PCA without position labels, from one frame to another at block 21 moves the reported position 0.86 of the way to the donor frame. No two-dimensional subspace we tested comes close.

No single leg of evidence suffices: a model can pass place-cell criteria, decode position or localize well without carrying a full place code.

Contributions.

1. **Behavior.** We introduce a novel-view localization task with controls that remove the simplest view-matching shortcuts. We test it across open VLMs, a fine-tuned model, a frontier model and a 3D pose model in procedural rooms and photoreal scenes.

2. **Representation.** We propose a strict heading-invariance test and show that a moving average of single-frame activations reproduces the invariance the walk adds, so the walk-driven rise alone is not evidence of an internal map.

3. **Causality.** In VGGT, a ten-dimensional subspace of the camera token, found by ordinary PCA without position labels, carries pose causally; no two-dimensional subspace we tested comes close. In the fine-tuned model, an input intervention in both runs we tested (two of the five) shows the decision uses where the walk visited more than what the frames show.

Section 2 describes the rooms, the walk and the evaluation; Sections 3–5 present the results; Section 6 situates the work; Section 7 discusses the findings and their limits.

2 Setup

2.1 Room and walks

Room A is a hand-built 10 m square, rendered with a ray caster at 256×256 pixels (1 m camera height, 90° horizontal field of view). Four walls carry distinct textures; six posters, a door, a window and two pillars make the room deliberately asymmetric, so that every position has a different view. We generate 48 rat-like foraging walks of 256 steps each (12,288 frames in total), with a forward step of 0.22 m. Every frame's position (x, y) and heading θ are known exactly.

We use nine room sets (Appendix Table A1). Room A (hand-built) serves the unit, probe and VGGT analyses and the first behavioral tests. Room B has a new layout but shares the same texture families as the training rooms. We generate the 1,136 training rooms and 64 validation rooms with a procedural generator that varies wall textures, poster contents and pillar positions. The 10 held-out rooms come from the same generator with different seeds. Photoreal scenes come from two sources: HSSD houses and Replica (3 apartments plus the FRL apartment, one floor plan in 5 furniture arrangements). In all, 34 photoreal scenes test generalization outside the training distribution. Appendix C lists each experiment's rule and when it was fixed.

2.2 Novel and familiar views

After each walk, we render **teleport views**: images taken near visited positions (a visited position plus Gaussian jitter, SD 0.15 m; median 0.10 m from the nearest walk frame) at random headings. A query view is **familiar** if some walk frame within 1 m faced within 45° of it; it is **novel** otherwise. Every walk frame facing within 45° of a novel view was taken at least 1 m away. The criterion therefore removes matching the query to a walk frame taken at the same place facing the same way. Two other shortcuts need separate controls. Lure-free trials balance pixel similarity between the options and the query. Frame-number association creates a frame-order cue; room A's secondary subset tests it, and room B balances it by design. Over all 48 walks we obtain 2,690 familiar and 382 novel teleport views from the 3,072 total.

2.3 Tasks

Nearest-frame question. The model sees the walk and a query image, and is asked which walk frame was taken closest to where the query view was taken. Answers within 1 m of the true position count as hits.

Two-choice question. The model is asked whether the query view was taken closer to Frame *a* or Frame *b*. One option is within 1 m of the query; the other is more than 3 m away. Chance is 50%. The original trial set has 104 novel and 48 familiar trials. The model receives all walk frames (each labeled with its number), then the query image, then the question, and answers with a frame number. Unparsed answers count as wrong.

No-walk condition. The model sees only the query view and the two options, with no walk frames. The **walk gain** is accuracy with the walk minus accuracy without it.

Shuffled walk. The frames appear in a random order instead of their temporal sequence. Each frame keeps its real frame number, so a model can recover the original order from the labels. The **shuffled + relabeled** condition also renumbers the frames in their new order, removing this cue. A shuffled walk without relabeling therefore separates use that depends on seeing the frames in temporal order from use that could reconstruct order from the labels. It does not show that a model ignores order (Section 3.3).

Lure-free trials. We construct 192 trials (102 novel) where both options face more than 90° from the query view and neither shows any of its posters or pillars. Pixel similarity favors the near option in 50% of all trials (54 of 102 novel trials, 52.9%), removing the landmark lure and balancing image similarity.

One cue remains in room A: the near option tends to be closer in frame number to the walk frame that looks most like the query view (66% of novel trials; a **frame-order cue**). We turn this cue into a rule and score it with four other families of retrieval rules on the same trials (Section 3.4). In the held-out rooms, every rule falls far below the fine-tuned model (numbers in Appendix B). As a secondary check in room A, on the 35 novel trials where that cue points to the far option, every run of the fine-tuned model scores 35 of 35. Room B has 192 trials (106 novel) and balances the frame-order cue by design: 50% favor the near option.

Matched and unmatched inputs. The fine-tuned model and the base model both see 64 frames per walk at 256 px (64 vision tokens per frame for the Qwen VLMs), so their comparison is matched. At test time, the prompt shows up to 64 images: 61 evenly spaced walk frames, the two options if not already among them, and the query view. Walk frames and options carry their real frame numbers; the query carries the label "New view:". Fine-tuning prompts show 64 evenly spaced frames. The novelty criterion is defined on all 256 frames, so it holds for any subset shown. For the base 8B, upsampling the renders to 512 px changes no-walk accuracy, but the walk adds nothing at either resolution (Appendix A). Claude Opus 5.5 and the other API models see the full 256-frame walk; Qwen3.5-397B sees 160 frames. These are cross-model comparisons, not controlled ones: the API models receive more frames, which may help or not. In the held-out rooms, we also match the fine-tuned model to Opus's 256-frame walk (Section 3.6).

2.4 Models

Appendix Table A2 lists every model and its role. **Open models** are those whose weights we can load, so we can record their internal activations; **API models** (Claude, Kimi, DeepSeek, GPT-6 Astra) are tested on behavior only. Open models use greedy decoding with no thinking unless stated; Claude models use adaptive thinking at a stated effort level. Prompt formats are in Appendix A. Key terms used throughout (token, MLP neuron, subspace, interchange patching, transfer, steering, camera token) are also defined there; place-unit scoring details are in Appendix E.

Scope. Every open-model failure we report holds at the inputs we tested, at 256 px (64 vision tokens per frame for the Qwen VLMs), and is not a general statement about a model or architecture. Our frame-swap interventions on the fine-tuned model act on its input (the walk frames it sees), so they show which inputs the decision depends on, not a stored map. The unit and population tests read MLP neurons at the recorded layers; a code elsewhere in the network, for example in the residual stream, is not excluded.

3 Behavior: which models localize novel views

3.1 Open VLMs match views

The first test is whether open VLMs can place a view they have never seen from that direction; if they rely on the shortcuts the task removes, novel views should defeat them. Open VLMs place familiar views but fail on novel ones. Within Qwen3-VL, nearest-frame accuracy on familiar views rises with scale, from 20% at 2B to 43% at 32B; Qwen3.8-27B reaches 58%. Novel views stay at 1%–3% across all sizes, against 6% for

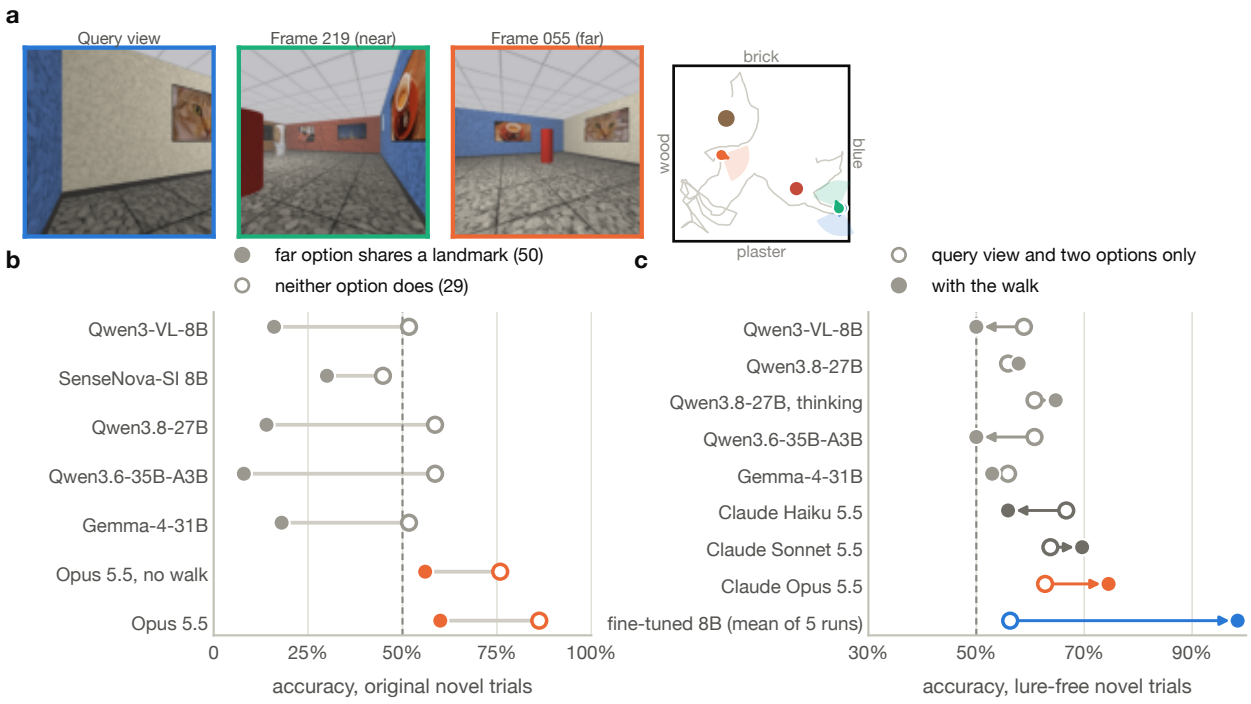


Figure 2. On lure trials, open VLMs mostly choose the far frame. (Lure $n = 50$; no landmark $n = 29$; dashed = 50% chance.) (a) A lure trial: the near frame (0.2 m away) shares no landmark with the query; the far frame (7.5 m away) shows the same cat poster. (b) Accuracy on lure trials (filled) and no-landmark trials (open); each pair is one model. (c) Lure-free novel trials: accuracy without the walk (open) and with it (filled).

a random frame. Most of the larger models' picks face within 45° of the query view, confirming that they choose the frame that looks most like the query.

SenseNova-SI 8B (Cai et al., 2026), a spatially fine-tuned version of the base architecture, scores 38% on the original novel trials, against the base model's 35% and Qwen3.8-27B's 38%. All three are below chance ($p = 0.024$ for Qwen3.8-27B): the lures pull them toward the wrong frame.

In 50 of the 104 original novel trials, only the far option shares a poster or pillar with the query view (**lure trials**; Figure 2a); in 29, neither does. Qwen3.8-27B falls to 14% on lure trials, against 59% without a lure (Figure 2b). On lure trials Claude Opus 5.5 with the walk is not distinguishable from chance (60%, $p = 0.20$; 86% without a lure).

The shared landmark marks these trials but does not cause the errors. We re-render the 38 qualifying lure trials with the shared poster removed. The base model changes only 4 of 38 answers, and it scores 26% on the edited trials, well below its 52% on no-landmark trials. It still picks the far option on 74% of the edited trials. The pattern is the same without the walk ($p = 0.38$). On most of these trials DINOv2 favors the far option (68%; exploratory); removing one shared poster does not change the model's behavior.

Open models choose by overall appearance, not just by the shared landmark. In the held-out rooms, the base model's accuracy depends on DINOv2 similarity by 15.5 points, while the fine-tuned model with the walk shows no such dependence (-0.7 to 0.8 points across runs, $p \geq 0.39$).

Without the walk, open models score above chance on lure-free trials given enough pixels (Figure 2c; Appendix Table G1), but no open model shows a detectable walk gain on novel trials in the procedural rooms. Every open model's walk-gain interval includes zero. A walk from another room gives 53%, not significantly different from no walk ($p = 0.34$), so the data neither show nor rule out an effect of extra

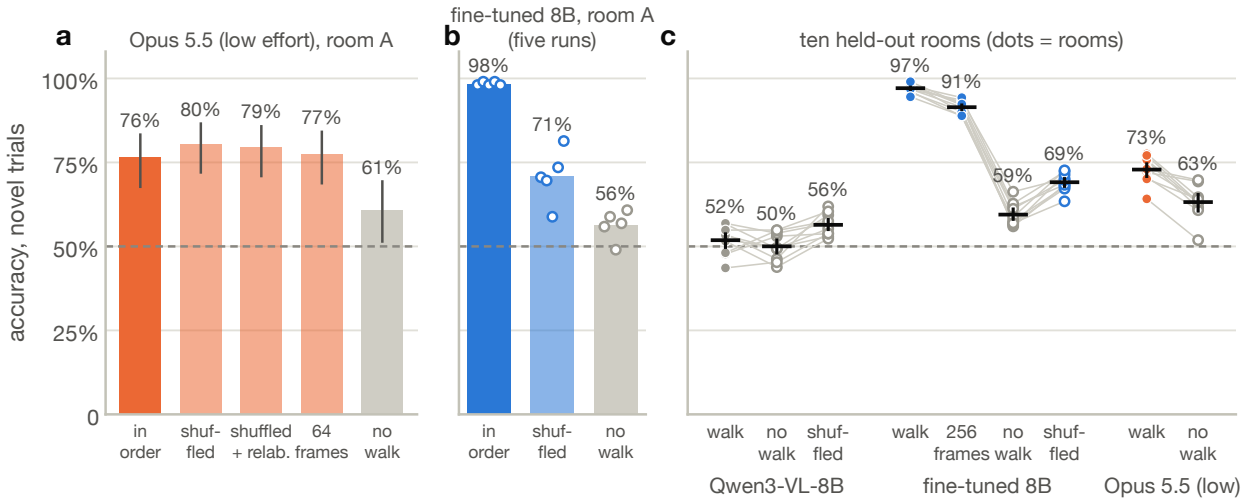


Figure 3. The walk helps Opus and the fine-tuned model; in room A, only the fine-tuned model depends on its order. (Novel trials; (a, b) 95% trial-level Wilson CIs, (c) 95% bootstrap CI over rooms; walk-clustered intervals are in Appendix Table L1; dashed = 50% chance.) (a) Opus, low effort, room A ($n = 102$; one run per bar). (b) The fine-tuned model, room A ($n = 102$; bars = mean over runs, circles = per-run accuracy). (c) Ten held-out rooms (from the training generator, the fine-tuned model's mean per room): per-room accuracy (dots) with mean and 95% bootstrap CI over rooms; "256 frames" = the same runs with the walk Opus sees. "Walk" means 64 frames for the base and fine-tuned Qwen models and 256 for Opus.

images. Qwen3.8-27B with thinking reaches 65% with the walk (Figure 1d), yet does no better trial by trial ($p = 0.64$).

3.2 The walk helps Opus and the fine-tuned model

We apply LoRA (Hu et al., 2022) to Qwen3-VL-8B: a low-rank update to about 0.5% of the weights, trained with answer-only loss on 1,136 procedural rooms and no coordinate or map supervision. We call the result **the fine-tuned model** and report its five training runs' mean throughout; Appendix F gives the run-to-run spread. Room A is not among the training rooms. A shorter two-epoch schedule gave unreliable runs, so the recipe we report uses four epochs.

The fine-tuned model reaches 97.1% in the 10 held-out rooms, with every run above the base model in all 10 rooms (Figure 3c). It leads Opus by 24.2 points, or +18.6 when matched on frames, order, labels and question (Section 3.6). The base model stays at chance. The recipe's validation bar, room-A accuracy and per-run spread are in Appendix F.

Claude Opus 5.5 also gains from the walk. Its held-out-room walk gain is 9.7 points, positive in 10 of 10 rooms. A shuffled walk helps it more than the ordered walk in all 10 rooms, by 5.9 points on average ($p = 0.002$). Its gain does not require temporal order, consistent with reference-view matching. Those shuffled walks keep the real frame numbers, and Opus's only relabeled run (room A) scores 79%, level with the ordered walk (Figure 3a). Appendix Table J1 lists every Opus accuracy and gain in the main text.

VGGT (Wang et al., 2025), a 3D pose model, serves as a behavioral positive control: it places 98% of novel views within 1 m (median error 0.30 m) and answers 104/104 two-choice trials correctly. Its internal analyses appear in Sections 4 and 5.

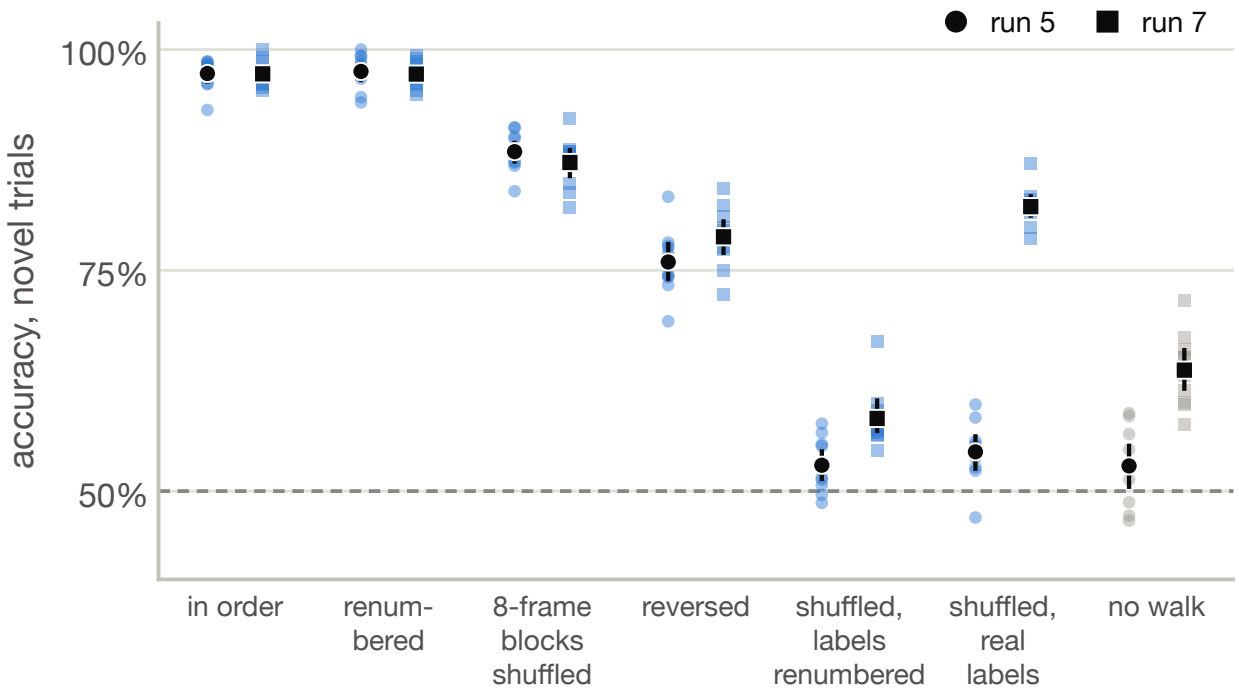


Figure 4. Renumbering costs nothing, shuffling 8-frame blocks costs less than reversing, and the relabeled shuffle removes the order effect. (Held-out rooms, novel trials, 64-frame walk; circles and squares = two of the five runs; small marks = rooms, black mark and line = mean and 95% bootstrap CI over rooms; dashed = 50% chance. The seven conditions are in order, renumbered, 8-frame blocks shuffled, reversed, shuffled with labels renumbered, shuffled with real labels, and no walk; the relabeled shuffle is the no-order control.)

3.3 What the walk contributes

The fine-tuned model depends on the walk's order. In the held-out rooms, the mean drops from 97.1% ordered to 69.1% shuffled ($p \leq 0.002$), with ordered ahead in every run. The shuffled walk keeps the real frame numbers, so the order can still be read from the labels.

We tested what two of the five runs use from the walk's order, in the 10 held-out rooms (Figure 4; the original two-epoch run in Appendix F shows the same pattern).

Renumbering the frames costs nothing (Figure 4), which rules out arithmetic on the frame numbers. Shuffling consecutive 8-frame blocks keeps most of the order effect in accuracy (0.74–0.80 of it across the two runs) but about half on the log-odds scale, which accounts for the ceiling near 100%. Reversing the walk removes more (0.47–0.48 of the effect), so the model is also sensitive to the direction of travel. These controls do not separate longer-range order from direction.

In one of the two tested runs, relabeling the shuffled walk drops it below the no-walk level, so the labels carry all of that run's shuffled-walk gain. In the other tested run, the shuffled walk is already near the no-walk level (54.4% against 52.9%), leaving little for labels to carry. The other three runs are untested (Appendix F).

3.4 Retrieval rules

A novel trial guarantees only that no walk frame within 1 m faces within 45° of the new view. A frame just past that radius can still look like the query, so a rule that retrieves the most similar frame might solve the

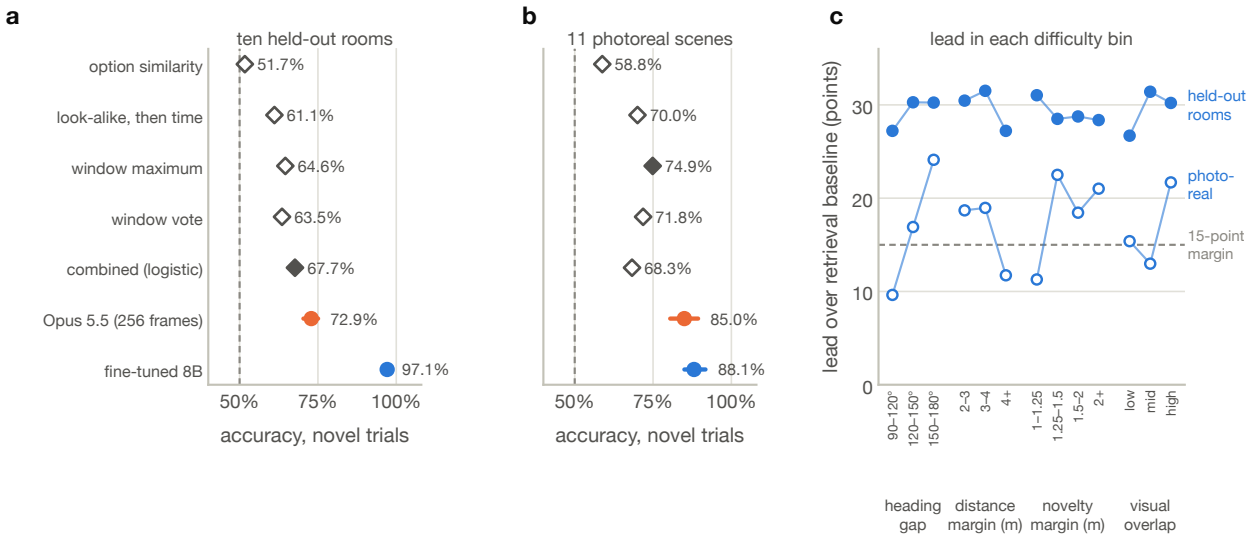


Figure 5. The fine-tuned model beats every retrieval rule in the held-out rooms, but not by the 15-point margin in the photoreal scenes. (149 rules in 5 families; novel trials; scene means in (a, b), trials pooled within each bin in (c); dashed = 50% chance in (a, b), the 15-point margin in (c).) (a) Held-out procedural rooms: the fine-tuned model (mean of its five runs) and Opus against the highest-scoring rule of each family (dark-gray diamonds; filled = the retrieval baseline). (b) Photoreal scenes: same layout. (c) The fine-tuned model's lead over the retrieval baseline in each bin of the four difficulty breakdowns (held-out rooms filled, photoreal hollow).

task with no sense of place. We tested 149 retrieval rules in 5 families on the same novel trials the models answered, using DINOv2 and pixel features (Appendix B). On any set of trials, the retrieval baseline is whichever rule scores highest there; choosing it on the same trials favors retrieval, and it can differ between sets of trials. On a given set of trials, the fine-tuned model beats retrieval when its mean leads the retrieval baseline by at least 15 points overall and in every bin where the retrieval baseline does best. We reject the retrieval account only if the model beats retrieval in both the held-out rooms and the photoreal scenes.

Figure 5a shows the results in the held-out rooms, Figure 5b the photoreal scenes and Figure 5c the per-bin breakdown.

Procedural rooms. The fine-tuned model leads the retrieval baseline by 29.4 points in the held-out rooms (97.1% vs 67.7% on 1,299 novel trials; Figure 5a). In every bin the lead is 26.7–31.5 points (Figure 5c), so the margin holds. The fine-tuned model's accuracy hardly depends on whether the retrieval baseline is right; Opus's does. Opus does not clearly beat the retrieval baseline: the 95% CI includes zero, and the lead is under the 15-point margin.

Photoreal scenes. The fine-tuned model leads the retrieval baseline by 13.2 points on the 424 novel trials across all 11 photoreal scenes (Figure 5b). The 95% CI excludes zero, but the lead is under 15, so the margin does not hold. In the apartments the retrieval baseline is level with the fine-tuned model and Opus (82.2%, 84.0% and 85.4%, respectively). Opus is above the retrieval baseline in the photoreal scenes. We do not read the photoreal results as localization: where the retrieval baseline does best, it comes within 9.6–15.0 points of the fine-tuned model.

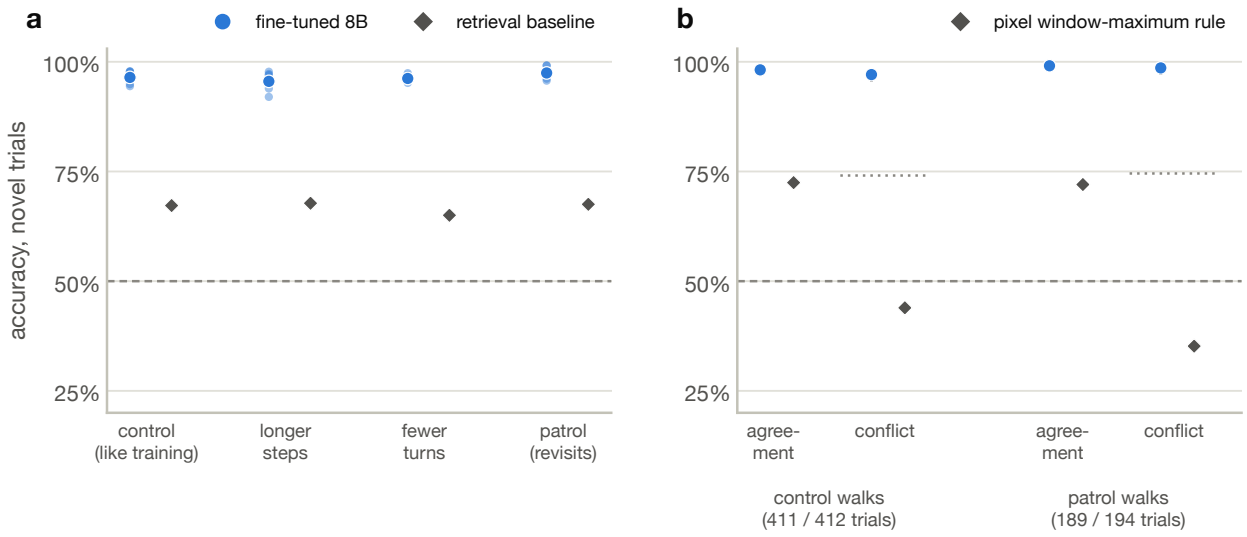


Figure 6. The fine-tuned model survives walks unlike the training walks, and geometric proximity wins. (Nine of the ten held-out rooms, novel trials, 64-frame walk; dashed = 50% chance. The retrieval baseline differs by condition.) (a) Accuracy under control, long-step, low-turn and patrol walks. Blue = five-run mean (small dots = rooms, black line = 95% CI over rooms); dark-gray diamond = retrieval baseline on the same trials (room mean). (b) Separation trials in control and patrol walks: agreement vs conflict (trial counts under each group). Blue = five-run mean; dark-gray diamonds = pixel window-maximum rule, which the conflict trials oppose by design; dotted = the midpoint between chance and agreement. Patrol separation rests on 6 rooms.

A trained sequence matcher. A small transformer trained on the fine-tuned model's own training-room trials, using one frozen DINOv2 summary per shown frame, narrows the gap. The order-free variant reaches 73.3% on the 1,299 novel trials in the held-out rooms, above the retrieval baseline by 5.6 and roughly at Opus. The fine-tuned model still leads by 23.8 points. Order gives the matcher nothing: ordered minus order-free is −1.0. The fine-tuned model loses 28.0 points when its walk is shuffled. The matcher does not transfer to the photoreal scenes (47.8%, at chance). A pixel variant narrows the gap to 17.1, so richer features, more data or larger matchers are not excluded.

Walks unlike the training walks. We gave nine of the held-out rooms new walks with longer steps, fewer turns or a patrol that revisits places (Figure 6a). On control walks drawn like the training walks, the fine-tuned model scores 96.4%. All three shifts survive: the lead over the retrieval baseline stays at least 15 points with the CI above zero, and the drop from control is at most 10 points (Appendix F).

In separation trials, the walk around the near option shows nothing of the new view's scene while the walk around the far option does (conflict trials; agreement trials reverse this). Geometric proximity wins: the fine-tuned model scores 97.1% on conflict trials against 98.2% on agreement (Figure 6b). The fine-tuned model leads the retrieval baseline on conflict trials by 32.0 points. The order-free matcher is just as flat across shifts, so surviving them alone does not indicate geometry. Every prompt keeps the ordered walk and its real frame numbers, so this test says nothing about order or labels (Appendix F).

3.5 Photoreal scenes

Three Replica apartments and eight HSSD houses (11 scenes, 424 novel trials; Figure 7a) test whether the result generalizes beyond the procedural rooms.

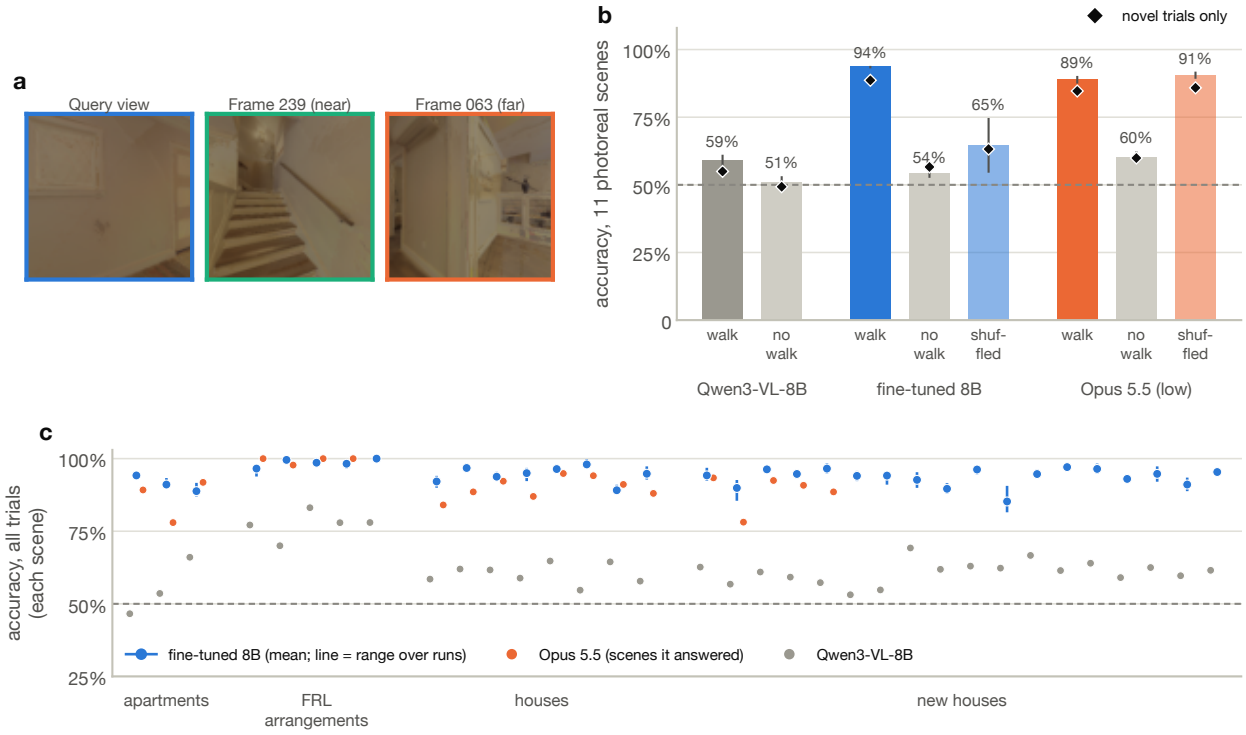


Figure 7. In the photoreal scenes the fine-tuned model and Opus both use the walk; shuffling with real labels costs the fine-tuned model and does not cost Opus. (1,953 trials from 11 scenes: 3 Replica apartments and 8 HSSD houses; dashed = 50% chance.) (a) A novel trial: query view, near frame (0.1 m away, facing 177° from the query) and far frame (4.0 m away). (b) Accuracy by condition, 11 scenes pooled: bars = the fine-tuned model's mean over runs (whisker = range), diamonds = 424 novel trials. Base and Opus keep trial-level CIs. (c) All 34 photoreal scenes, all trials with the walk: the 3 apartments, the 5 FRL arrangements, HSSD1–8 and the 18 new houses. Per scene: the fine-tuned model's mean (blue dot) and range (line), Opus where it answered (orange) and base 8B (gray).

Over all 11 photoreal scenes, the fine-tuned model reaches 93.7% on all trials (88.6% on novel; Figure 7b), above the base model in every scene. Without the walk, accuracy drops to 54.4%. Shuffling the walk order drops it to 64.5%. Every run leads Opus on all trials. The one exception is the novel trials of the three Replica apartments (58), where Opus leads every run by 1.7–5.2 points; the difference is not significant. Opus's pooled novel accuracy across all 11 scenes is 84.7%.

Opus reaches 86.3% in the apartments, with a walk gain of 30.6, about three times its procedural-room gain on all trials (11.5). The base model stays near chance (55.1%).

In the houses, Opus reaches 89.8% with a walk gain of 28.1. A shuffled walk helps Opus more than the ordered walk in the apartments. In the houses the difference is not significant; across all 11 scenes the shuffled walk is slightly better. On novel house trials, the fine-tuned model still leads Opus.

Unlike in the procedural rooms, the base model gains from the walk in the houses (8.8 points), mostly on familiar trials. Houses have several rooms, so room-level cues may contribute. Among house trials whose two options share the same room, the fine-tuned model still leads the retrieval baseline by 19.8 points, so room recognition alone does not explain the lead. VGGT scores 82.7% in the houses.

In 18 more houses and 5 more apartment arrangements, the fine-tuned model's accuracy holds (Figure 7c). In the new houses, the five runs' scene means range from 92.9% to 94.0%, against 94.5% in the original eight. Opus's walk gain is 29.3 points over the 9 new scenes it answered. The FRL apartments are at ceiling: the

task cannot separate the fine-tuned model from Opus there. Over the 13 houses and 7 apartment arrangements Opus answered, every run leads Opus on all trials combined; over the apartments, no run's lead is significant.

Over all 34 photoreal scenes (Figure 7c), the fine-tuned model averages 94.4%. On novel apartment trials across all arrangements, Opus scores higher, not significantly.

In the new houses, the fine-tuned model leads the photoreal retrieval baseline by 18.9 points overall, but the margin does not hold in the strongest-retrieval bin (Appendix B). The retrieval baseline is the rule chosen on the 11 earlier scenes, carried over unchanged.

3.6 Scope

No other open-weight model we tested shows a detectable walk gain on novel trials in the procedural rooms. Kimi K3 and DeepSeek V4.1 Flash show no walk gain, and of 8 newer open-weight VLMs screened, the 3 that accept the walk show none either (Appendix I). The largest open Qwen (Qwen3.5-397B), tested with a 160-frame walk, also shows no gain.

GPT-6 Astra gains in room A, with a walk-clustered gain of 20.6 (Appendices I and L). Astra's shuffled walk scores 77%, not significantly different from the ordered walk, the same direction as Opus.

The fine-tune adds little general spatial ability on two standard benchmarks. On VSI-Bench (Yang et al., 2025) (5,130 questions), the fine-tuned model changes by -2.1 points relative to the base model: the fine-tune hurts. The loss concentrates in metric tasks while view-relating tasks are unchanged. On MindCube-Tiny (Wang et al., 2026b), the fine-tuned model gains 4.1 points, but every run stays near chance (30.5–37.7) and the gain sits mostly in one question type (Appendix F).

Gemma 4 12B (Gemma Team, 2026), the only other model family we fine-tuned, does not learn the task reliably. Its three-run mean is 69.2% in the held-out rooms, under Opus's 72.9%. Of three runs, one reaches the task, one learns part of it and one does not (48.4%–88.6%). The best Gemma run leads Opus in rooms A and B and in the held-out rooms, but not on all trials in the photoreal scenes. Every Gemma run falls below every run of the fine-tuned model on every with-walk measure. The family difference is not separated: Gemma-4-12B has no vision encoder, and the schedule was set on Qwen.

With a 256-frame walk, the length Opus sees, we matched the frame count, order, labels and question. The fine-tuned model still leads Opus by 18.6 points on the 1,299 novel trials of the ten held-out rooms. In its own 64-frame format the fine-tuned model scores 97.1%, so the longer walk changes its accuracy by -5.7 points; it retains 0.77 of its 64-frame lead. This cost is a finding about the fine-tuned model, which learned on 64-frame walks. Opus scores about the same with 64 frames and 256 in room A ($p = 1.0$). The match covers frames, order, labels and question but not image encoding or training, and Opus ran at low effort. It covers the ten held-out rooms only; every photoreal comparison stays at 64 against 256 frames.

3.7 Human baseline (provisional)

Two participants (naive, not authors) took the same 45-novel-trial task. One participant watched the walks and scored 37 of 45 novel trials correctly (82%). The other barely watched and scored at chance (42%). No exclusion rule was set, so both are counted; per-walk details are in Appendix D. On the same trials, Claude Opus 5.5 scores 33 and the base model 14.

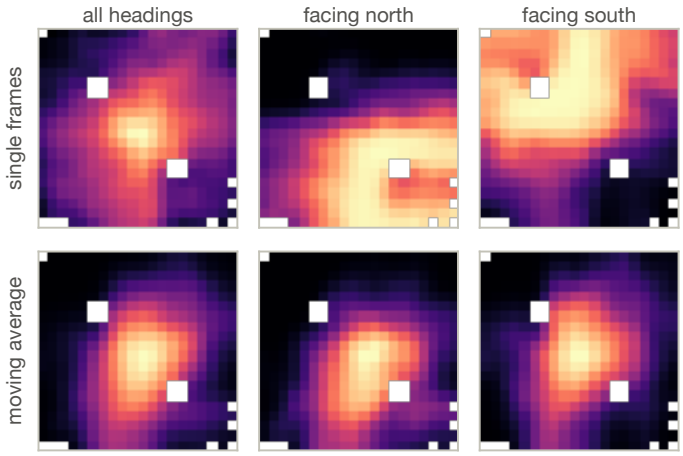


Figure 8. A raw-pixel place unit becomes heading-invariant after temporal averaging. Top: rate maps on single frames (all headings, north, south; heading correlation −0.72, stability 0.85). Bottom: after an exponential moving average with a half-life of 8 frames (heading correlation 0.85). Bright = high activity; white = fewer than 3 visits.

4 Representation: is there a heading-invariant place code

4.1 Rodent criteria are not enough

The behavioral tests show that open models match views, the fine-tuned model uses the walk's order and Opus gains from the walk (its shuffled walks keep the real frame numbers). The question now is whether units that pass rodent place-cell criteria have heading-invariant maps or whether they reflect what is in view.

A **place unit** is a model neuron whose activity varies with position in a way that resembles a rodent place cell. More precisely, we score the positive and negative parts of each neuron separately (signed "poles," because activations can be negative while firing rates cannot). We build a **rate map** for each pole: its mean activity at each position on a grid. A pole qualifies as a place unit when its spatial information (Skaggs et al., 1993) exceeds the 99th percentile of 100 circular shifts. We also require split-half stability above 0.5 and one to three firing fields covering at most 40% of the room. We score on single frames.

Units passing these criteria appear everywhere. Raw pixels produce 72% place units out of 6,144 signed poles of 3,072 z-scored features (8×8-pixel block averages, 32×32 per RGB channel). The circular-shift null tests association with position, not whether the underlying mechanism is visual or spatial. Random-weight Qwen3-VL-2B passes at 79%–81%, above pixels; random-weight DINOv2 (Oquab et al., 2024) passes at 61%–63%, below pixels. Passing alone says little about a spatial code; the test that separates view-tuned units from genuinely place-like ones is whether their maps hold when the camera turns around (validated in Appendix E).

Figure 8 shows the problem concretely. A raw-pixel place unit's north and south maps correlate only −0.72 (stability 0.85). After the exponential moving average, the heading correlation rises to 0.85. The average creates the invariance.

4.2 A strict heading test and its validation

Heading invariance is the test that separates view-tuned units from genuinely place-like ones. We build rate maps from frames in each of four 90° sectors centered on N, S, E and W. We correlate each of the six sector pairs (N–S, N–E, N–W, S–E, S–W, E–W) and take the minimum. A place unit is **strictly heading-invariant**

when this minimum correlation exceeds 0.5. The test checks whether a unit's field stays in the same place across all headings: a unit whose firing rate changes but whose field stays put still passes. Because rodent place cells can become directional with experience (Navratilova et al., 2012), heading invariance is an operational test here, not a universal property.

The six-pair minimum is stricter than the average of N–S and E–W alone. Axis-split synthetic units, which fire in one field when facing N or S and another when facing E or W, pass the average at 100.0% but the minimum at only 6.9%. Genuinely invariant units pass both at 98.9% under the minimum. The path-integrating RNNs receive only step-by-step displacements and a start state taken from place-cell activity at the true position. They are trained with place-cell targets and never see a view (Appendix E), so their units are heading-invariant by construction. They meet the bar at 97% or above under all six pairs, which validates the statistic: the tests detect a known place code.

Synthetic validation. We test the statistic on four kinds of synthetic unit generated along the real walks: invariant place fields, gain-modulated units with rate R times higher at one heading, conjunctive place$\times$heading units, and view-dependent units. At the real units' reliability (stability 0.81–0.91), invariant place fields pass the six-pair test at 98.9%, conjunctive units at 0.0%, and view-dependent units at 0.0%. Gain-modulated units pass at 73.9% for a fivefold rate change ($R = 5$); half pass at about $R \approx 7.4$ (Figure 9a,b,c). The path-integrating RNNs themselves have 8%–28% of the units that the looser two-pair average counts invariant (Figure 9c) with more than a twofold rate change between opposite headings ($G > 1$). Heading-dependent rate does not rule out a position code, which is why the test uses correlation rather than rate. Reliability matters: invariant synthetic units pass at 21% with stability 0.5–0.6 and 90% at 0.7–0.8. Estimator details and additional conditions are in Appendix E.

4.3 Units across models

On single frames, at most 0.3% of VLM place units pass the strict test, and at most 1% in any model scored without anchors (Figure 10a,b). VGGT with 8 anchors reaches 34%.

No model shows a grid population at spacings the scorer can detect (at most 0.3% of units flagged; Appendix E).

The moving-average account. On a foraging walk, heading changes faster than position, so averaging recent frames blends many headings at nearly the same place. This is the empirical counterpart of Franzius et al. (2007), who showed that slow feature analysis gives orientation-invariant place fields when rotation is fast.

With the walk in context, Qwen3-VL-2B's heading-invariant share rises to 25% (Figure 10b). The base 8B reaches 2%. A moving average of single-frame activations with a half-life of 8 frames and no model in the loop reproduces the same pattern. For Qwen3-VL-2B, 84% of place units become heading-invariant. For raw pixels, 95% become invariant. We do not claim the model computes this average internally; the point is that temporal smoothing alone is enough to explain the walk's effect on heading invariance.

VGGT's units. VGGT computes position, so its units should carry a position code. The question is how far that code generalizes across headings.

Its behavioral scores are in Section 3.2. We score 4,096 MLP neurons per block at the **camera token**, the per-frame summary vector that a small readout network, the **camera head**, turns into a predicted pose. We score 7 of 24 blocks (blocks 2, 5, 9, 12, 15, 19 and 23; block 21 is added with anchors).

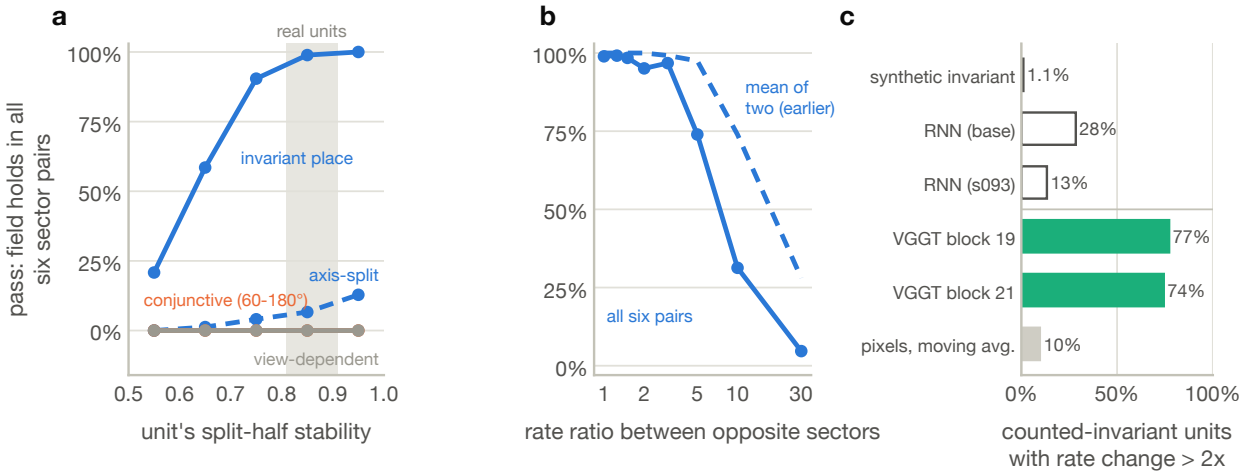


Figure 9. The strict test (field in all six sector pairs) counts invariant place units and rejects view-dependent, conjunctive and axis-split units at the real units' reliability, but it is less tolerant of rate changes than the mean of two. Synthetic units whose answer is known, scored like the real units. (a) Share passing the six-pair test against each unit's split-half stability, for invariant place units, axis-split units (one field facing N–S, another facing E–W, dashed), conjunctive units (60–180°) and view-dependent units. The gray band is the real units' stability; there invariant units pass at 98.9% and axis-split units at 6.9% (mean of two: 100.0%). (b) Invariant units whose rate differs between opposite headings, by rate ratio: the six-pair test (solid) passes half at about 7.4-fold, against 18-fold for the mean of two (dashed). (c) Share of counted-invariant units (mean of two, as in the synthetic validation) whose rate changes more than twofold between opposite headings: synthetic invariant units (1.1%), the path-integrating RNNs (bars: the higher of each RNN's two modes; all four 8%–28%), VGGT blocks 19 and 21 (77%, 74%) and pixels with the moving average (10%).

The strictly invariant share is 34% on a single frame with anchors, at block 19. This is intermediate between VLMs on single frames, at most 0.3%, and path-integrating RNNs at 97%–100% (Appendix E). At that block, the looser two-pair average counts 45% of place units invariant, and 77% of those change their rate more than twofold between opposite headings ($G > 1$). With the walk, the strictly invariant share rises to 52%.

VGGT's place units code position in the coordinate frame the anchors define, not in the room's own frame. When all anchor poses are rotated 45° about the room center at block 19, the rate maps turn with them ($r = 0.95$ against the turned map, vs 0.68 against the unturned map). This parallels in form the classic finding that hippocampal place fields turn with a rotated cue card (Muller and Kubie, 1987). Details are in Appendix K.

A linear readout reads VGGT's position almost perfectly (R^2 0.99 at block 21); Section 5 tests whether the camera head uses the directions that readout finds.

4.4 The fine-tuned model

The fine-tuned model has no full place code in the units we record. Only 8%–17% of its place units pass the strict test with the walk at layer 35 (the path-integrating RNNs, which validate the statistic, reach 97%–100%). These shares overlap those of base models with the walk (2%–29%), so the unit test does not separate the model that solves the task from those that do not. The walk shares are undercounts: at layer 35's stability, a truly invariant synthetic unit passes only 59% of the time, and the shares are not reliability-matched against the RNNs, whose units are near-perfectly reliable. Single-frame shares are 0.1%–0.3% (Figure 10). A moving average of single frames, which computes nothing spatial, reaches a higher share (82%–84%): this is the

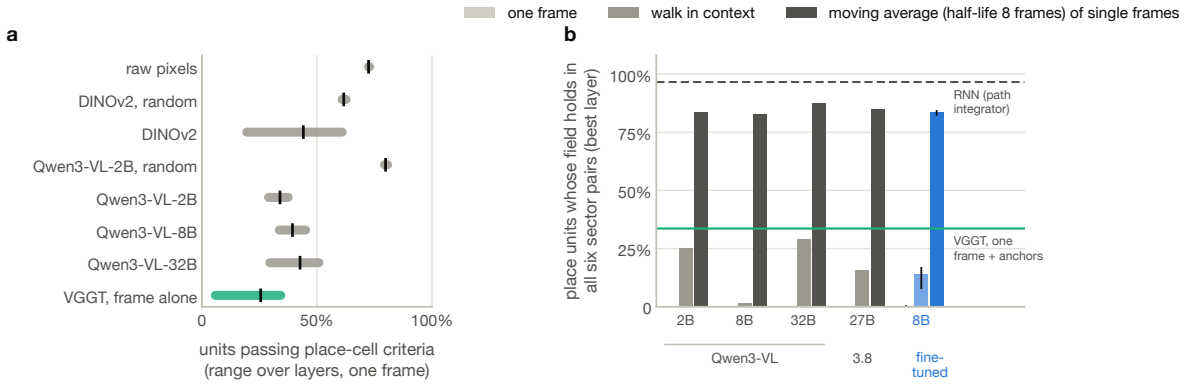


Figure 10. Place-cell criteria pass at every layer; heading invariance is the test. (Best layer per condition; strict test = the field holds in all six sector pairs. Short labels: 3.8 = Qwen3.8-27B, fine-tuned = the fine-tuned model (Qwen3-VL-8B).) (a) Place-cell pass rate on single frames, range over layers; VGGT (green) scored without anchors. (b) Share of place units passing the strict test, for one frame (light gray), the walk in context (gray) and a moving average (half-life 8 frames) of single frames (dark gray). The fine-tuned model's single-frame share is under 1% (no visible bar). The fine-tuned model's group shows bar = mean over its five runs, whisker = range (the walk shares are undercounts at that layer's stability: a truly invariant unit passes 59% of the time). Green line: VGGT, one frame with 8 anchors (block 19, 34%); dashed: the path-integrating RNNs (97%).

reference for how much invariance a walk gives for free. The partial invariance is confined to the last layers: at layers 3–29 the strict share stays at most 6.8%.

Task readout. The decision becomes readable from about layer 20 in all five runs, and position does not transfer across rooms in room coordinates (best cross-room R^2 at most 0.04). The test that the answer accounts for almost all the readout does not hold, though narrowly. Within a room, position is readable from the visual tokens at 1.43–1.53 m and from the last token at 1.90–2.04 m (chance 3.53 m; Appendix F).

4.5 Population assay

Temporal smoothing explains walk-frame HG in every Qwen-family model we recorded, and training gives the late layers a partial code for teleport views. For the population assay, we render 64 teleport views per walk at random headings. All teleport views go into one prompt after the walk frames, in random order; in the no-walk condition they appear alone. Of the 3,072 views scored, 2,690 are familiar and 382 novel (Section 2.2). The unit-level test counts individual neurons; the heading-held-out population decoding test asks whether the whole-layer representation carries position at headings it was not trained on. The metric is the heading-held-out gain

$$\mathrm{HG} = \frac{\text{chance error} - \text{error on an unseen heading}}{\text{chance error} - \text{error on the seen headings}}.$$

$\mathrm{HG} = 1$ means position fully carries across headings; $\mathrm{HG} = 0$ means chance on the unseen heading; values below zero mean worse than chance, for example mirrored predictions. As a calibration, HG 0.25 corresponds to about 11% invariant units, and 0.5 to about 24%. Values above 0.5 are positive, 0.25 to 0.5 partial.

Validation. Both path-integrating RNNs reach HG 0.98 (positive control). Every negative control (single-frame pixels, VGGT without anchors, single-frame Qwen3-VL-2B and Qwen3-VL-8B) gives at most -0.21.

On synthetic populations the assay rises steadily with the invariant fraction, so it is validated. In a population whose invariant units are spread across rotated directions, HG reads 0.46. A unit count on the same population (mean-of-two criterion) finds only 5%, so the assay finds a distributed code the unit test misses (Appendix E).

Walk frames carry position across headings in every model we recorded. In the base 8B, the walk gives HG 0.73 at layer 3. Across the Qwen family, every walk condition is positive at an early layer (0.70–0.87). Single frames are below zero in every trained Qwen-family model (-0.36 to -0.16). Random-weight 2B reaches 0.07, above zero but below the partial threshold. A moving average of single-frame features reaches 0.98, above the walk in all five base models by 0.11–0.28, and pixels with it reach 0.94. HG grows with the averaging length (0.82 at half-life 2 to 1.01 at 32). A shuffled walk is below zero (-0.12). The moving-average account now holds at the population level as well as at the unit level: temporal smoothing of single-frame features is enough to produce a heading-general population code.

VGGT reaches HG 0.57 on a single frame with anchors and 0.70 with the walk (layer 19).

Training changes the late layers. At layer 3, the fine-tuned model sits where the base model does (0.70–0.74). At layer 35, the base model has HG 0.58 and the fine-tuned model reaches HG 0.81, with 1.40–1.68 m on unseen headings against 0.96–1.18 m on seen ones (chance 3.74 m). Single frames stay below zero (-0.13 to -0.03). The moving average of single frames reaches the walk's HG in all 5 runs. In all five runs the assay is positive (Figure 11a). That late-layer walk code depends on the walk's order: on shuffled walks, HG at layer 35 falls to 0.28–0.35 (partial in every run), a drop of 0.47–0.50. The signal falls by more than half but is not erased, and what remains is close to the teleport-view values below. Like the walk condition, this is room A only.

For teleport views, the population code is partial. After the walk, HG is 0.39–0.49 at layer 35 (partial in every run). The base model's teleport-view HG is at chance (0.00 at its best layer with the walk, -0.27 at layer 35; Figure 11b), so the code comes from training.

Most of the teleport-view code does not depend on the walk's order. After a shuffled walk: 0.36–0.40 (partial). Without the walk: 0.20–0.30 (2 below partial, 3 partial), so much of the code is present before any walk is shown. Ordered minus shuffled is 0.06 on average, against the 28.0-point behavioral cost of shuffling in the held-out rooms. The matched cost, with the walk's order destroyed under identical labels, is 44.4 and 39.0 points in the two runs tested. Either way, most of this code does not depend on what the task needs.

Four limits apply. First, the thresholds were set on walk frames, so their application to teleport views is descriptive; a single frame alone scores higher on the teleport-view sample than on walk frames in every model. The base model on the same sample is therefore the better reference. Second, under the secondary split (training on the opposite sector only) the teleport-view HG is 0.12–0.23, against -0.29 for the base model. Third, the assay is within one room and does not say whether the code carries between rooms; the cross-room readout answers that (Section 4.4). Fourth, the assay scores teleport views recorded 64 at a time, not the task's single novel query, so the partial code may differ under task conditions.

Reading. Training gives the late layers a partial, within-room, heading-general position code for teleport views, which the untrained model lacks and which mostly survives shuffling the walk. The population assay and the place-by-view swap (Section 5.3) ask different things: the assay reads a partial code at the last layers; the swap shows the decision depends on where the walk frames were taken. Within a room, the population carries position across headings on walk frames and, partly, for teleport views. Across rooms, no position

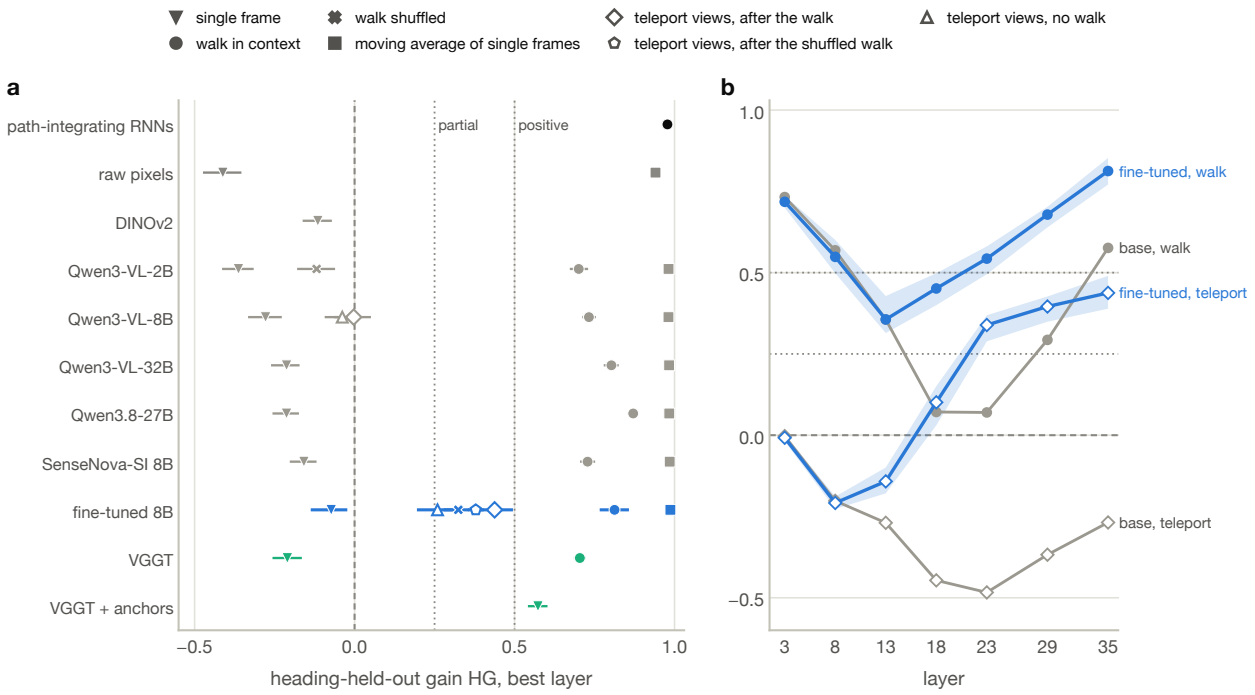


Figure 11. Training gives the late layers a within-room population code for teleport views that the base model lacks. (Dashed = 0 (chance on the unseen heading); dotted = the call thresholds, partial 0.25 and positive 0.5.) (a) Heading-held-out gain HG at each condition's best layer, one row per model. Filled markers: triangle = single frame, circle = walk in context, X = walk shuffled, square = moving average of single frames. Hollow markers = teleport views: diamond = after the walk, pentagon = after the shuffled walk, triangle = no walk. Lines are 95% CIs; in the fine-tuned model's row the mark is the mean over its five runs and the line the range. (b) HG by layer: base 8B (gray) vs the fine-tuned model (blue; line = mean, band = range), walk frames (filled circles) and teleport views after the walk (hollow diamonds). Run 0 in Appendix F.

code that a linear readout finds in room coordinates transfers, with at most a weak walk-relative component (Appendix F).

5 Causality: does a model that localizes use a position code

A probe can read position from activations the model never consults; a causal intervention is the stronger test. We apply it to VGGT, the 3D pose model that serves as a positive control (Section 4.3; Figure 12a), and to the fine-tuned VLM.

5.1 Pose subspace

We use **interchange patching** (Geiger et al., 2021; Vig et al., 2020; Zhang and Nanda, 2024): activations at the camera token from a **donor** frame replace those in a **target** run. **Transfer** is the share of the way the camera head's prediction moves toward the donor: 0 means not at all, 1 means all the way (128 targets; donors median 5.3 m away). Subspaces are fit on 2,048 other frames; for position PCA, we take the top principal components of the mean residual-stream token per 1 m position bin.

Patching the whole camera token transfers 0.41 at block 19 and 0.87 at block 21, but at most 0.001 through

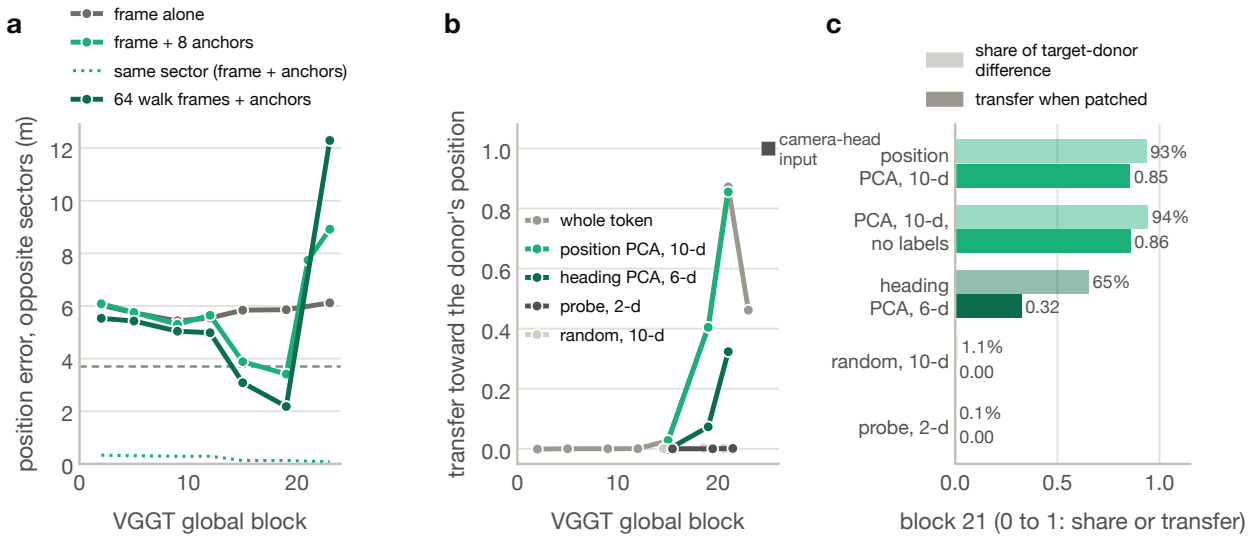


Figure 12. VGGT's dominant 10-d variation carries pose: copied between frames it moves the camera head, whether found with position labels or without. (128 targets; dashed = chance 3.70 m.) (a) Cross-sector decoding error by block (the older evaluation, which fails its positive control; Appendix E): frame alone, one frame with 8 anchors, walk with anchors. (b) Patching transfer by block (the drop at block 23 is because only the global half of the token is patched, 0.46). (c) Block 21 share (light) and transfer (dark) for each subspace, including PCA 10 with no position labels.

block 12 and 1.00 at the camera-head input (Figure 12b; Appendix Table K1). At earlier blocks, later blocks recompute position from tokens the patch did not touch (which tokens was not tested), so patching moves the output only from block 19 on.

The 10-d pose subspace. A 10-d position PCA subspace carries most of the causal signal, transferring 0.85 at block 21 and accounting for 93% of the target–donor difference (Figure 12c). Its complement and a random 10-d subspace transfer far less, and erasing it raises the camera-head error, the validations Makelov et al. (2024) recommend (Appendix Table K1).

The subspace carries heading as well as position: patching it turns the predicted heading by 77° at block 21. It is a causal pose subspace, not a pure position one. Position is just as decodable at block 15 (probe R^2 0.98), but patching there transfers only 0.03: decodability and causal use come apart (Hewitt and Liang, 2019; Belinkov, 2022; Ravichander et al., 2021; Elazar et al., 2021).

Position labels are not what finds the subspace. Ordinary PCA of the same activations, fit with no position labels, transfers 0.86 at block 21, against 0.85 for position PCA 10. A control subspace fit to the activation differences between other target and donor frames transfers 0.86. Ordinary and position PCA overlap 0.85: the result reflects the token's low-rank, pose-dominated variation, not something special about position labels (Figure 12c).

A heading-balanced fit changes the transfer by −0.05, within our margin of 0.1, so heading leaking into the position bins is not what makes the subspace move position. No two-dimensional subspace we tested comes close: position PCA 2 transfers only 0.18.

What the ten directions are. An exploratory analysis identifies the ten directions as three pairs of pose features (heading, camera-frame position and room-frame position) plus second-order terms. The six pose features explain 84.5% of the token's variance at block 21 (room A; Figure 13a–d; Appendix K). The same features explain 35.2% of raw pixels, and the camera-frame signal is also in pixels (20.9%), so that part is not specific to the token.

We patch each family alone and leave it out of the ten, testing which parts the camera head reads (Appendix Table K3; Figure 13e). L is the share of the transfer lost when a family is left out; A is the share it reaches alone. The label-free arm uses ordinary PCA directions; the feature-fit arm uses regression-derived directions.

Heading drives the predicted heading and camera-frame position drives translation; both are needed at both blocks in both arms. The room-frame family is needed at block 19 in both arms, but the two arms disagree at block 21: $L = 0.19$ in the label-free arm, $L = 0.32$ in the feature-fit arm. We draw no conclusion about this family at block 21. From block 19 to 21 the room-frame share of the token's variance falls from 19.8% to 9.9%.

The four losses sum to 1.16 times the ten's transfer (0.86), because position comes from heading and translation together, $C = -R^\top t$; L and A are therefore not shares of a whole. "Needed" means the head's output depends on that part at that point, not that the network builds its pose from it. All results are in one room, with one anchor set, at two blocks, and say nothing about the fine-tuned VLM.

Probe-derived directions do not recover the causal subspace. The readout from Section 4.3 reaches R^2 0.99 at block 21, yet patching along its directions transfers only 0.00. Raising the regularization strength to the largest value tested raises the transfer to 0.20, still well below the 10-d subspace's 0.85. The Haufe pattern (Haufe et al., 2014) and MLP-neuron interventions fare no better (Appendix K). Multi-dimensional realizations of geometric variables have been reported independently in vision-language-action policies (Jacobs et al., 2026) and video encoders (Joseph et al., 2026).

All interventions are at the camera token; other tokens are untested. The causal subspace carries over to room B and survives heading-matched donors and a fixed alignment (Appendix K).

5.2 Units and the subspace

VGGT carries the causal pose code in its residual stream. The write-vector test groups neurons by the mean-of-two heading criterion (the strict set was not tested; Appendix K). Invariant neurons put 0.057 of their writes' squared norm in the position PCA 10 directions, against 0.041 for non-place neurons; they also read from these directions. The steering tests in Appendix K used the older 180° halves, which pass 92% of synthetic place-by-heading (conjunctive) units. The conclusion is the same under either criterion: swapping the invariant neurons, or all 4,096 MLP neurons, transfers at most 0.002. The units point into the pose subspace but do not carry it.

5.3 The fine-tuned VLM

Attention cuts. The near neighbors matter most. In two of the five runs, we block every token outside a set of 8 walk frames from attending to those frames, testing which walk frames the decision reads. Tokens inside the set still attend to each other, and the frame labels stay visible. Hiding the near-option neighbors costs more than hiding random frames (near minus random -6.0 and -5.5 points). The near neighbors also hurt more than the far-option neighbors. Whether the look-alike cost exceeds half the near cost is inconclusive: in both runs the log-odds difference between the two has a CI that includes zero (Figure 14a; per-run values in Appendix F).

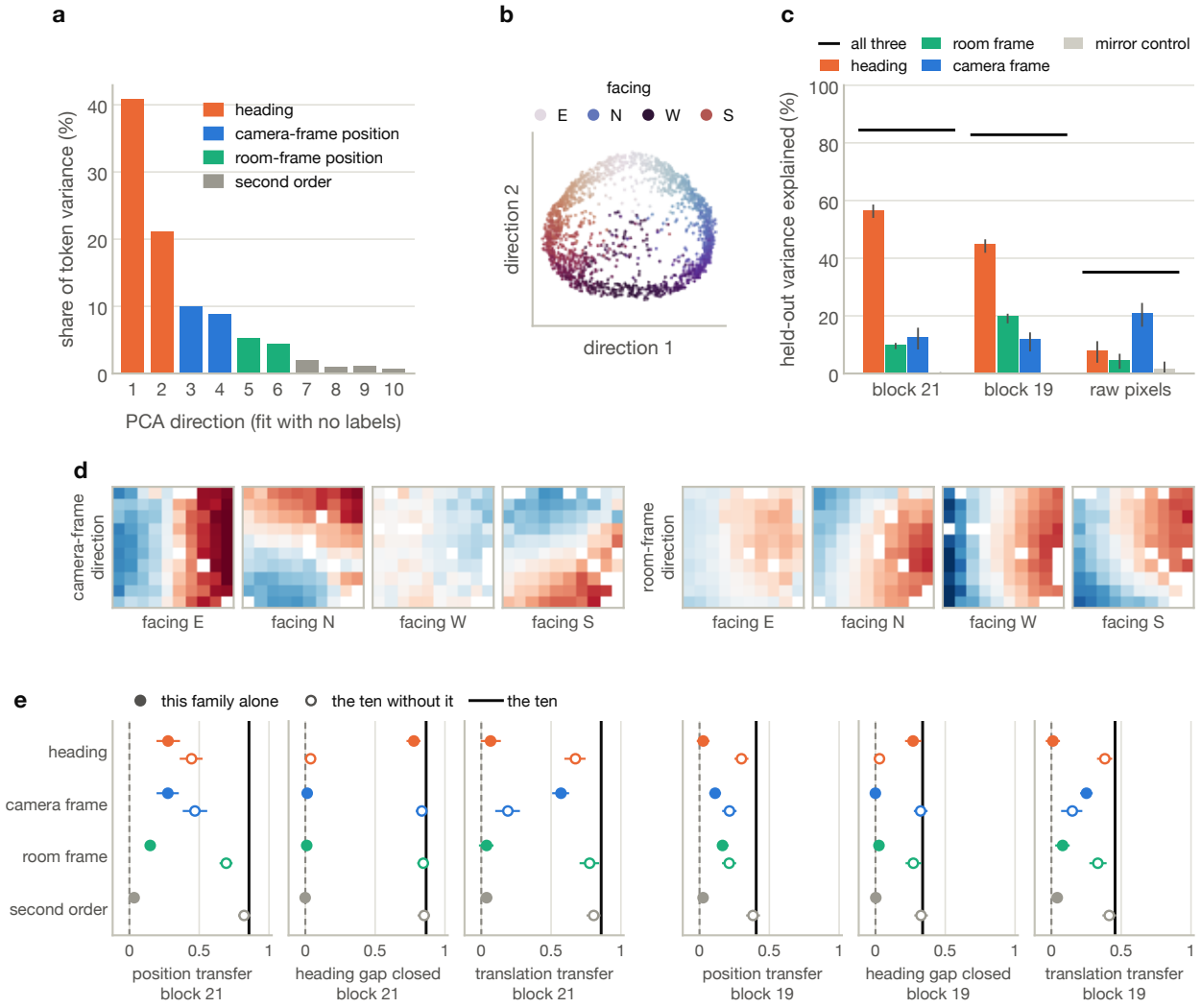


Figure 13. Six pose features explain most of VGGT's camera token, and the head's two outputs depend on different parts of it. Room A. Panels (a)–(d) are descriptive and scored on frames the PCA did not see; (e) is the label-free arm of the patching test. (a) Share of the token's variance in each of the ten PCA directions at block 21, colored by the feature family that explains each one better. (b) Directions 1 and 2 place the camera's heading around a ring. (c) Held-out variance explained by each family at two blocks and in raw pixels (all three = heading, room frame and camera frame together; mirror control = the camera-frame products with the rotation reversed; whiskers = 95% intervals over walks). (d) Mean score at block 21 along the direction of a pair that best follows one feature, one map per heading sector, the room seen from above in 1 m bins (red = high, blue = low, white = fewer than three frames). The camera-frame gradient turns with the camera and is weakest facing W; the room-frame gradient stays put. (e) Each family patched alone and left out of the ten, read on three outputs at two blocks (whiskers = 95% intervals over the 128 pairs; dashed = no movement).

The place-by-view intervention. We re-render the near option's 8 shown frames in a 2×2 design, crossing place with view direction. Here/there means taken near or far from the option; toward/away means facing toward or away from the point the new view looks at. The swap acts on the input: the frames that showed the near option's neighborhood are replaced with renders from a different location, a different heading, or both.

The decision depends more on where the walk frames were taken than on what they show (1,149 novel trials across nine rooms in two of the five runs). Place accounts for 0.76–0.79 of the swap gap; view accounts for

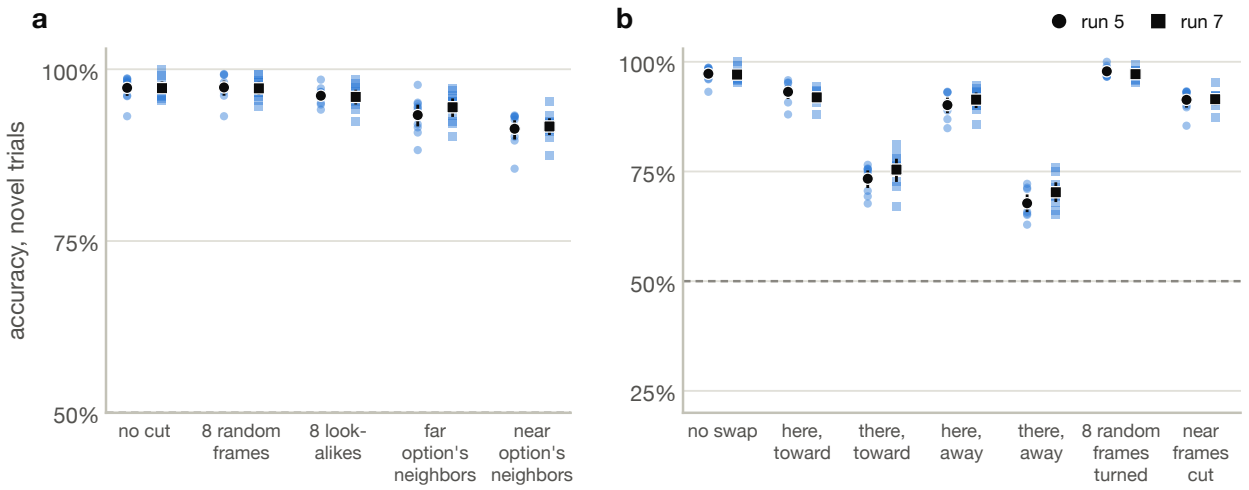


Figure 14. The near neighbors matter more than the far ones, and the decision depends more on where the walk frames were taken than on what they show. Circles and squares = two of the five runs; small marks = rooms, black mark and line = mean and 95% bootstrap CI. The two panels have different y ranges: (a) starts at 50%, (b) at 20% with a dashed chance line. (a) Attention cuts: no cut, 8 random frames, 8 look-alikes, the far option's neighbors, the near option's neighbors. (b) Place-by-view swap on the 1,149 included trials: no swap, here-toward, there-toward, here-away, there-away, 8 random frames turned away, and the near frames' attention cut. Place carries 0.76–0.79 of the swap gap, view 0.19 (two of the five runs).

0.19. Moving the frames to a different place costs 21.7–23.9 points even when they still face the point the new view looks at. Turning them away in place costs 5.7–7.1 (Figure 14b; per-run values in Appendix F).

Most of the place effect is harm from wrong-place frames rather than help from right-place ones: there-away is 21.2–23.6 points worse than hiding them, while here-away is level with hiding. No swap restores clean accuracy: here-toward stays 4.1–5.2 points below clean. The there-away frames face away from what the new view shows. A view-similarity rule (the "view agent") checks that the swapped frames do not accidentally match the new view; "where they were taken" holds as far as that check holds.

Broken-walk control. In both runs tested, the cost comes from where the frames go, not from breaking the walk. A separate control moves frames other than the near option's (678 trials in 9 rooms, in the same two runs). Moving a block of frames away from both options costs nothing; moving the far option's set while it stays far costs nothing either. Moving the far set to within 1.5 m of the new view pulls the decision toward the far option by 10.2 to 11.1 points (Figure 15). The two conditions make the same break in the walk and differ only in where the far set ends up, so the cost is not a broken-walk penalty.

The pull is not visual: the view agent's score stays under the 5-point limit, and the far-close frames look less like the new view than the far-far frames do. The pull is partial: accuracy stays well above 50%, so moving an option's neighbors toward the new view pulls the decision but does not settle it (per-run values in Appendix F).

6 Related work

Place and grid cells in networks. Path-integrating recurrent networks produce grid-like units (Banino et al., 2018; Cueva and Wei, 2018), and the Tolman-Eichenbaum Machine (Whittington et al., 2020) and

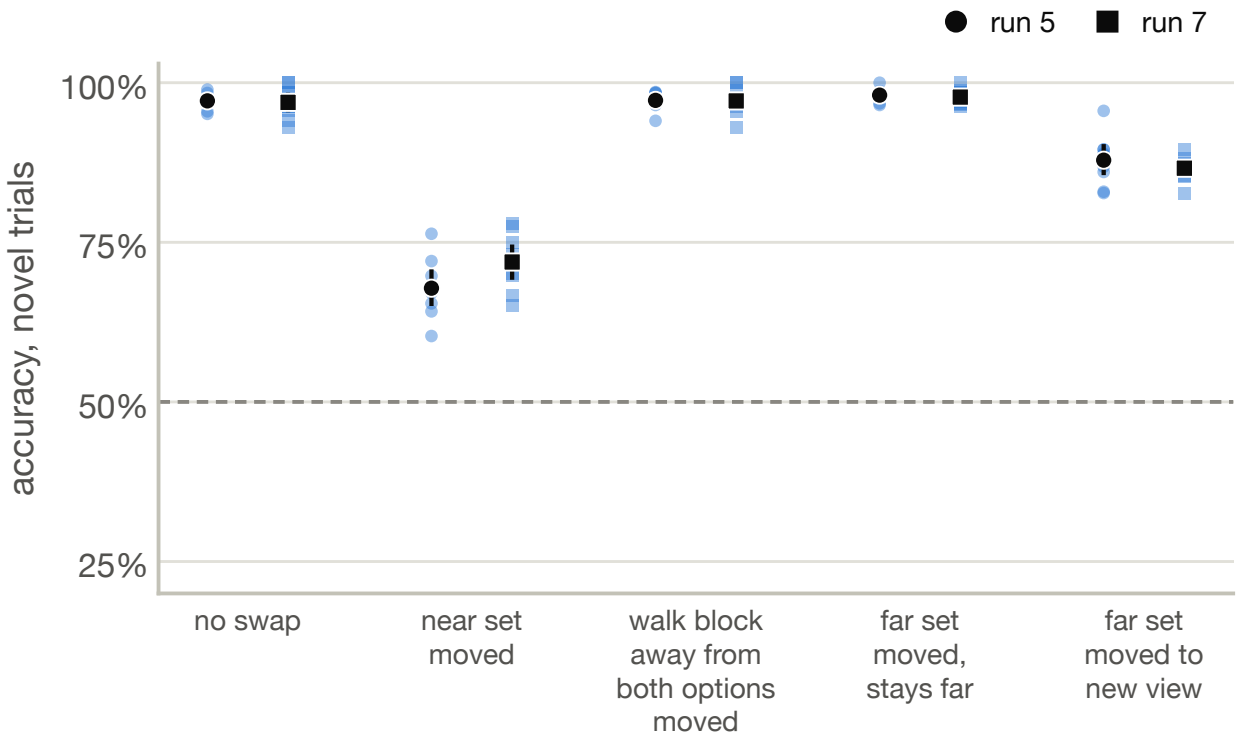


Figure 15. The cost comes from where the frames go, not from breaking the walk. All moved frames face away from what the new view shows. On the control's own 678 trials: moving a block away from both options (W) or moving the far set while it stays far (F) costs nothing; moving the far set near the new view pulls the decision toward the far option (P, 10.2 and 11.1 points). Swap gap G: 29.3 [26.8, 31.9] and 25.0 [22.9, 27.7] (9 rooms, two of the five runs; circles and squares = the two runs).

transformers with recurrent position encodings (Whittington et al., 2022) produce both place- and grid-like units, with a theory linking hexagonal grids to the training objective (Sorscher et al., 2019, 2023). These criteria are fragile. Schaeffer et al. (2022) showed that grid cells depend on the readout, the targets and the hyperparameters, and Luo et al. (2025) found such units in random-weight networks. We extend this to raw pixels and add a heading-invariance test. Franzius et al. (2007) derived orientation-invariant place fields from slowness in a simulated rat's visual input. We test for such invariance in pretrained VLMs, DINOv2 and VGGT, with pixel and random-network baselines and a causal test in VGGT. Vijayabaskaran and Cheng (2022) applied a similar heading split to a deep Q-network.

Place-like representations also appear outside spatial navigation networks. Gurnee and Tegmark (2024) recovered real-world coordinates from an LLM, and Gurnee et al. (2026) found place-cell-like tiling of a non-spatial variable with causal validation in a language model. In the vision and embodiment setting, Uria et al. (2022) found place-like units in a visual prediction network and Wijmans et al. (2023) found map-like representations in blind navigation agents. Martorell (2025) found position-correlated units in a text-only grid world but no place-cell-like units.

Spatial behavior of VLMs. VLMs fail when the query view differs from the reference. This pattern appears in VSI-Bench (Yang et al., 2025), MindCube (Wang et al., 2026b), cross-view benchmarks (Deng et al., 2026; Zhang et al., 2026b; Lin et al., 2025a; Khangaonkar et al., 2026) and trajectory studies where context barely helps (Ravi et al., 2025); chain-of-thought lowers spatial accuracy slightly (Kancheti et al.,

2026). Mahendran et al. (2026) reach the same "recognition, not place" reading from behavior on km-scale real video. We add a controlled novel-view test with lure trials, order controls, units, probes and interventions. Behavioral failure alone need not mean the model lacks a map (Beckmann et al., 2026); our negative claim for open VLMs rests on units and probes as well as behavior.

Several groups fine-tune VLMs on spatial tasks: GST-Bench (Zhang et al., 2026a), SAT (Ray et al., 2025) for simulator spatial QA, and Same Space or Not (Cao et al., 2026) for four-view consistency with answer-only LoRA. We use answer-only loss, a different task and an order test. Agentic pipelines instead hand the 3D structure to explicit multi-view geometry and keep the VLM for the query's semantics (Wang et al., 2026a). Mechanistic interpretability of multimodal models is less developed than that of language models (Lin et al., 2025b). Kang et al. (2026) decode and steer image-plane object positions. Wang and Gao (2026) find a causal linear subspace for objects' 3D layout in VLMs and improve spatial benchmarks by regularizing it during fine-tuning. Our test differs: we decode the viewer's own position across opposite headings, not the layout of objects, and find that open VLMs carry no such code for the query view. Shi et al. (2026) show a causal subspace for object coordinates in a viewer-centered frame. Liao et al. (2026) show that some decodable axes are priors ungrounded in the image, the same "decodable is not used" lesson in a different form from the VGGT block-15 dissociation we find. Sharma et al. (2026) train a conditional diffusion model to generate a bird's-eye map and read localization from the generated marker, which they describe as a functional analog of a cognitive map. Our fine-tuned model shows that localization can succeed with only partial heading invariance, so the unit test here could answer whether such models carry a heading-invariant place code.

Probing vs causal use. Probes can read information the model does not use (Hewitt and Liang, 2019; Ravichander et al., 2021; Elazar et al., 2021). The gap between decodability and causal use has been confirmed empirically. Huang and Chang (2025) found that probes and activation patching disagree in a ViT, the same pattern we find in VGGT at block 15. Xun (2026) and Billa (2026) found similar gaps across training stages and layers. Subspace patching can give false positives (Makelov et al., 2024); we run such validations.

In a continuous world, Brodjian et al. (2026) find that probes decode the state but steering along probe-derived directions fails, the pattern we find in VGGT. Wang et al. (2026c) report that the best probing and steering layers differ in multimodal LLMs, a layer-level analog of the split between decodable block 15 and causally active block 19 in VGGT. Our interchange patching builds on Geiger et al.'s causal abstraction framework (Geiger et al., 2021, 2024); the Haufe pattern (Haufe et al., 2014), which recovers source directions rather than filter weights, transfers 0.20 at block 21, against 0.85 for the 10-d subspace.

Inside 3D pose models. Probing studies on 3D vision models have begun to appear. Bratulić et al. (2026) probed and knocked out correspondence heads in DUSt3R, VGGT and Depth Anything 3 (causal, but on pairwise geometry rather than an absolute position code). Ziliotto et al. (2026) probed co-visibility in VGGT without interventions. You et al. (2026) showed that deep cross-frame attention is essential for VGGT's pose, consistent with the late emergence we find. Chen et al. (2026) recovered relative camera motion nonlinearly from many encoders without a causal test. Bahmani et al. (2025) inferred causal use from probes, with ablations of the conditioning rather than of the probed activations; our patching shows that inference can fail. Whether probes and causal tests agree depends on the model: in Zhu et al.'s video diffusion model (Zhu et al., 2026), ego-motion probes and causal scores rise together; in VGGT they do not.

Table 1. Models × tests. Place units: peak pass rate. Heading-invariant share: place units that pass the strict six-pair test, on single frames unless noted (Appendix Table E1 gives the looser mean-of-two shares). Novel view: lure-free two-choice accuracy with the walk; a check means above chance ($p < 0.05$). Population code: the heading-held-out gain HG at the best layer (0.5 and above positive, 0.25 to 0.5 partial, lower values below partial; 1 = full transfer across headings, 0 = chance). Retrieval: leads every tested rule by the 15-point margin in the procedural rooms; in the photoreal scenes the lead is under the 15-point margin. The fine-tuned model's lead survives all three shifted-walk conditions, which rules out, as tested, a local sequence association fitted to the training walks. By row: Qwen3.8-27B passes the novel-view test only with thinking ($p = 0.0019$, one-sided; direct 58% does not). SenseNova and VGGT are scored on the original two-choice trials, which contain lures. For the fine-tuned model, the novel-view value is the room A mean of five runs; in the held-out rooms it is 97.1%. Its heading share is the walk at layer 35, an undercount at that layer's stability. Its population values are walk frames and teleport views, which come 64 per prompt, are mostly familiar and are not the task's single query. Its causal entry is an intervention on the input whose control rules out a broken-walk penalty in the two runs tested; the pull toward an option's neighbors is partial. In the Summary column, "teleport −" means that the base 8B is at chance on teleport views; the other base models were not scored on them. Opus is not clearly above the retrieval baseline in the held-out rooms. VGGT's place-unit value is its peak pass rate without anchors (block 2), separate from its six-pair share with anchors, which is also 34%; its causal entry is position PCA-10, which ordinary PCA without labels largely recovers. "Walk pop. +" is not evidence of a learned place code: a moving average (half-life 8 frames) of single frames matches or beats that walk gain (0.98; Section 4.5). — = not tested.

Model	Place units	Heading-inv. share	Novel-view task	Beats tested retrieval	Population code	Causal interv.	Summary
Path-int. RNNs	✓	97%–100%	—	—	HG 0.98 (positive) below partial	—	Positive control
Raw pixels	72%	0%	—	—	HG −0.41 below partial	—	View
Random Qwen3-VL-2B	81%	0%	—	—	HG 0.07 below partial	—	View
Random DINOv2	63%	0%	—	—	HG −0.02 below partial	—	View
DINOv2 (trained)	61%	1%	—	—	HG −0.11	—	View
Qwen3-VL-2B	37%	0%	— ×	—	positive	—	View; walk pop. +
Qwen3-VL-8B	✓	0%	50%	—	walk 0.73	—	View; walk pop. +; teleport −
Qwen3-VL-32B	✓	0%	— ×/✓	—	positive	—	View; walk pop. +
Qwen3.8-27B	✓	0%	58%/65% ×	—	positive	—	View; walk pop. +
SenseNova-SI 8B	✓	0%	38%	—	positive walk 0.77–0.85 (positive) teleport 0.39–0.49	—	View; walk pop. +
Fine-tuned Qwen3-VL-8B	✓	8%–17% (walk, L35)	✓ 98%	✓	(partial)	place	Behavior; partial inv.; walk pop. +, teleport partial; place (input)
Claude Opus 5.5	—	—	✓ 73%	×	—	—	Behavior only (API)
GPT-6 Astra	—	—	✓ 72% (room A)	—	—	—	Behavior only (API)
VGGT	34% (block 2, no anchors)	34% (intermediate)	✓ 104/104	—	positive HG 0.57	✓ 0.85	Intermediate; causally used

7 Discussion

Table 1 summarizes the results across models by the three legs of the question. **Behavior:** the fine-tuned model localizes novel views in every procedural room, beating every retrieval rule we tested (Section 3.4). Opus gains from the walk but does not clearly beat the retrieval baseline in the held-out rooms. In the photoreal scenes, the fine-tuned model's lead over retrieval falls short of the required margin. **Representation:** place-cell criteria are easy to satisfy (raw pixels produce 72%), so heading invariance and a population code are the sharper tests. On single frames, at most 0.3% of VLM place units are strictly invariant; a temporal moving average is enough to produce the invariance the walk adds (Section 4.3). On walk frames, every Qwen model we recorded shows a positive heading-held-out gain, and the same gain appears in the untrained models. For teleport views, the base model is at chance, but the fine-tuned model carries a partial population code that mostly survives shuffling (Section 4.5). **Causality:** VGGT's 10-d position-PCA subspace carries pose causally, and ordinary PCA without labels finds a largely overlapping subspace with the same transfer (overlap 0.85). In the fine-tuned model, in the two runs tested, swapped walk frames show that the decision depends more on where they were taken than on what they show (Section 5.3). In the same two runs, a broken-walk control confirms this is not a generic penalty.

The fine-tuned model localizes with only a modest heading-invariant share in the units we record. Few place units pass the strict test (Section 4.4), fewer than a moving average of single frames passes and well below the path-integrating RNNs that validate the statistic. At the population level, the heading-held-out gain is positive on walk frames and partial for teleport views, but only within a room (Section 4.5). No position code that a linear readout finds in room coordinates transfers across rooms. What supplies the remaining accuracy is an open question: the fine-tuned model is not a pure place reader, and the place-by-view design cannot isolate what else it uses.

The fine-tuned model and Claude Opus 5.5 differ in how they use the walk's order. Every run of the fine-tuned model does better with the ordered walk, with margins that vary between runs. Opus does better with the shuffled walk in the held-out rooms and the apartments; in the houses the difference is not significant. These shuffled walks keep the real frame numbers, so a model can recover the original order from the labels. GPT-6 Astra also gains from the walk in room A (72% vs 51%) and, like Opus, scores higher with the shuffled walk, though not significantly. Among the open-weight models we tested, none shows a detectable walk gain on novel trials in the procedural rooms.

In the procedural rooms, every run of the fine-tuned model leads Opus in every room, and the lead holds when the frame count is matched (Section 3.6). In the photoreal scenes the fine-tuned model leads on all trials, though not significantly in the apartments, where Opus scores higher on novel trials (not significant). Room-level cues may contribute in these multi-room environments. Whether either model builds an internal map remains open.

VGGT's heading invariance is intermediate, between VLMs and path-integrating RNNs (Section 4.3). A 10-d subspace at block 21 carries pose causally; no 2-d subspace we tested comes close (Section 5.1). Adding the "1 m east" direction moves the prediction east, about 0.18 m per meter of offset, but only 11% of the 10-d variance is linear in x, y. The subspace needs more than two dimensions, and leaving out the second-order directions loses little transfer.

In open environments, a hippocampal place cell is largely heading-invariant and updated by self-motion, its field rotating with a rotated cue card (Navratilova et al., 2012; Muller and Kubie, 1987; Moser et al., 2008); cells can become directional (Navratilova et al., 2012). No model we tested shows all of these properties.

The frontier models' walk gain cannot be traced to a mechanism, because their internals are not available. Opus does not clearly beat the retrieval baseline in the held-out rooms (Section 3.4), and both Opus and Astra gain from the shuffled walk, consistent with reference-view matching. In the two runs tested, the fine-tuned

model's decision depends on local order and on the direction of travel (Section 3.3). Hiding the near-option neighbors costs more than hiding random or far-option frames (Section 5.3). Yet no position code that a linear readout finds in room coordinates transfers across rooms (Section 4), and the training adds little general spatial ability on the two benchmarks we tested (Appendix F.4). Whether a richer code or a path-integrating component would do more stays open.

VGGT raises a different question. Its unit test and causal code come apart: the place-like units are intermediately heading-invariant, but the MLP adds no transferable position; the pose code is already in the residual stream (Sections 4.3, 5.1 and 5.2). A negative unit test in the VLMs is therefore a statement about hippocampal-like units at the sites we record, not proof that the model has no spatial code.

Open VLMs match views rather than places: they pick the visually similar frame and carry no heading-invariant position code for the query view. Training on walks closed this gap in the procedural rooms, and a VLM-based agent may learn to localize from its walk with only a partial heading-invariant code in the units we record.

What these results do and don't show. The fine-tuned model beats every retrieval rule we tested in the procedural rooms. This does not mean it uses no retrieval at all: retrieval plus local frame order, as implemented by the 149 rules, does not account for its accuracy. In the separation trials, geometric proximity wins over visual similarity, but this does not show a place code and does not exclude strategies that chain views across the whole walk (Section 3.4). Position is decodable from VGGT's residual stream at block 15, yet patching there barely moves the output (0.03); decodable is not the same as used. The place-by-view swap acts on the input, so it shows that the decision depends on where the frames were taken, not how the model computes that. The population assay is within one room; a separate cross-room readout finds no transfer in room coordinates. In the photoreal scenes, the fine-tuned model's lead over retrieval is under the required margin in the apartments and in the houses' strongest-retrieval bins, so those results do not separate localization from view matching.

Limitations

Every open-model failure we report holds at the inputs we tested and is not a general statement about a model or architecture. Most experiments use procedural rooms of one family; the held-out rooms come from the same generator. The eight Replica scenes use per-vertex-colored meshes, and the far option in the apartments and houses is often in another room. In the apartments the fine-tuned model's lead over retrieval is under the required margin; in the houses it clears the margin overall but not in the bins where retrieval does best (Section 3.4). Only one model family learned the task reliably (Qwen3-VL-8B). A second family (Gemma 4 12B) did not, in three runs (Section 3.6); whether other open VLMs learn the same skill, and with the same partial heading invariance, is untested. The training recipe narrows the run-to-run spread, but ordered-over-shuffled margins still vary between runs. The shuffled walk keeps the real frame numbers, so its accuracy includes any use of the labels; the relabeled shuffle removes this cue (Section 3.3). The two-epoch runs and the GPU-control experiment are reported in Appendix F.

The frame count is matched with Opus only in the held-out rooms; every photoreal comparison uses 64 frames for the fine-tuned model against 256 for Opus. On the longer walk, 256 minus 64 frames is -5.7 (Section 3.6).

The renumbering, block-shuffle and reversal controls, the attention cuts and the place-by-view intervention, with its broken-walk control, ran on two of the five runs; the ordered-versus-shuffled comparison ran on all five. The place-by-view intervention acts on the model's input. It shows that the decision depends on where the walk frames were taken, but not how the model computes that, and does not show a stored map.

Moving an option's neighbors toward the new view pulls the decision, but accuracy stays well above 50% (Section 5.3). Under the two-choice prompt, the decision reads out from layer 20 in every run, but position does not transfer across rooms (R^2 at most 0.04, room coordinates). Removing the shared poster changes only 4 of 38 answers. Gemini 4 Argon was not tested. The population assay records teleport views in a multi-view prompt, not the single novel query of the two-choice task; the partial code may differ on that task. The unit test and the population assay both read MLP neurons at the recorded layers; a code elsewhere in the network, for example in the residual stream, is not excluded. VGGT anchors are tested at block 19 in room A only, and all interventions are at the camera token. The unit test and the patching test act on different objects: MLP neurons and directions in the residual stream. At block 19 they connect (Section 5.2), but the MLP adds no transferable position.

API models provide behavior only, from single runs, and Opus's walk gain varies across rooms. Trial counts are modest (102 novel); no-walk runs lean to the second-listed option. The human baseline rests on two participants, one of whom barely watched most walks. Heading invariance is measured across 90° sectors: the strict unit test takes the minimum correlation over all six sector pairs. The population test is a linear readout, validated on the path-integrating RNNs (Section 4.5). Both tests use one token per frame. The older cross-sector decoder can fail even with an allocentric code (Appendix E).

Use of AI assistants. This project was completed with the assistance of Claude Opus 5.5, Claude Opus 4.6, and GPT-6 Astra.

Code and data

Code, trial data and result plots are at https://github.com/molinzhao27/vlm-place-cells (code MIT, figures and trial data CC BY 4.0). Large files (walk frames for the Claude test, raw generation records) are archived on Hugging Face: the dataset at https://huggingface.co/datasets/molinzhao/vlm-place-cells and the LoRA adapters at https://huggingface.co/molinzhao/vlm-place-cells-lora.

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A Prompts and trial construction

Experiment numbers in the appendices follow the numbering of the registration file; Table C1 maps each number to the name the main text uses.

Walk prompt.

> These are consecutive first-person views as you walk around a room. Each view is labelled with its frame number.

Followed by 256 labeled frames ("Frame 001:" through "Frame 256:"), then "New view:" with the query image, then the question.

No-walk prompt.

> These are two first-person views from a walk around a room, labelled with their frame numbers.

Followed by the two option frames with their labels, then "New view:" and the question.

Question.

> This new view was taken somewhere in the same room, possibly facing a different direction. Was it taken closer to where Frame *a* was taken, or to where Frame *b* was taken? Answer with one of the two frame numbers only.

where *a* and *b* are the two option frame numbers.

Lure-free trials. Both options face more than 90° from the query view's heading, neither shows any of the query view's posters or pillars, and pixel similarity favors the near option in exactly 50% of trials. Near-option median distance: 0.6 m; far-option median: 4.1 m. The frame-order cue (Section 2.3) favors the near option in 66% of novel trials. A **look-alike-then-time baseline** formalizes this cue: for each trial it finds the walk frame whose image is most similar to the query (cosine on z-scored 32×32 pixels or DINOv2 features) and answers whichever option is closer in frame number. In room A its eight variants reach 58%–69% (pixels 61%–66%, DINOv2 58%–69%). In the ten held-out procedural rooms the same rule falls to 54.6% (pixel, maximum room 57.3%) and 59.2% (DINOv2, maximum room 67.5%), far below the four-epoch model's 97.1%. The cue inflates room A accuracy and explains little of the held-out result.

Room B balancing. Room B trials (192 total, 106 novel) balance frame order by design: 50% of all room B trials have the frame-order cue favoring the near option.

Model settings. Open models use greedy decoding with no thinking unless stated. Claude models use adaptive thinking at a stated effort level (Claude Opus 5.5 cannot run with thinking off). Claude Sonnet 5.5's "thinking off" condition uses the `between_tools` setting (342 median output tokens). High effort: 1,160 median output tokens (Claude Sonnet 5.5). SenseNova-SI 8B (Cai et al., 2026), a spatial fine-tune of Qwen3-VL-8B trained on 8 million spatial QA samples, reports VSI-Bench 64.8 and MindCube-Tiny 73.8 for this 8B model. Hardware: a 96 GB RTX PRO 6000 for early open-model inference; a rented RTX 5090

Table A1. Room sets.

Room set	How made	Used for
Room A	Hand-built	Units, probes, VGGT, first behavior
Room B	New layout, training texture families	Behavior (held-out)
1,136 training rooms	Procedural generator	Fine-tuning
64 validation rooms	Same generator, separate seeds	Fine-tuning (validation)
10 held-out rooms	Same generator, seeds 5000–5009	Registered test
3 Replica apartments	Habitat-Sim, vertex-colored mesh	Photoreal (Exp. 12)
8 HSSD houses	Habitat-Sim, object-id labels	Photoreal (Exp. 16)
5 FRL apartments	Habitat-Sim, same floor plan	Photoreal (Exp. 19)
18 new HSSD houses	Habitat-Sim, object-id labels	Photoreal (Exp. 19)

Table A2. Models. VLM = vision-language model. "Record" = internal activations can be read. "API" = tested through a provider's interface (behavior only).

Model	Type	Access	Why included
Qwen3-VL-2B	VLM	Record	Scale ladder
Qwen3-VL-8B	VLM	Record	Base for fine-tuning
Qwen3-VL-32B	VLM	Record	Scale ladder
Qwen3.8-27B	VLM	Record	Best open VLM
Qwen3.6-35B-A3B	VLM	Record	Thinking model
Gemma-4-31B	VLM	Record	Non-Qwen family
Qwen3-VL-8B + LoRA	VLM	Record	Fine-tuned model
Gemma-4-12B	VLM	Record	Second fine-tune family
Qwen3.5-397B	VLM	API	Largest open Qwen
Path-integrating RNNs	RNN	Record	Positive control
Sequence matcher	Transformer	Trained here	Retrieval comparison
SenseNova-SI 8B	VLM	Record	Spatial fine-tune
DINOv2	Image network	Record	No language
VGGT	3D pose model	Record	Positive control
Claude Haiku 5.5, Claude Sonnet 5.5	VLM	API	Effort sweep
Claude Opus 5.5	VLM	API	Walk gain
Kimi K3	VLM	API	Other provider
DeepSeek V4.1 Flash	VLM	API	Other provider
GPT-6 Astra	VLM	API	Other provider
MiniMax-M3	VLM	API	Screen (amend. 5)
MiMo-V2.6-Pro	VLM	API	Screen (amend. 5)
Muse Spark Open	VLM	API	Screen (amend. 5)

for four-epoch training and most four-epoch inference; an RTX 4070 for the VGGT λ-sweep and later VGGT runs. Room B, the base and run-0 tests in the held-out rooms, and some scenes ran on Apple MPS (192/192 agreement with CUDA).

Without the walk, the base 8B scores 59% at 256 px and 71% when the same renders are upsampled to 512 px (more tokens, no new detail); with the walk it scores 49% at 512 px and 50% at 256 px. So the input setting moves no-walk accuracy, and the walk adds nothing at either setting.

Key terms.

- **Token.** One input slot in a transformer: a word piece, or one image or image patch. Each token carries a vector of internal activations that later layers read from and add to (the **residual stream**).
- **MLP neuron.** One dimension of the feedforward block inside a transformer layer. In VGGT each block has 4,096 such neurons. We score each neuron by building a **rate map** (its mean activity at each position on a grid) and applying the rodent place-cell criteria to it.
- **Subspace.** A small set of directions in the residual stream (for example, the top 10 principal components of the mean token per 1 m position bin).
- **Interchange patching.** Copy part of the activations from a run on a **donor** frame (at a different position) into a run on the **target** frame, and check whether the output follows.
- **Transfer.** How far the output moves toward the donor position: 0 = not at all, 1 = all the way.
- **Steering.** Set a neuron's or a subspace's activity to its value at a chosen location and measure whether the output moves there.
- **Camera token and camera head.** VGGT produces one summary vector per frame (the camera token); a small readout network (the camera head) turns it into a predicted camera pose.

B Retrieval rules (Experiment 20)

Experiment 20 scores 149 retrieval rules in 5 families on the same novel two-choice trials the models answered, using only the images and frame numbers (no position). The families are: (1) *option similarity*, which picks the option whose image is more similar to the query; (2) *look-alike, then time*, which finds the most similar walk frame and picks the option closer in frame number; (3) *window maximum*, which for each option takes the most similar walk frame within k frames of it and picks the higher match; (4) *window vote*, which counts how many of the most similar frames fall within k frames of each option; and (5) *combined*, a leave-one-scene-out logistic regression over all other rules' scores. Features are 4: pixels, DINOv2 CLS (z-scored and raw) and DINOv2 patch mean. Frame sets are all 256 walk frames (what Opus sees) and the 64 shown frames (what the fine-tuned model sees). Per family: option similarity 4 variants (one per feature); look-alike then time 24 (with $m = 1, 3, 5$ most similar frames); window maximum 40 (window sizes $k = 2$ to 32 frames, doubling); window vote 80 (same k with $m = 1$ or 5); the combined regression 1. Total: 149. The retrieval baseline in each group is the rule with the highest novel accuracy, chosen on the same trials, which favors retrieval. The registered rule requires the four-epoch mean to beat it by at least 15 points overall and in the bin where it scores best, across every breakdown. The 15-point margin was fixed at registration, before any new rule was scored; the registration gives no further rationale.

Overall numbers in this section are scene means over novel trials; bin and right/wrong numbers pool the trials, with CIs over scenes. The per-run table (Table F1) pools trials only for the apartments.

Highest-scoring rule per family. Procedural rooms (novel, scene mean): option similarity 51.7%, look-alike then time 61.1%, window maximum 64.6%, window vote 63.5%, combined 67.7%. Photoreal scenes: option similarity 58.8%, look-alike then time 70.0%, window maximum 74.9%, window vote 71.8%, combined 68.3%.

Best-bin breakdown (procedural rooms). The four-epoch mean leads the retrieval baseline (a logistic regression combining the other rules, fitted leave-one-scene-out) by 29.4 points [24.9, 34.2] overall. In the bin where the retrieval baseline scores best, the leads are: heading gap 90–120°: 27.2 [20.2, 33.4] (baseline 70.5%, model 97.7%); distance margin 4 m or more: 27.2 [22.4, 32.0]; novelty margin 1.5–2 m: 28.8 [23.8, 34.4]; visual overlap lowest third: 26.7 [20.9, 31.6].

Best-bin breakdown (photoreal scenes). The four-epoch mean leads the retrieval baseline (window maximum (DINOv2 CLS, all 256 frames, $k = 32$)) by 13.2 [5.6, 19.4] overall, under the 15-point margin. In the bin where the retrieval baseline scores best: heading gap 90–120°: 9.6 [3.9, 15.2]; distance margin 4 m or more: 11.7 [3.8, 18.3]; novelty margin 1–1.25 m: 11.3 [6.9, 15.8]; visual overlap middle third: 13.0 [4.4, 21.0]; rooms (options in different rooms, new view in the near option's room): 15.0 [8.3, 21.8], which meets the margin.

In the new houses, the fine-tuned model leads the photoreal retrieval baseline by 18.9 points on 947 novel trials. Where the far option is 4 m or more farther, the lead is 11.7, so the margin does not hold in the new houses.

Dependence on the retrieval baseline. The four-epoch model's accuracy depends little on whether the retrieval baseline is right: in the procedural rooms, 97.7% on trials it gets right ($n = 879$) vs 95.8% on those it gets wrong ($n = 420$). Opus's accuracy does depend on it: 76.7% vs 66.0%. In the photoreal scenes, the model scores 91.6% when it is right ($n = 287$) vs 81.2% when it is wrong ($n = 101$; 36 ties excluded). Opus: 89.2% vs 78.2%.

Apartments and houses. Apartments alone (58 novel trials): the retrieval baseline (window maximum (DINOv2 CLS, all 256 frames, $k = 32$)) scores 82.2%, level with the model (84.0%) and Opus (85.4%). Houses alone (366 novel): the retrieval baseline 72.1% vs the model 89.6%, a lead of 17.5 [13.1, 22.0]. Opus 84.8%. On the 176 house trials whose two options share a room, the retrieval baseline scores 64.5% vs the model 84.3% (lead 19.8 [9.9, 26.7]).

64-frame restriction (exploratory, not registered). Restricting retrieval to the 64 shown frames: the retrieval baseline (window maximum (DINOv2 patch mean, the 64 shown frames, $k = 32$)) scores 62.7% in the procedural rooms and 68.1% (window maximum (DINOv2 CLS, the 64 shown frames, $k = 32$)) in the photoreal scenes. The model leads by 20.0 [14.1, 25.6] in the photoreal scenes; against this restricted rule, too, the registered criterion does not hold in the photoreal scenes.

Sequence matcher (Experiment 28). The sequence matcher is a pre-LayerNorm transformer encoder with one token per shown walk frame and one for the new view, trained on frozen DINOv2-base CLS features z-scored within the prompt. Each walk-frame token combines a linear map of the feature, the cosine similarity to the new view, and a shared option marker. The ordered variant adds a frame-number embedding: sines and cosines at 16 periods, spaced geometrically from 16 to 1,024 frames. The order-free variant has no position input. We train on the fine-tuned model's own training trials: 9,083 trials from 1,136 rooms, with 64 validation rooms and 512 trials. The optimizer is AdamW (weight decay 0.05, batches of 64, 60 epochs, a two-epoch warm-up, cosine decay). We search a grid of layers $\in \{2,4\}$, width $\in \{128,256\}$, dropout $\in \{0.1,0.3\}$ and learning rate $\in \{3\times10^{-4},10^{-3}\}$ on validation accuracy. The best configuration is retrained with four more seeds, and the five checkpoints' mean accuracy is the matcher's score.

C Registration

Each experiment listed below was registered in `docs/experiments.md` with its date before that experiment's models saw any of the new data, with one exception, Experiment 25's amendment 1 (below). Experiments 22 and 24–32 were committed to the public repository before they ran (including Experiment 24's and Experiment 29's amendments). Experiments 15 and 16 were committed before they ran; the commit still reads "not yet run." Experiments 10–14 were first committed after their runs (for Experiments 11 and 14, after their first arms), as were Experiments 9c and 17. Experiments 20, 21 and 23 were first committed on 2026-10-28, after their runs. Experiments 18 and 19 were first committed on 2026-10-28, after their first runs had started. The dated log entries record the order of registration and running; the public commit history confirms the timing for Experiments 15, 16, 22 and 24–32. Room A, room B and the analyses developed on them (Experiments 1–9) were examined before the registered experiments. The registered experiments listed below are the confirmatory tests; analyses labeled exploratory or post hoc are follow-ups.

Table C1 gives each experiment's rule, its outcome and its timing; the entries after the table give the details and the amendments.

Table C1: Each experiment's rule, outcome and timing. Logged: the date on the experiment's entry in `docs/experiments.md` (2026). Committed: when that entry first reached the public repository, relative to the experiment's runs. "After" means logged before the runs and first committed publicly after them; "during" means committed after the first runs had started. Name: what the paper calls the experiment; the main text doesn't use the numbers. "No numeric rule" marks entries that fix an outcome and report it either way; am. = amendment.

Exp.	Name	Rule and threshold	Outcome	Logged	Committed
9c	four-epoch recipe	The four-epoch recipe is reliable if every run reaches at least 85% on the validation rooms, the five-run mean on room A's novel trials is at least 85%, and no run is below 80% there.	Rule met	10-25	after
10	held-out rooms	Opus's walk gain in the held-out rooms holds up if the 95% CI of its mean excludes 0.	Holds: +9.7 points [7.0, 12.1]	10-22	after
11	other providers	Walk gain: two-sided sign test on novel trials, $p < 0.05$; the claim that Opus alone gains stands only if no tested model passes. Open-weight screen: $p < 0.00625$.	GPT-6 Astra gains, so the claim does not stand; no gain for Kimi K3, DeepSeek V4.1 Flash or Qwen3.5-397B; no model passes the screen	10-23	after its first arms
12	Replica apartments	No numeric rule: accuracy on all lure-free trials in each scene and pooled.	Reported	10-23	after
13	order controls (a), irrelevant walk (b), patching controls (c), appearance cue (d), strict heading split (e), RNN control (f)	13c: complement transfer below 0.05; fixed-alignment and heading-matched transfers within 0.1 of the originals. 13f: both RNNs keep a clear majority of place units on the strict split, with cross-heading error well below chance. The other parts: predictions with no numeric rule.	13c holds. 13f: the unit test passes; cross-sector decoding fails in the base RNN. The other parts reported	10-23	after

Exp.	Name	Rule and threshold	Outcome	Logged	Committed
14	task readout (a), Opus shuffled walk (b), poster removal (c), cross-room patching (d), anchor controls (e)	14b: "Opus does not use the walk's order" only if the 95% CI of ordered minus shuffled lies within ± 3 points. 14d: room-A-fit transfer in room B within 0.1 of room-B-fit transfer. The other parts: no numeric rule.	14b not claimed: the CI lies outside ± 3 points because the shuffled walk is better. 14d holds. The other parts reported	10-23	after its first arms
15	attention cuts, run 0	Gate: the clean run gives Experiment 10's answer on at least 99% of trials. Predictions for the near, far and look-alike cuts, with no numeric rule.	Gate passed; cuts reported	10-24	before
16	HSSD houses	No numeric rule: accuracy on all lure-free trials, per house and pooled.	Reported	10-25	before
17	standard spatial benchmarks	A contrast helps if its scene-bootstrap CI lies above 0, hurts if below 0, and is no change if within ± 2 points (VSI-Bench) or ± 5 (MindCube-Tiny). Contrasts: run 0, the two-epoch mean and the four-epoch mean, each against the base model. A benchmark transfers if a mean helps and no contrast hurts, and hurts if a contrast hurts and none helps.	VSI-Bench hurts; MindCube-Tiny transfers, with the models near chance	10-26	after
18	Gemma fine-tune	As Experiment 9c, on Gemma 4 12B, for five runs. Amendment 2 stopped at the three runs already trained, using a three-run form of the rule first written for a slow fit check.	Rule not met	10-27	during
19	new scenes	Transfer if each four-epoch run beats the base model, no walk and the shuffled walk (new houses: scene bootstrap CI above 0; FRL apartments: pooled sign test, $p < 0.05$). Opus's walk gain replicates at $p < 0.05$ (pooled sign test). LoRA minus Opus is reported with no threshold.	Houses: met; FRL apartments: met; Opus's gain: met	10-27	during
20	retrieval rules	In each group (held-out rooms, photoreal scenes), the four-epoch mean beats the retrieval baseline by at least 15 points overall and in the bin where it does best in each breakdown. Retrieval is rejected if both parts hold in both groups.	Held-out rooms: holds; photoreal scenes: does not hold, so retrieval is not rejected. New houses, reported apart: does not hold	10-28	after
21	heading-statistic validation	Validated if, at the real units' reliability, at least 90% of invariant units score above 0.5 and at least 90% of conjunctive and of view units score at most 0.5 (mean of two). Amendment 1, all six pairs: closed if each RNN condition keeps at least 95% of its place units.	Statistic validated; amendment 1 closed	10-28	after; am. 1 before

Exp.	Name	Rule and threshold	Outcome	Logged	Committed
22	four-epoch units and task readout	Units carry over if every run's walk share is at most 25% and its single-frame share at most 6%, do not if any walk share reaches 85%, and otherwise carry over partly (mean of two). No room-general position code if the best cross-room R^2 is at most 0.10. Decision layer 18–22 in every run. The readout is mostly the answer if the residual stays at most 0.10 at layers 20–30.	Units: partly (all six pairs: 8%–17%); no position code: carries over; decision layer: carries over; mostly the answer: does not hold, narrowly	10-28	before
23	units and the subspace	At block 19, connect if the invariant neurons' median fraction in the subspace is at least twice chance and above the other neurons' (CIs above 0). They carry the causal code if their interchange transfer is at least 0.1 and above each comparison set's.	The neurons connect; no measurable causal role (no neuron set at this token reaches 0.1)	10-28	after
24	256-frame comparison	The lead survives if the five-run mean at 256 frames minus Opus at 256 frames is at least 10 points with its CI over rooms excluding 0, shrinks if the CI excludes 0 under 10 points, and is gone if the CI includes 0.	The lead survives	10-28	before
25	population assay	Heading-held-out gain HG: positive at 0.5 or above, partial from 0.25, negative below that; no position code if the seen-heading readout beats chance by less than 0.5 m. Validated if both anchored RNNs are positive, every negative control is negative and the synthetic mixtures rise monotonically.	Assay validated; fine-tuned model on walk frames positive in all six runs. New-view probes are outside the rule	10-28	before
25, am. 1	four-epoch shuffled walk	D is walk HG minus shuffled-walk HG at layer 35, per run. Depends on order if in every run D is at least 0.25 and shuffled HG is not positive; does not depend on order if in every run D is under 0.10 and shuffled HG is positive; otherwise mixed.	Result: depends on order in all five runs ($D = 0.47$–0.50; shuffled HG 0.28–0.35)	11-01	before
26	walk-relative probes	Best cross-room R^2 at most 0.10: absent. At least 0.30 and at least 0.20 above its walk-shuffled controls: present. At least 0.30 without that: not walk-specific. Otherwise weak.	First-frame position: base absent, four-epoch runs weak. In those runs heading weak, last-frame position absent	10-28	before
27	subspace controls	Block 21 is primary. Position-specific if position PCA 10 transfers at least 0.2 more than ordinary PCA 10 and than the difference-matched control, both CIs above 0. Generic if the difference from ordinary PCA 10 is under 0.2 or its CI reaches 0.	Subspace generic at block 21, generic at block 19	10-28	before

Exp.	Name	Rule and threshold	Outcome	Logged	Committed
28	trained sequence matcher	D, the four-epoch mean minus the matcher in the held-out rooms: at most 5 points, learned matching suffices; at least 15, it falls short; otherwise partial.	Result: learned matching falls short ($D = +23.8$)	10-28	before
29	place-by-view intervention	Valid if the swap gap G is at least 5 points with its CI above 0 and the random cost is at most $G/2$. Uses place if $R_{\text{place}} \geq 0.5$ with the place effect's CI above 0, $R_{\text{view}} < 0.5$ and the conflict's CI above 0.	Uses place in runs 0, 5 and 7	10-28	before
29, am. 1	broken-walk control	Not a generic penalty if $W \leq G/4$ and the upper end of W's CI is below $G/2$. Directional if $P \geq 5$ points with its CI above 0. Place confirmed if both.	Result: place confirmed in runs 5 and 7; run 0 not tested	10-30	before
30	shifted walks	A shift survives if the lead over the retrieval baseline on the same trials is at least 15 points with its CI above 0 and the drop from control is at most 10; it fails if the lead is under 5 or its CI includes 0; otherwise it is reduced. Geometric proximity wins if K is at most 10 points; local association if conflict accuracy lies at most halfway from chance to agreement accuracy.	Long-step survives, low-turn survives, patrol survives; control and patrol walks: geometric proximity wins	10-28	before; am. 1 after the run
31	order controls and attention cuts, four-epoch	A criterion replicates if its CI lies wholly on run 0's side of its threshold, differs if wholly on the other side, and is inconclusive if it contains the threshold. Order controls: a gate and four criteria. Attention cuts: three criteria.	Order controls: replicates; attention cuts: inconclusive	10-28	before
32	per-part patching	L is the share of the transfer lost without a family, A the share it reaches alone. Needed if $L \geq 0.25$ with its CI above 0; not read if $L < 0.10$ and $A < 0.12$, each CI's upper end under 0.25; read but redundant if $L < 0.10$ and $A \geq 0.25$. The headline is the room-frame family at block 21, label-free arm; if the two arms disagree, the claim follows the weaker verdict.	Heading and camera frame needed and second order not read at both blocks. Room frame needed at block 19; no verdict at block 21, where the arms disagree	10-31	before

The "Logged" column gives the date each experiment's entry was written in the registration file; the "Committed" column records when that entry first reached the public repository, relative to the experiment's runs. Every experiment was registered before the models saw the data it analyzes, with two exceptions: Experiment 25's amendment 1, whose activations already existed but had never been scored, and Experiment 30's amendment 1, submitted after the rerun.

Experiment 10 (ten held-out rooms): registered 2026-10-22, run 2026-10-23. Amendment 1 (before any result from rooms P5001–P5009): the Claude budget forced Opus to drop the shuffled-walk condition, so only walk and no-walk were run. Amendment 2 (from trial counts only, before any model saw any room but P5000): rooms with fewer than 100 novel trials at 16 walks were re-rendered with 32 walks, capped at 6 trials per walk. Deviation (found by external review): the runner defaulted to 16 walks, so Opus's first pass

covered only walks 0–15 in six rooms (859 of 1,299 novel trials); the remaining walks have since been run. On the first-pass trials, the walk gain was 9.6 [6.8, 12.2]. The base model's shuffled walk scores 56% [55%, 58%], above chance and higher than in order in 9 of 10 rooms (we report this without explanation beyond the data). No-walk runs lean to the second-listed option (63% base, 78% LoRA). Run 0 was the registered model for the held-out rooms; later runs are in Table F1. Mean of the three two-epoch runs: room A 82% (70%–92%), room B 79% (62%–95%).

Experiment 11 (other providers): registered 2026-10-23. Amendments narrowed the model list before the corresponding runs; the registration lists them with reasons. Final scope: Kimi K3, DeepSeek V4.1 Flash and GPT-6 Astra. Gemini 4 Argon was dropped before any trial (amendment 4). Amendment 5 screened 8 newer open-weight VLMs via OpenRouter; 3 accepted the walk (MiniMax-M3, MiMo-V2.6-Pro, Muse Spark Open), none gained ($10.18); 5 were untestable (context or image limits). Amendment 6 tested Qwen3.5-397B separately with a 160-frame walk (the OpenRouter request-size estimate blocked the full 256-image walk); no walk gain (52.9% vs 56.9%, $p = 0.63$). Astra's registered shuffled-walk follow-up in room A was run; the held-out rooms were not (budget).

Experiment 12 (photoreal): registered 2026-10-23. Amendment: VGGT stride changed to every 8th frame (32 walk frames) before any VGGT result. Deviation: Opus's first pass covered only 16 walks (the same runner bug as Experiment 10); the remaining walks were then run.

Experiment 13 (controls): registered 2026-10-23. Experiments 13d (DINOv2 appearance cue) and 13e (strict opposite-sector heading split) have results, as do 13a–c.

Experiment 13a (order controls, ten held-out rooms): ordered minus renumbered 0.6 [−0.8, 1.8]; reversed walk 67.8%; window-shuffled 81.9%. **Experiment 13b** (irrelevant walk): interval −5.9 [−15.2, 3.7].

Experiment 14c (poster removal): without the walk, the base model scores 24% on edited trials vs 16% on originals.

Experiment 15 (attention cuts): registered 2026-10-24, run 2026-10-24/25. Clean accuracy 92.5% (10 rooms, 1,299 trials). Near vs random cost 8.6 [7.3, 10.0]; near vs far 6.1 [3.8, 8.2]; look-alike vs random 1.2 [−0.2, 2.7]. Near-option frames are read in both an early window (layers 1–18: 86.5%) and a late window (layers 19–36: 88.8%). The layer-window analysis uses the 3 rooms that have enough trials per layer (clean 91.9%, $n = 399$). Without overlap: near vs random 6.4 [3.3, 9.6] on the 503 trials whose near and look-alike sets share no frame; near vs far 7.1 [4.7, 9.4] on the 1,028 trials whose near and far sets share none.

Experiment 16 (HSSD houses): registered 2026-10-25. The Habitat-Sim scene dataset lists 168 default houses (not 211 as in the full HSSD catalog); 13 were drawn at random, 8 accepted after excluding five: four whose largest navigable island fell outside 40–250 m^2 and one with too few lure-free trials. Qwen ran on Apple MPS for houses 1–3 and on a rented RTX 5090 for houses 4–8.

Experiment 20 (retrieval baselines): registered 2026-10-28, before any new baseline was scored. 149 rules in 5 families, scored on the same novel trials the models answered; registered 15-point margin. Section 3.4 and Appendix B.

Experiment 21 (heading-statistic validation): registered 2026-10-28, before any synthetic unit was scored. Invariant, conjunctive, view-dependent and gain-modulated synthetic units at the real units' reliability. Appendix E.

Experiment 25, amendment 1 (registered 2026-11-01, before the assay ran): the shuffled-walk activations had been recorded on 2026-10-30 for Experiment 22 and never scored.

Experiment 26 (walk-relative probes): registered 2026-10-28, before any probe was run. Probes predict the new view's position and heading relative to the first walk frame and its position relative to the last; walk-shuffled controls with 3 seeds. Decision rule: "absent" below R^2 0.10, "present" at 0.30 or above and at

least 0.20 above its controls. Appendix F.

Experiment 29 (place-by-view intervention): registered 2026-10-28, before any model saw a swapped walk. The near option's 8 shown frames are re-rendered in a 2×2 design (place: here or there; view: toward or away from the point the new view looks at). Validity gate: swap gap ≥ 5 points with CI above 0 and random-away cost at most half the gap. Amendment 1 (registered 2026-10-30, after Experiment 29's results but before any control frame was rendered): a broken-walk control moves frames other than the near option's. Section 5.3.

Experiment 30 (trajectory shifts): registered 2026-10-28, before any shifted walk was rendered or any model saw one. Four walk conditions (control, long-step, low-turn, patrol) in nine held-out rooms (P5000–P5009 without P5005); conflict and agreement separation trials test whether the model relies on geometric proximity. Decision rule: four-epoch mean leads the retrieval baseline by at least 15 points with drop from control at most 10. Amendment 1: run 3's first file was garbled by a machine fault (most replies were "!", return code 0); the rerun was submitted after all five runs' accuracies had been seen. All 5,177 + 1,206 trials were answered by each run; 3 of run 3's replies name neither option and count as wrong. Without run 3, the verdicts are the same (95.6%, 96.1%, 97.5%; K +1.0 on runs 4–7). Section 3.4.

Experiment 31 (four-epoch order controls and attention cuts): registered 2026-10-28, before any four-epoch run saw a reordered walk or a cut. Repeats Experiments 13a and 15 on four-epoch runs 5 and 7 with the same trials and frame draws. Sections 3.3 and 5.3 and Appendix F.

Experiment 24 (256-frame walk): registered 2026-10-28, before any four-epoch run saw a 256-frame walk. The fine-tuned model answers the same held-out-room trials with all 256 frames instead of 61. Amendment 1 (registered 2026-10-30, before any run): moved from Apple MPS to CUDA. Section 3.6.

Experiment 32 (pair patching): registered 2026-10-31, before any pair was patched. It tests which families of Experiment 27's ten PCA directions the camera head depends on, at blocks 21 and 19. Already seen before registration: the unregistered token PCA, which fixed the families, and Experiment 27's rank sweep. The headline is the room-frame family at block 21 in the label-free arm; if the label-free and feature-fit arms disagree, the claim follows the weaker verdict. No amendments. Section 5.1 and Appendix K.

The amendments note which results, if any, existed at the time.

D Reproducibility

API model identifiers.

- Claude Opus 5.5: `claude-opus-5-5`
- Claude Sonnet 5.5: `claude-sonnet-5-5`
- Claude Haiku 5.5: `claude-haiku-5-5`
- Kimi K3: `kimi-k3` (Moonshot)
- DeepSeek V4.1 Flash: `deepseek-flash`
- GPT-6 Astra: `gpt-6-astra`
- MiniMax-M3, MiMo-V2.6-Pro, Muse Spark Open: via OpenRouter (provider routing)
- Open models (Qwen3-VL-8B, Qwen3.8-27B, Qwen3.6-35B-A3B, Gemma-4-31B): `transformers`, `torch.bfloat16`, single GPU

Decoding. Open models: greedy (no sampling), except thinking runs, which sample with the model's own settings. Claude: adaptive thinking at the stated effort level ($T = 1$, set by the API). Kimi K3: reasoning `low`,

$T = 1.0$ (fixed by provider). DeepSeek: thinking on, reasoning `low`. GPT-6 Astra: reasoning `low`. Open models' answers are the last option number in the reply. For API models the answer is the option the reply states: a bare option number on the last line, an `<answer>` tag, an option number opening the reply, a last line saying the view is closer to an option, or the option in bold or named alone. Replies that state no option (106 of 22,423) count as wrong.

Run dates. Room A behavior (Claude, open models): 2026-10-15 to 2026-10-17. Room B: 2026-10-22. Ten held-out rooms: 2026-10-23. VGGT probes and patching: 2026-10-16 to 2026-10-17. λ-sweep: 2026-10-23. Kimi K3: 2026-10-23. DeepSeek: 2026-10-23. GPT-6 Astra: 2026-10-25. OpenRouter screen: 2026-10-25.

Costs. Claude API spend is tracked per run in `results/api_costs.csv` and summarized by `scripts/cost_log.py`. Total Claude spend: $395.59. Other providers: Kimi K3 $14.92, DeepSeek $1.27, GPT-6 Astra $82.69, OpenRouter screen $10.18, Qwen3.5-397B $14.13 (total $123.19).

Human baseline details. Study time per walk is measured from the previous walk's last answer (or the run's start) to the current walk's first answer, minus that answer's response time.

	Walk 8	Walk 9	Walk 11	Walk 15	Run
Study time (s)					
Participant 1	—	21	56	53	16.5 min
Participant 2	57	13	3	3	6.4 min
Novel accuracy (correct / trials)					
Participant 1	10/13	8/9	12/13	7/10	37/45
Participant 2	6/13	4/9	6/13	3/10	19/45

Participant 1's walk 8 study time is not measurable (it began in an earlier session). The two participants make the same choice on 23 of 45 novel trials. Run 0 scores 42 on these trials; the four-epoch runs did not answer them. Participants were adult volunteers. The page recorded each participant's answers, study times and optional initials, with no names, contact details or demographics. The UC Davis IRB office advised that, based on a description of the task, the human baseline is not human subjects research and does not require IRB oversight (an informal opinion, not a formal NHSR determination).

LoRA adapter reproducibility. Run 0 used torch's unseeded default generator, so its LoRA initialization and dropout stream cannot be replicated exactly; its results reproduce from the saved adapter, which is archived with the paper. Runs 1 and 2 seed the generator and retrain deterministically up to GPU nondeterminism.

Code and archive. The code repository is at `https://github.com/molinzhao27/vlm-place-cells`. Walk frames, the training rooms and every model's per-trial answer are archived at `https://huggingface.co/datasets/molinzhao/vlm-place-cells`, and the LoRA adapters (runs 0–2, the GPU control and the four-epoch runs 3–7) at `https://huggingface.co/molinzhao/vlm-place-cells-lora`. The photoreal renders are not redistributed because of the Replica and HSSD licenses; their trial lists and the render scripts are included.

E Unit scoring, heading statistic and population assay

Unit scoring. Units are neuron poles: the positive and negative parts of each neuron, scored separately. Poles silent in more than 99% of frames are dropped. Position is binned at 0.5 m resolution. Information content follows Skaggs et al. (1993); each unit's threshold is its own 99th percentile over 100 circular-shift shuffles, each shifting a walk's activity against its own positions by at least 32 steps and wrapping within the walk (the pooled per-layer percentile is used only for gridness). Split-half stability divides the 48 walks into

two groups (even/odd) and correlates their rate maps ($r > 0.5$). A firing field is a connected set of at least 3 bins above 50% of the way from the map's minimum to its maximum. Place units are those exceeding their shuffle threshold with stability above 0.5 and 1–3 firing fields covering at most 40% of the room. Heading invariance builds rate maps from frames in each of four 90° sectors (N, S, E, W). The strict test correlates all six sector pairs across bins valid in both and takes the minimum; a unit passes when the minimum exceeds 0.5. The mean of two, registered first, averages the N vs S and E vs W correlations and applies the same threshold. Table E1 gives both shares for each model. Rate maps are computed on a 20 ×20 grid of 0.5 m bins; activity and occupancy are smoothed with a Gaussian of width 0.5 m (7-bin kernel); a bin needs at least 3 frames in both maps to be included. Split-half stability divides the 48 walks into even and odd groups and correlates their rate maps. Cross-heading and ridge decoding are trained on even walks and tested on odd walks. Features are z-scored on the training walks. The ridge penalty is selected from 5 values (10^1 to 10^5) on a walk-level validation split inside the training walks; the test walks are not used for selection. Best-layer values report the best of the recorded layers on the test walks, so they are optimistic: for negative results (a readout failing across headings), this favors the positive reading, making those results conservative; for positive results (within-sector decoding), treat them as upper bounds.

Recording sites. For the Qwen VLMs, we record the SwiGLU MLP neurons (the input to each layer's down projection) in 7 language-model layers spread through depth, read at the image-end token (`<|vision_end|>`) that closes each frame's 64 visual tokens. For DINOv2, we record post-GELU MLP neurons at the CLS token in all 12 layers. For VGGT, we record the MLP neurons (fc2 input) at the frame's camera token.

Grid-cell scorer. Our grid-cell scorer flags a unit only if its autocorrelogram has hexagonal periodicity above a shuffle threshold (Langston et al., 2010).

Synthetic grids. We generate 40 hexagonal rate maps at each spacing (1 to 5 m) and noise level. At 2 m spacing with no noise the scorer detects 100%; at 1.5 m, 2.5%, limited by the 0.5 m bins and smoothing. Place-like distractors (1–3 random fields) are flagged at 0.1%.

Path-integrating RNN. Following Sorscher et al. (2019), we train a 4,096-unit recurrent network with 512 difference-of-Gaussians place-cell targets for 100,000 steps. Its only inputs are the step-by-step displacement (dx, dy) and a start state set from place-cell activity at the true position; it never sees a view. It is tested in two modes: anchored, where the state is reset from the true position every 20 steps (the training regime), and free, where it path-integrates the whole walk. With 0.093 m steps, 145 units are flagged (3.6%); with 0.22 m steps (our walks), 0; grid cells depend on training choices (Schaeffer et al., 2022). Sensitivity against dense-map gridness: 24% at gridness above 0.3, 46% above 0.5 and 93% above 0.8; false flags 0.

At most 0.3% of units pass the grid criterion in any layer of any model (VGGT 0.1%). Figure E1a shows the scorer's positive control on synthetic grids; Figure E1b shows example RNN units with grid-like autocorrelograms.

Heading-statistic validation (Experiment 21). To test whether the mean-of-two statistic classifies units whose answer is known, we generate four kinds of synthetic unit along room A's real walks: (a) invariant place fields, (b) gain-modulated units whose rate is R times higher at a preferred heading, (c) conjunctive place×heading units that fire only within a heading window of 60–180° and (d) view-dependent units that fire when a wall point or pillar is in view. Each kind is generated at several noise levels; the results below compare synthetic and real units within the same split-half stability window (0.81–0.91, centered on the real

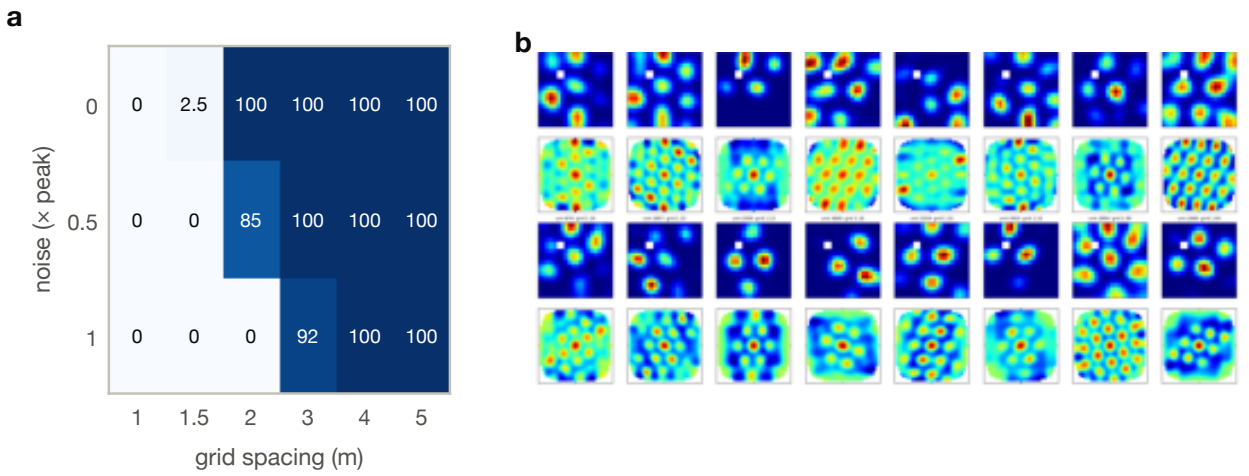


Figure E1. The grid-cell scorer detects synthetic grids but flags at most 0.3% of units in any model. (a) Synthetic detection (% of 40 grids) by spacing and noise level. (b) Dense rate maps and autocorrelograms of example RNN units (0.093 m run).

median 0.86). A **gain index** G measures the $\log_2$ ratio of a field's amplitude between opposite headings, combined over both pairs; $G = 1$ means a twofold change.

At real reliability the statistic counts 100% of invariant units ($n = 376$), 0.9% of conjunctive units ($n = 436$) and 0% of view-dependent units ($n = 251$). Gain-modulated units pass as invariant: 100.0% at $R = 2$ (G median 1.05), 99.2% at $R = 3$ (1.60), 97.5% at $R = 5$ (2.48), 73.9% at $R = 10$ (3.41) and 28.0% at $R = 30$ (4.63). The pass rate halves near $R \approx 18$. Invariant synthetic units have median $G = 0.29$, and 1.1% exceed $G = 1$.

Reliability matters: invariant units pass at 49% with stability 0.5–0.6, 85% at 0.6–0.7 and 98% at 0.7–0.8, while conjunctive and view units pass at most 0.5% below stability 0.9. The older 180° halves count 92% of 180° conjunctive units as invariant, against 3.8% under the mean of two on 90° sectors. At very high reliability (stability above 0.9), 24% of 180° conjunctive units pass, relevant to conditions with very reliable units (the RNNs, VGGT block 19); a conjunctive population would still read far below the RNNs' 98%–100%, so the positive control stands.

Real units, under the mean of two: VGGT-single block 19 counts 45% invariant, of which 77% have $G > 1$ (median rate ratio about 3.4), against 0.0% expected from pure place units. Block 21: 34% invariant, 74% with $G > 1$. Raw pixels: 1.2% invariant, 75% with $G > 1$. The pixel moving average (half-life 8 frames): 97% invariant, of which only 10% have $G > 1$. Most of the units VGGT counts under the mean of two change rate more than twofold; the six-pair test, which asks whether a field holds across all four sectors, passes 34% of VGGT's place units at block 19 (Section 4.3); the moving average's invariant units are rate-stable.

Population assay (Experiment 25). Units are z-scored over all frames, with no positions used. For each of the four 90° test sectors and each even/odd walk fold, two ridge regressions of (x, y) are trained on the training walks and tested on the test walks' frames facing that sector. The unseen-heading readout trains on frames at least 90° from the sector's center, which leaves a 45° gap to the test sector; the seen-heading readout trains on frames within 90° of the center. The penalty is chosen on held-out walks inside the training arc, so the test heading never touches the unseen-heading readout's fit. Errors are median Euclidean errors over all frames, and chance is the median error of predicting each readout's mean training position. HG is $(\text{chance}_{\text{unseen}} - \text{error}_{\text{unseen}})$ / $(\text{chance}_{\text{seen}} - \text{error}_{\text{seen}})$: 1 means the position information transfers fully to the

Table E1. Heading-invariant share of place units, strict and looser test. Single frames, maximum over layers, unless noted. The six-pair test (the paper's statistic) asks that the field hold in all six sector pairs; the mean of two averages the N vs S and E vs W correlations and passes more units.

Model	Six-pair test	Mean of two
Path-integrating RNNs	97%–100%	98%–100%
Raw pixels	0%	1%
Random Qwen3-VL-2B	0%	1%
Random DINOv2	0%	2%
DINOv2 (trained)	1%	4%
Qwen3-VL-2B	0%	0%
Qwen3-VL-8B	0%	0%
Qwen3-VL-32B	0%	0%
Qwen3.8-27B	0%	0%
SenseNova-SI 8B	0%	0%
VGGT, one frame with 8 anchors (block 19)	34%	45%
Fine-tuned model, walk at layer 35	8%–17%	18.2%–31.4%

unseen heading, 0 means chance there, and below 0 means worse than chance. Its 95% interval comes from 1,000 bootstrap resamples of the walks. Amendment 1 runs the same assay on the four-epoch shuffled-walk frames; D is walk HG minus shuffled-walk HG at layer 35, per run. Teleport views are recorded 64 per prompt, all from one walk, in random order: after the walk frames, after the shuffled walk, or alone with no walk. Each teleport view can therefore attend to the ones before it. The assay scores all 3,072 teleport views, 2,690 of them familiar and 382 novel by the definition in Section 2.2; it does not measure the single novel query of the two-choice task.

HG calibration and the rotated population. Synthetic populations of 2,000 units mix a share of heading-invariant place units into a heading-dependent background (conjunctive, view and head-direction units), with autoregressive noise at k times each unit's SD; $k = 3$ matches the real units' median stability. HG rises with the invariant share (Figure E2). At $k = 3$, HG 0.25 corresponds to about 11% invariant units, and 0.5 to about 24%. A random rotation of the same population makes every unit a mixture of the original ones. It leaves HG unchanged (0.46 at 20% invariant units), while the unit test, scored here with the mean of two, drops from 43% to 5%. The population test therefore reads a code that the unit test can miss when the code is spread across units.

Cross-sector decoding (an older test). This test fails its own positive control: on the base path-integrating RNN it gives 7.57 m although 98% of its place units pass the six-pair test (Table C1, Experiment 13f). Its failures in VLMs are therefore not evidence that a code is absent.

Walk-frame decoding is a ridge regression from all MLP neurons in one layer to (x, y), trained on frames facing one set of headings and tested on frames facing the opposite. Chance is 3.70 m. For pixels, the error is worse than chance (5.11 m vs 3.70 m; the predictions are mirrored, not random), a pattern that also appears in VGGT, whose code carries heading (Section 5.1).

With the walk in context, Qwen3-VL-2B's walk-frame decoding drops to 2.38 m (Figure E3). Shuffled frames stay at 4.68 m, showing that the improvement needs the temporal order. A moving average of single-frame activations drops decoding to 0.60 m.

Within a heading sector, position decoding in VGGT is precise: 0.08 m–0.13 m at blocks 15–23. Across opposite 90° sectors, the picture changes. Walk-frame decoding gives 3.41 m at block 19 from a single frame

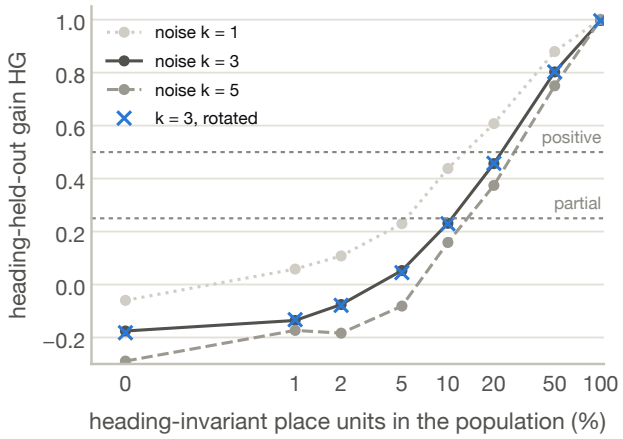


Figure E2. HG rises with the share of heading-invariant place units and does not depend on how the code is spread across units. Synthetic populations of 2,000 units at three noise levels ($k = 1, 3, 5$; $k = 3$ matches the real units' stability). Crosses: the $k = 3$ populations after a random rotation, which makes every unit a mixture. Dotted lines: the partial (0.25) and positive (0.5) thresholds.

with 8 anchors (chance 3.70 m; Section 5). At blocks 21 and 23, the across-heading error rises to 7.74 m and 8.92 m, about twice chance: the predictions are mirrored, not random. With the walk in context, decoding improves to 2.18 m (0.17 m within a sector) but stays far from the within-sector precision on a single frame.

Query-view probes: cross-sector evaluation. These probes use the same cross-sector evaluation, so the caveat above applies. Within a heading sector, probes at the query token reach 0.92 m for Qwen3-VL-32B and 1.13 m for Qwen3-VL-2B. Across opposite sectors, every base model's error stays high (4.16 m–4.44 m vs chance 3.62 m; Figure E4), and the walk, shuffled walk and no walk differ by at most 0.19 m. A **recall baseline** reads the moving-averaged pixels at the walk frame most similar to the teleport view. It gives 1.95 m across headings: the position is available in the context, but a linear probe at the query token does not read it under this evaluation.

F The fine-tuned model

F.1 Fine-tuning details

Procedural rooms. We generate 1,200 rooms of the same 10 m size, each with 1–3 posters per wall and 1–3 pillars. Room A and room B are not among them. 9,595 lure-free two-choice questions (42% novel) across all 1,200 rooms (1,136 training, 64 held out), each with a 64-frame walk. Walls and pillars draw from 7 surface patterns and floors from four of them, in random colors. Posters are crops of 25 bundled images (20% plain colored panels), each wall panel 0.80 m–2.50 m wide, and 12% of panels are door-like. Pillars have radius 0.25 m–0.45 m and height 1.30 m–2.70 m, centered in 1.80 m–8.20 m on each axis, at least 1.20 m apart; 70% carry a surface pattern and the rest a poster image.

Walk process. Each walk starts at a random free position with a random heading. At each step, the heading drifts by a Gaussian with SD 15 °, and the agent steps forward by 0.22 m times a uniform factor from 0.7 to 1.3. With probability 15% per step, an on-the-spot head scan of 30–90 ° is inserted. When a step would hit a wall or pillar, the agent turns away by 60–180 °.

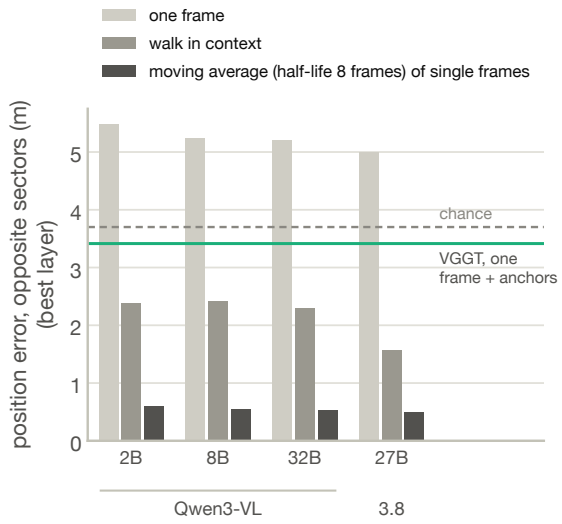


Figure E3. The older cross-sector decoder, shown for completeness; it fails its own positive control. Walk-frame position error across opposite 90° sectors at the best layer, for one frame, the walk in context and a moving average (half-life 8 frames) of single frames. Green line: VGGT, one frame with 8 anchors. The decoder also fails on the base path-integrating RNN (7.57 m), so a high error here is not evidence that a code is absent.

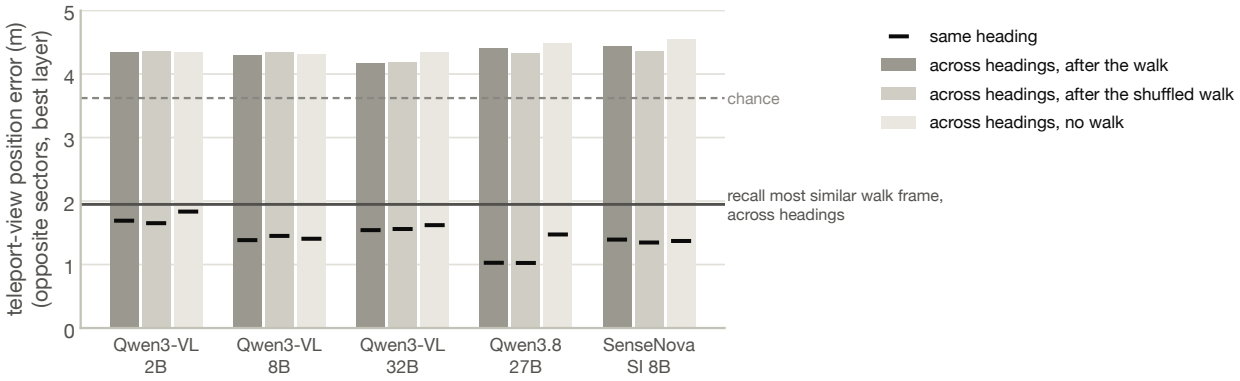


Figure E4. Under the older cross-sector evaluation, a linear probe at the query token does not read position across headings. The evaluation fails its own positive control (the base path-integrating RNN), so this is not evidence that a code is absent. Ridge regression from one MLP layer to (x, y), trained on even walks, tested on odd walks. (Dashed = chance 3.62 m; best layer per condition.) Dark-gray line: the recall baseline (ridge readout of the moving-averaged pixels at the walk frame most similar to the teleport view), across headings.

Training. LoRA (Hu et al., 2022) rank 16, alpha 32, dropout 0.05, on every attention and MLP projection of the Qwen3-VL-8B language model (vision tower and merger frozen): 44 M trainable parameters (0.5% of the model). AdamW with peak learning rate 10^{-4}, no weight decay, 30 warm-up steps then cosine decay, 8 rooms per update, bf16 with gradient checkpointing. Answer-only cross-entropy loss; no coordinate or map supervision. 2 epochs, 284 updates, 96 minutes on one GPU (peak 29 GiB). The loss stayed near 0.17, the coin-flip level ($\ln 2$ per answer, spread over its tokens), for about 130 updates, then held-out accuracy rose to 69% after epoch 1 and 92% at the end. Run 0 used torch's unseeded default generator, so its LoRA initialization and dropout stream cannot be replicated exactly; its results reproduce from the saved adapter, which is archived with the paper. Runs 1 and 2 use seeds 1 and 2; runs 3–7 use seeds 3–7. All seeded runs retrain deterministically up to GPU nondeterminism.

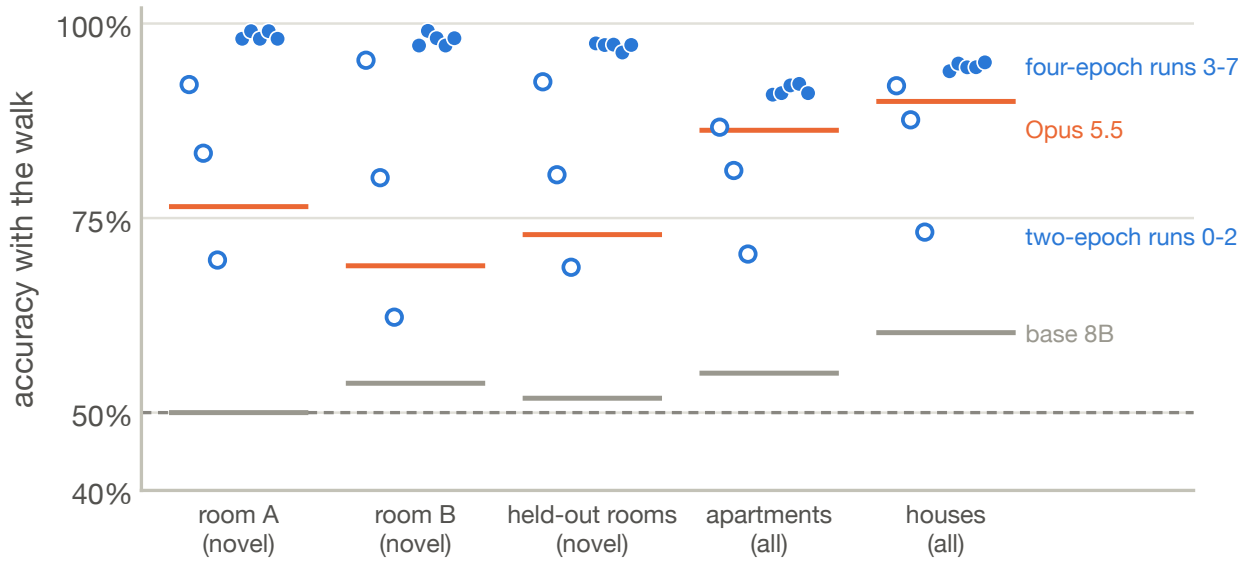


Figure F1. The four-epoch recipe is reliable; the two-epoch runs were not. Accuracy with the walk for every training run. Filled circles: the five four-epoch runs (3–7); hollow circles: the three two-epoch runs (0–2). Orange line: Opus 5.5 on the same trials; light-gray line: the base 8B; dashed: chance. Rooms A and B (102 and 106 trials) and the held-out rooms (room mean) are novel trials; apartments are all trials, pooled; houses are all trials, mean over houses.

Four-epoch recipe (Experiment 9c). The recipe and decision rule were registered after the two-epoch runs' spread was known and before any four-epoch run. Five additional runs (runs 3–7) use 568 updates (four epochs) with torch seeded. Every run leaves the loss plateau by update 235 (range 140–235 of 568), compared with 135–215 of 284 for the two-epoch runs. The registered decision rule is met. It required every run to reach at least 85% on the validation rooms (actual 95.3%–98.4%), a five-run room-A novel mean of at least 85% (actual 98%), and no run below 80% in room A (minimum 98%). Room B ranges 97%–99%. Every four-epoch run beats the base model, the no-walk condition and the shuffled walk in every held-out room (10/10) and every house (8/8). In the apartments, where 3 scenes are too few for a scene-level test, every trial-level p for these contrasts is at most 0.019. Ordered beats shuffled by 15.1–42.9 in the held-out rooms and 17.7–40.2 in the houses. Shuffled-walk accuracy varies between runs (54.4%–82.1% in the held-out rooms, higher than the two-epoch runs' 51.3%–63.4%); run 7 gains least from the order (15.1 points).

Per-run and per-room accuracy. All five runs reach 96.3%–97.4% in the 10 held-out rooms (Figure 3c), every run above the base model in all 10 rooms and above Opus by 23.4–24.6 points. Room A novel accuracy averages 98% (98%–99%; Figure 3b); room B 98% (97%–99%). The base model stays at chance (52% with the walk, 50% without; room A 50%, room B 54%).

Per-run behavior (two-epoch). Walk-clustered intervals for room A: run 0 vs base 42.2 [29.4, 53.8]; ordered vs shuffled 42.2 [29.3, 57.0]. Room B: run 0 vs base 41.5 [29.8, 53.1]. Run 1's room A order $p = 7 \times 10^{-5}$. Ordered minus shuffled in the held-out rooms: 7.8–41.2 across two-epoch runs. In the houses: 4.4–30.7 (novel). Run 2 does not significantly beat the base model in room B ($p = 0.18$); in room A, every run beats it ($p \leq 0.0017$). The order effect in room A is significant for runs 0 and 1 but not for run 2 ($p = 0.23$).

Per-run table. Table F1 shows the accuracy of every run and Opus across all room sets (Figure F1).

Table F1. Per-run accuracy (held-out rooms and houses: mean over scenes; apartments: pooled over trials) for the two-epoch (runs 0–2) and four-epoch (runs 3–7) recipes, and Claude Opus 5.5 at low effort with the walk. Rooms A and B and the held-out rooms report novel trials; apartments and houses report both all-trial and novel-trial accuracy. Room A $n = 102$; room B $n = 106$; held-out rooms: 10 rooms; apartments: 3 scenes; houses: 8 scenes.

Run	Room A	Room B	Held-out rooms	Apts (all)	Apts (novel)	Houses (all)	Houses (novel)
Two-epoch recipe (284 updates)							
0	92%	95%	92.5%	86.7%	79.3%	92.0%	86.4%
1	83%	80%	80.6%	81.1%	77.6%	87.6%	81.6%
2	70%	62%	68.7%	70.4%	72.4%	73.2%	66.3%
Four-epoch recipe (568 updates)							
3	98%	97%	97.4%	90.9%	84.5%	93.9%	88.7%
4	99%	99%	97.2%	91.1%	84.5%	94.9%	89.5%
5	98%	98%	97.3%	92.0%	87.9%	94.4%	90.5%
6	99%	97%	96.3%	92.2%	84.5%	94.4%	88.9%
7	98%	98%	97.2%	91.1%	86.2%	95.0%	90.4%
Opus	76%	69%	72.9%	86.3%	89.7%	90.0%	84.8%

GPU control. A second seed-0 run on a different GPU matches the original (room A novel 94%, room B 94%), which is consistent with the run-to-run spread coming from the seeds rather than the hardware; it is a single draw.

F.2 Behavior controls of the fine-tuned model

Order controls in two four-epoch runs. A further control asks what the frame labels carry. Run 7's shuffled walk (82.2%; a rerun of P5005 on another device moves these numbers by 0.1 from the per-run ones) drops to 58.2% when relabeled (S − R 24.0 [21.7, 26.1]), below its own no-walk baseline (63.7%; R − no walk −5.5 [−8.7, −1.4]): the labels carry the shuffled walk's accuracy. Run 5 shows the same direction but its shuffled walk was already at the no-walk level (54.4% vs 52.9%), so there was little to carry (1.5 [−0.4, 3.5]). Against the relabeled floor, the two runs' order effects are close (44.4 [42.6, 46.4] and 39.0 [37.0, 40.7] points); against the labeled floor they look far apart (42.8 and 15.0). Only runs 5 and 7 were relabeled, so whether the other three runs' shuffled accuracy also depends on the labels is untested.

Attention cuts in two four-epoch runs (Experiment 31). In runs 5 and 7 (the same trials and frame draws as run 0), attention is blocked in all decoder layers; tokens inside the set still attend to each other and the frame labels stay visible. Near minus random is −6.0 [−7.7, −4.1] and −5.5 [−7.2, −4.1] points, with excess log-odds costs of 1.30 [0.88, 1.67] and 1.18 [0.82, 1.59]. The near neighbors also hurt more than the far-option neighbors (near minus far: −2.0 [−3.7, −0.3] and −2.8 [−4.4, −1.2] points). Whether look-alike frames cost more than half the near cost is inconclusive in both runs (excess-logit difference −0.18 [−0.40, 0.08] and −0.19 [−0.40, 0.03]; both CIs include 0). Look-alike minus random is −1.2 [−2.3, −0.2] and −1.3 [−2.6, −0.1] points. On the 503 trials whose look-alike set shares no frame with the near set, look-alike minus random is 0.1 [−0.9, 1.0] and −0.2 [−1.6, 1.1] points, while near minus random stays at −5.6 and −4.6. This near cut reproduces the near-cut condition of the intervention below answer for answer (1,149 trials in each run).

Shifted walks (Experiment 30). Experiment 30 tests the fine-tuned model on walks it was not trained on, in nine held-out rooms (P5000–P5009 without P5005; P5005 is left out because two of its textures

were drawn without a seed and cannot be reproduced). Four walk conditions change one property of the training distribution: *control* draws new walks from the training distribution itself; *long-step* doubles the step length (to 0.44 m); *low-turn* reduces heading drift and removes head scans (median turn 3.6° against 13.0° in control); *patrol* walks a route from the training distribution, returns to its start and walks the same route again in the same direction until step 256 (frames revisited after 64 or more steps: 86%). 5,177 novel trials across the four conditions use the standard trial rules. Separation trials (1,206 in control and patrol walks) type each pair by its options' neighborhoods (the 4 shown frames before and 4 after each option): in a conflict trial, no frame in the near option's neighborhood shows a landmark the new view shows and at least one frame in the far option's neighborhood does; agreement trials are the reverse.

Accuracy rises with the distance margin: 89.0% to 95.8% at 2–3 m, 98.5% to 99.5% at 4 m or more. The smallest lead over the retrieval baseline in any bin is +18.0 (rule 81.5%). The learned sequence matcher's route reproduced its stored scores exactly on 119 trials.

A walk shift counts as survived when the lead M over the retrieval baseline on the same trials is at least 15 points with its CI above 0, and the drop from control is at most 10 points. Control walks, drawn like the training walks: 96.4% [95.6%, 97.2%] (against 97.0% for the original walks in the same rooms, so fresh walks cost 0.6 points; retrieval baseline 67.2%, M +29.2 [26.1, 32.1]). Longer steps: 95.5% (M +27.8 [25.0, 30.8], change from control −0.9 [−1.9, 0.0]). Fewer turns: 96.2% (M +31.2 [29.2, 32.8], change −0.2 [−1.1, 0.6]). Patrol: 97.5% (M +29.9 [26.9, 33.1], change +1.0 [−0.1, 2.1]). All three shifts survive, in the five-run mean and on their own in each of the five runs (5 of 5 each).

In separation trials (412 conflict, 411 agreement) the walk around the near option shows nothing of the new view's scene while the walk around the far option does (conflict), or the reverse (agreement); the pixel rule opposes the right answer as designed (conflict 44%, agreement 72%; on these trials, the pixel rule scores 43.9% on conflict and 72.5% on agreement). Geometric proximity wins, in the mean and in each of the five runs, on control and patrol walks: control conflict 97.1% [95.5%, 98.4%], agreement 98.2% [96.8%, 99.3%], midpoint 74.1%, cost K +1.1 [−1.3, 3.3]; patrol 98.6% vs 99.1%, K +0.5 [−0.3, 1.5] (6 rooms; three rooms have fewer than 20 conflict trials). The retrieval baseline on conflict trials reaches 65.0%, and the model leads it by +32.0 [27.6, 37.7]. A local view-sequence association fitted to the training walks does not explain the accuracy, as tested here. This does not show a place code and does not exclude strategies that chain views across the whole walk.

Three limits. First, conflict trials remove shared landmarks, not all shared appearance: the retrieval baseline on conflict trials reaches 65.0%, and the order-free sequence matcher scores 76.9%, so neither is at chance. Second, the cost is small, not zero: K is the smallest of the gaps; on standard trials typed by the same rule, the gap runs 1.9 to 7.3 points by condition (94.4% against 98.2% pooled); those trials are not balanced for pixel similarity within each type and are not the test behind K; all stay under 10 points. Third, these are procedural rooms only; the photoreal scenes do not separate localization from retrieval by the 15-point margin (Section 3.4).

Being flat across the shifts is not a mark of geometry: the sequence matcher is as flat (74.5%, 73.8%, 75.0%, 73.1%) and conflict trials cost it nothing (76.9% on conflict, 73.6% on agreement). It uses no frame order, so it differs from the local-sequence strategy this test targets; view-based accuracy can also ignore these shifts. The model leads it by +21.2 to +24.3 points, in all nine rooms each time, so level separates them, not shape. Every prompt keeps the ordered walk and its real frame numbers, so this test says nothing about order or labels (the order controls above do).

Place-by-view intervention (Experiment 29). Experiment 29 re-renders the near option's 8 shown frames (the 4 shown frames before and 4 after the near option, never an option frame) in a 2×2 design: place (here, the original positions, or there, all 8 positions moved by one translation that puts every one at least 3 m from

the new view) × view (toward, facing the new view's target point, or away, facing directly away from it). The four conditions, a random-away baseline and a near-cut reference are run on four-epoch runs 5 and 7 and on run 0. Two agents with known strategies serve as the stimulus check: the place agent picks the option whose 8 frames have a mean position nearer the new view, and the view agent picks the option whose 8 frames include the frame most similar to the new view (DINOv2). The stimuli pass if the rule calls the place agent "place" and the view agent "view plus walk."

Amendment 1's broken-walk control (678 trials in 9 rooms) moves frames other than the near option's: a block of 8 frames moved away from both options, the far option's set moved while staying far, and the far option's set moved near the new view. Amendment 1's stimulus check: the place agent must come out "place confirmed"; the view agent must show a pull P below 5 points; and a synthetic broken-walk agent, starting from run 5's clean answers and flipping each answer wrong with probability equal to run 5's G on these trials, must come out "broken walk."

The stimulus gate passed. Amendment 1's control gates: the place agent's verdict is place confirmed (P 81.3); the synthetic broken-walk agent's verdict is broken walk (W 28.1, P 0.5).

Run 0 leans more on view than the four-epoch runs: 0.42 of the gap against 0.19, conflict effect 5.1. The four-epoch shuffled-walk condition was not tested, so whether the decision's place dependence requires the walk's order is untested for this experiment.

The following observation is exploratory, chosen after the verdict: on there-away trials, the per-SD logit coefficient for moved distance is -0.31 [-0.50, -0.14] and -0.19 [-0.35, -0.05]; on here-away it is about zero (-0.04, -0.03).

The validity gate requires the swap gap (clean minus there-away) to be at least 5 points with its CI above 0, and the random-away cost to be at most half the gap. The gate is met in both four-epoch runs (5 and 7): swap gap 26.8–29.5, random cost about zero (-0.6 and -0.1). The place effect is 21.1–22.4 points, the view effect 5.2–5.6, and the conflict (here-away minus there-toward) 15.9–16.8. On the 908 trials whose near and far frame sets share no frame, the same pattern holds (R_{place} 0.76 and 0.80). Near-cut cost 5.6–6.0; here-toward is 4.1–5.2 below clean. The decision links the new view to the near option without any neighboring frame that shows what the new view shows, provided those frames were taken near the option; frames that show the new view from elsewhere do little. The there-away frames face away from what the new view shows, so "where they were taken" holds as far as the view agent's check holds.

Broken-walk control (Experiment 29, amendment 1). In four-epoch runs 5 and 7, the swap gap on these trials is G = 29.3 [26.8, 31.9] and 25.0 [22.9, 27.7]. A block of 8 consecutive shown frames around an unmoved center frame, moved away from both options, costs nothing (W: -0.1 [-1.0, 0.8] and -0.2 [-1.1, 0.6]). Moving the far option's set while it stays at least 3 m from the new view costs nothing either (F: -0.9 [-2.0, 0.3] and -0.8 [-2.3, 0.7]). Moving the far set to within 1.5 m of the new view pulls the decision toward the far option: P = far-far minus far-close, 10.2 [7.8, 12.4] and 11.1 [9.7, 12.7] points, in 9 of 9 rooms in both runs (77 to 7 and 81 to 6 trials). The view-agent check gives -5.7 (must stay under 5), and the far-close frames look less like the new view than the far-far frames do (DINOv2 cosine 0.61 against 0.63; best frame 0.70 against 0.72). The cost follows where the frames go: the intervention's limit on a broken-walk penalty is lifted for these two runs. Far-close puts the far set 1.1 m from the new view on average, and accuracy stays at 87.9% and 86.6%, well above 50%. A pure place reader would fall toward 50% (the place agent loses 81.3 points). Run 0 was not tested, so the broken-walk limit stays in run 0's appendix entry.

256-frame comparison (Experiment 24). Experiment 24 tests the four-epoch model on all 256 walk frames instead of the 61 evenly spaced frames at test (training prompts show 64 frames). The same held-out-room

trials and novel trials are used; Opus's 256-frame accuracy (72.9%) serves as the comparison. Amendment 1 moved the runs from Apple MPS to CUDA (vast-3 RTX 5090) before any run, so the 256-minus-64 contrast (-5.7 points) is same-device (all CUDA); the supplement against Experiment 9c's own files gives -5.7. CUDA parity with MPS was checked on P5005's novel trials pooled over runs (99.9%; this decides nothing). 0.1% of replies name neither option and count as wrong.

Per run (256 / 64 frames, lead over Opus, 256 minus 64, right where Opus is wrong / wrong where Opus is right): run 3: 90.8%/ 97.4%, +17.9, -6.7, 275/ 47; run 4: 92.3%/ 97.2%, +19.4, -4.9, 278/ 29; run 5: 90.6%/ 97.3%, +17.7, -6.7, 271/ 44; run 6: 90.8%/ 96.3%, +17.9, -5.5, 271/ 43; run 7: 92.7%/ 97.2%, +19.8, -4.5, 288/ 34. Rooms: accuracy 88.8% to 94.3%, drop -3.1 to -8.5.

The following observations are unregistered and carry no thresholds. Against the procedural retrieval baseline (67.7%), the recipe at 256 frames leads by +23.8 [19.1, 28.8], in all 10 rooms. The extra errors lean to the earlier frame: the runs name the later option on 45.6%–46.9% of trials at 256 frames and 49.5%–50.3% at 64, and the near option is the later one on 50.9%. At least three of five runs miss 105 trials at 256 frames (31 at 64), all five miss 46, and Opus is wrong on 69.5% of those 105 against 26.8% overall. The drop shrinks as the options lie farther apart in walk time: -7.6 (under 32 frames), -7.3 (32–63), -5.1 (64–127), -3.3 (128+).

Fine-tuned model versus Opus. In the procedural rooms, every four-epoch run leads Opus in all 10 rooms, by 23.4–24.6 points on average (Opus 72.9%). When the frame count is matched (256 frames), the lead is +18.6 [16.6, 20.8] over rooms (+17.7 to +19.8 across runs). In the photoreal scenes Opus answered, every four-epoch run also leads on all trials (13 houses by +4.7 to +5.4, apartments by +2.4 to +3.4, though in none of the five runs does the CI over the 7 apartments stay above zero). On the 67 novel apartment trials, Claude Opus 5.5 reaches 91.0% against 86.6%–89.6% (not significant). In the held-out rooms (low effort, novel trials), Opus shows little dependence on DINOv2 appearance similarity with the walk (1.6, $p = 0.53$) but some without it (7.8, $p = 0.0039$).

F.3 Internals of the fine-tuned model

Unit analysis. Base (single frame): 0% of place units are strictly heading-invariant under the six-pair test (maximum over layers). Fine-tuned (single frame): 0%. With the walk, late layers (≥ 23), mean of two: base 0%–1% of all units, fine-tuned 1%–3% (Figure F2a); strict cross-sector decoding base 3.41 m–4.58 m, fine-tuned 2.73 m–3.59 m (Figure F2b; this test carries a caveat: a path-integrating RNN with 99% invariant units gives 7.57 m; Appendix E). From layer 23 on, place units in the walk condition have median stability 0.64–0.67; at that reliability, the mean of two counts only about 85% of truly invariant units and the six-pair test 59% (Experiment 21), so those fractions are underestimates. They are still far below what a mostly invariant population would give. Grid units: 0.2% at most.

In the four-epoch runs (3–7), the walk shares under the mean of two are 18.2%–31.4%. The noise-normalized ratio (the median, over place units, of the mean opposite-heading correlation divided by split-half stability) is 0.44–0.63 at layer 35 (base 0.37, moving average 0.82–0.88). At layer 29 the ratio drops to 0.23–0.38. The median place-unit stability at layer 35 is 0.62–0.67, and in that bin a truly invariant synthetic unit passes the six-pair test 59% of the time [85% under the mean of two]. The units count as only partly invariant: on the mean of two, a walk share of at most 25% in every run would mean few invariant units, and 85% in any run would put it near the moving average. The walk shares are above 25% in every run except run 6 and well below 85%, and the moving average beats the walk in every run.

The test that the answer accounts for almost all the readout does not hold, narrowly: the residual after the answer is 0.030–0.102, with one run at 0.102 exceeding the 0.10 threshold by 0.002. Nearer-option accuracy from the last token reaches 97.4%–97.9%. The distance-difference readout peaks at R^2 0.82–0.84, of which

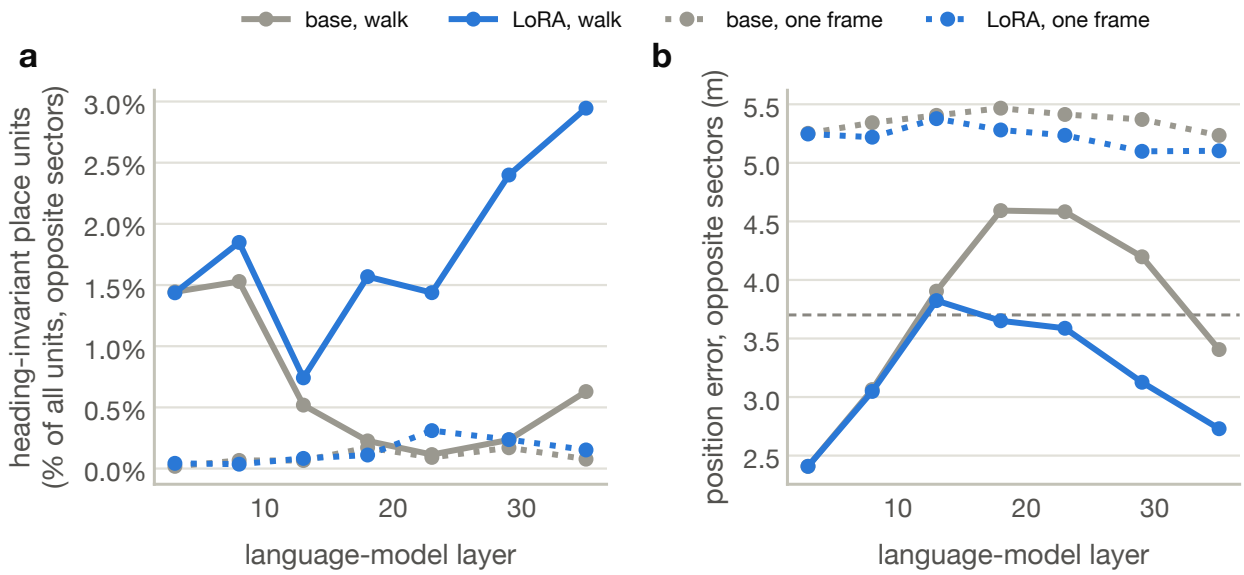


Figure F2. Fine-tuning changes the place code a little. (a) Heading-invariant place units (mean of two) as a % of all units by layer, base vs run 0, walk and one frame. (b) Cross-sector decoding error by layer, base vs run 0, walk vs one frame.

the answer alone gives 0.80–0.82. Under the two-choice prompt, the base model's cross-sector probe error is 4.53 m, and its distance-difference readout peaks at 0.02.

Walk-relative probes (Experiment 26). Experiment 26 probed whether the new view's position or heading is coded relative to the walk's own starting point. In addition to room-coordinate probes, ridge and one-hidden-layer MLP readouts (288 cells) were trained to predict the new view's (x, y) and heading with frame 1 placed at the origin facing $+x$, and its (x, y) relative to the last frame. Each target was tested with 3 walk-shuffled controls (seeds that randomly reassign walk-frame coordinates). The registered rule: a target is "absent" when its best cross-room R^2 stays below 0.10 and "present" when it reaches 0.30 and beats its walk-shuffled controls by at least 0.20.

In the base model, the best first-frame-relative position R^2 is 0.088 (median error 3.71 m vs chance 4.01 m; verdict absent). The walk-shuffled control gives R^2 -0.009. Heading relative to frame 1 reaches R^2 0.216 (verdict weak); the walk-shuffled control stays at -0.006. Run 0 gives similar values: first-frame-relative position R^2 0.092 (median 3.64 m), heading R^2 0.157. The four-epoch recordings (Experiment 22) are detailed below; the rerun gives R^2 0.088 beside the original 0.092.

Exploratory: the position R^2 splits into forward (0.22, the position component in the walk's direction of travel) versus left (-0.05); the heading R^2 into cosine 0.34 versus sine 0.09. DINOv2 similarity between the new view and frames 1 and 256 gives at most R^2 0.03, so the position signal is not explained by visual similarity to the walk endpoints.

In the four-epoch runs (3–7), the fine-tuned model's signal is front-back only, so it reads as a comparison with frame 1 (forward R^2 0.21–0.34, left -0.01 to 0.02; exploratory). The heading relative to frame 1 is weak (R^2 0.175–0.210). Last-frame position is absent (at most R^2 0.042). Room-coordinate position stays absent (at most R^2 0.038). The walk-shuffled control stays at or below zero in every four-epoch run (seed means at most -0.003).

In the four-epoch runs, the first-frame-relative position R^2 is 0.108–0.172: above the absent threshold (0.10)

but below the present one (0.30). The query view's position in room coordinates is not decodable across rooms in the fine-tuned model (best R^2 0.04).

F.4 Standard spatial benchmarks

Standard spatial benchmarks (Experiment 17). The protocol and transfer rule were registered before any model saw benchmark data. We evaluate the base model and all eight fine-tuned runs on two benchmarks that test spatial reasoning from video. **VSI-Bench** (Yang et al., 2025): 5,130 questions across eight task types, using the official lmms-eval protocol (32 frames at 640×480, greedy decoding, 16 output tokens). Our base-model score (55.4) is within the registered tolerance (no more than 8 points below the published 32-frame score of 57.9; Qwen's reported figure is 59.4). **MindCube-Tiny** (Wang et al., 2026b): 1,050 questions with official raw-QA scoring; chance is 33.0, and the base model scores 30.1.

Fine-tuning lowers VSI-Bench scores slightly. Run 0 scores −1.8 [−2.9, −0.8] relative to the base model; the two-epoch mean −1.2 [−2.1, −0.3]; the four-epoch mean −2.1 [−3.1, −1.1]. All intervals resample scenes, following the registered decision rule; the verdict is "hurts" (7 of 8 runs individually labeled "hurts"). The Debiased subset (2,362 questions, base 48.9) gives the same picture: run 0 −3.0, two-epoch mean −2.1, four-epoch mean −2.8. The loss concentrates in metric tasks: counting −5.4 [−7.3, −3.5] and absolute distance −6.4 [−8.3, −4.5] for the four-epoch mean. View-relating tasks (relative direction, relative distance, route planning and appearance order) are unchanged: −0.9 [−2.7, 1.0]. Without the video input, the fine-tuned models score slightly above the base model (2.1 points for the four-epoch mean). The base model gains 20.4 from the video, and fine-tuning changes that gain by −4.2 [−5.7, −2.8].

On MindCube-Tiny, the two-epoch runs do not change the score (run 0 −0.1, mean 0.2). The four-epoch mean gains 4.1 [1.0, 7.3], meeting the registered transfer rule: a change counts as transfer when the scene-resampled 95% interval excludes zero, and the 5.0-point margin defines "no change" (verdict "transfers"). The fine-tuned models stay near chance, however: the four-epoch runs score 30.5–37.7 and the two-epoch runs 30.0–30.8. The gain concentrates in the "around" setting (14.9 [8.0, 21.9]). The following observations are exploratory (chosen after inspecting the results). Of the 87 "around" yes/no questions, the correct answer is always "yes"; the base model answers yes 33 times, and the four-epoch runs 62–69 times. Without the yes/no questions (chance 27.9), the base model scores 23.9 and the four-epoch gain is 5.1 [1.6, 8.5]. Most of the gain therefore removes below-chance biases in the base model rather than adding spatial reasoning.

Two VSI-Bench runs were rerun after a GPU fault produced invalid output; the rerun scores are used throughout.

F.5 The original two-epoch run (run 0)

Run 0 behavior. In the held-out rooms, run 0's accuracy shows no dependence on DINOv2 similarity with the walk (1.9 points, $p = 0.21$). It gains 36.6 points [33.9, 39.6] from the walk and leads the base model by 40.7 [38.2, 43.5], ahead in all 10 rooms and above Opus in all 10 by 19.6 points [17.3, 22.1]. The two-epoch three-run mean is 80.6% (68.7%–92.5%); the two-epoch mean minus Opus is 7.7 (room mean).

In the apartments, run 0 reaches 86.7% [83.4%, 89.5%] (pooled, all trials; gain 34.2 [29.2, 38.9]; novel 79%; shuffled 53.5%). The two-epoch mean is 79.4% (70.4%–86.7%); minus Opus −6.9.

In the houses, run 0 reaches 91.9% [90.4%, 93.3%] (no walk 51.0%; shuffled 54.2%), above the base model by 31.7 [28.4, 34.8] in all 8 houses. The two-epoch mean is 84.3% (73.2%–92.0%); minus Opus −5.7. Over the 11 photoreal scenes of Experiments 12 and 16, run 0's pooled novel accuracy is 84.2% ($p = 0.91$ vs Opus).

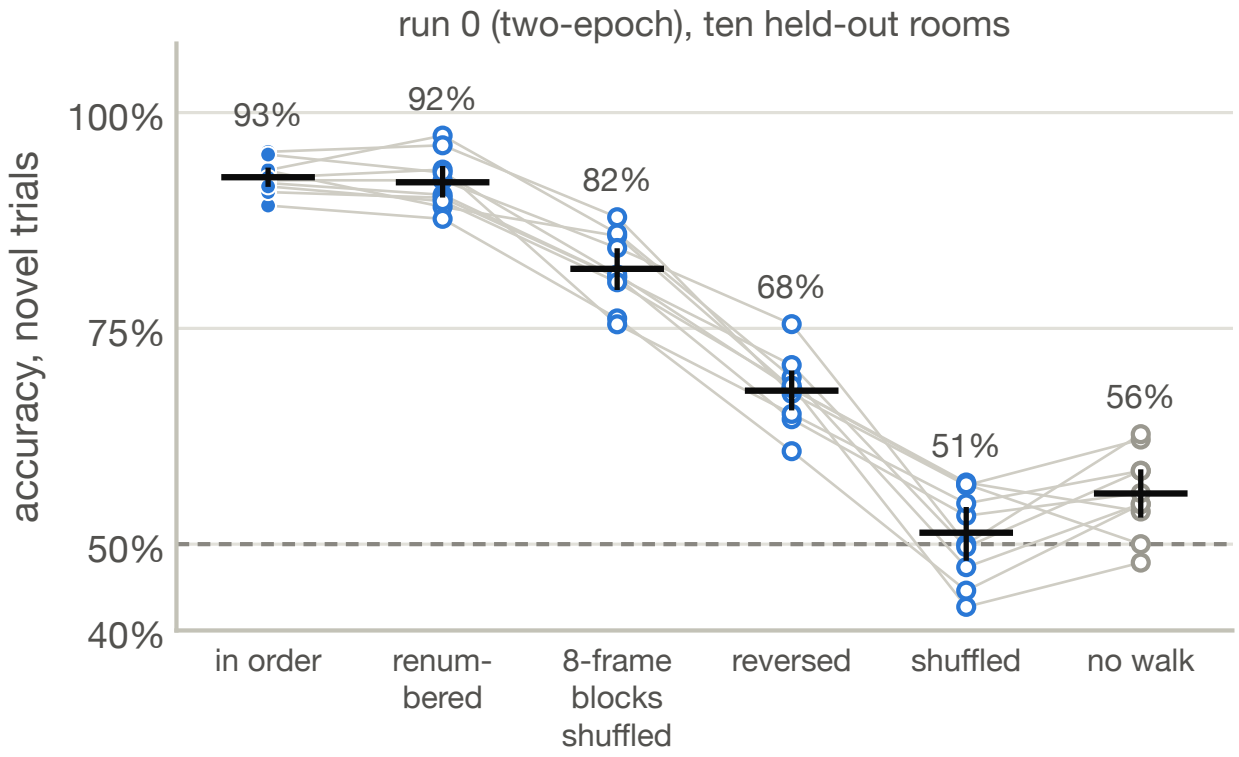


Figure F3. Run 0's accuracy drops under shuffled and reversed walks (Experiment 13a). Accuracy under shuffled, reversed and window-shuffled walks, compared with the ordered walk and the no-walk baseline.

Run 0 order controls (Experiment 13a). In room A, shuffling drops run 0 to 50%; in the held-out rooms (Figure F3) it drops to 51%. Shuffling consecutive 8-frame blocks keeps 74% of the order advantage in the held-out rooms: most of the information is local. Reversing the walk costs 24.7 points but stays 16.5 above a full shuffle, so the model is also sensitive to the direction of travel; these controls do not separate longer-range order from direction. On Experiment 31's definitions (ten held-out rooms, novel trials), run 0's local share is 0.74 (log-odds 0.60); reversal share 0.60 (log-odds 0.72). For run 0, floors S and R coincide (registration). Run 7's shares against run 0's floor S read 0.33 (local) and 1.23 (reversal); this is the wrong floor for run 7, as Experiment 31c explains (floor R removes the real frame numbers that floor S retains).

Run 0 attention cuts (Experiment 15). Cutting attention to the walk frames around the near option costs 8.6 [7.3, 10.0] points more than cutting random frames ($p = 0.002$, all 10 rooms; Figure F4). Cutting the look-alike frames costs no more than random ($p = 0.51$). The decision rests on the stretch of walk around where the view was taken, not on frames that look like it. The excess log-odds cost is 0.80 [0.66, 0.94]; look-alike minus half the near cost is -0.28 [-0.49, -0.11]. The near-option frames are read in both an early window (up to layer 18: -5.5 points) and a late window (-3.1 points). In Experiment 31b (four-epoch), the same early/late split gives run 5 -3.0/ -0.2 and run 7 -3.8/ -0.3 (399 trials).

Run 0 task readout (Experiment 14a). Under the two-choice prompt, run 0's decision becomes readable at the answer token from about layer 18: the nearer-option accuracy is 82% at layer 20, peaking at 92.5% at layer 27; the distance-difference R^2 reaches 0.54 at layer 20, up from 0.14 at layer 18. The cross-sector probe error is 4.55 m. Most of the readable decision signal is the chosen answer itself: a predictor given only the

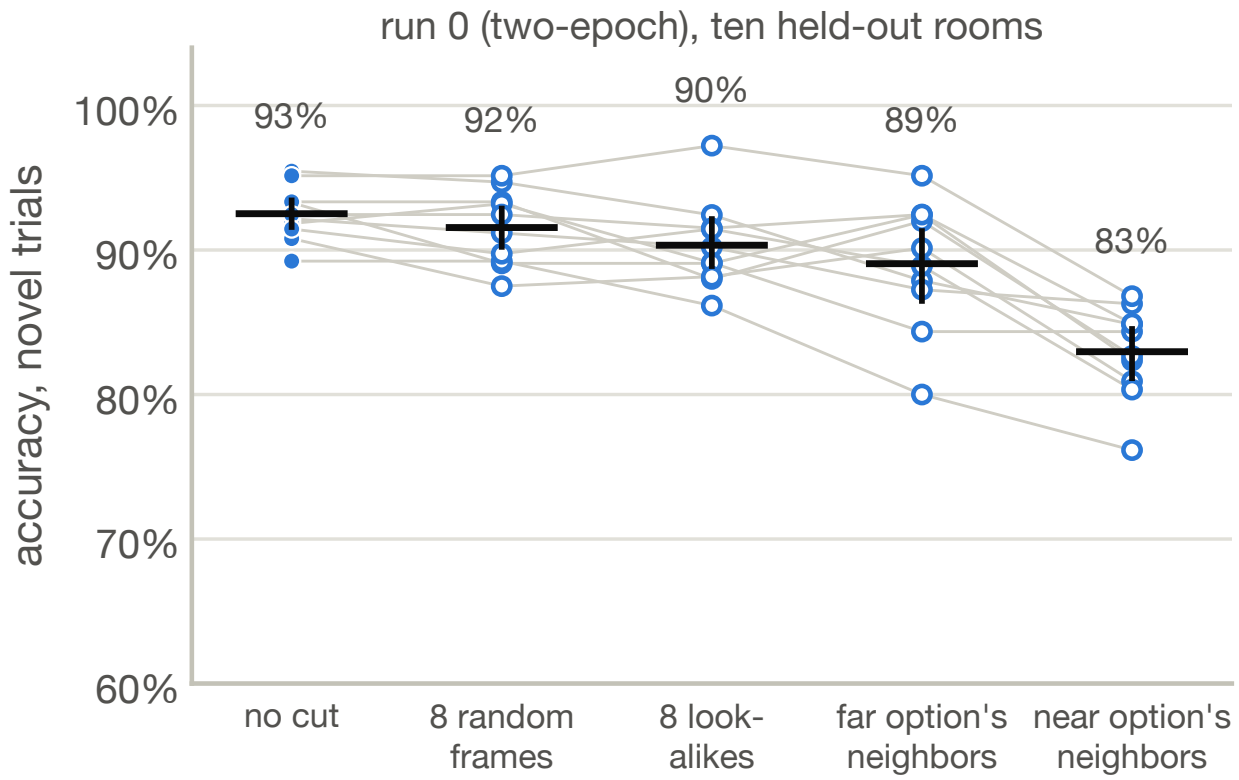


Figure F4. Run 0 attention cuts (Experiment 15). Accuracy when every token outside a set of 8 walk frames is blocked from attending to them, across the held-out rooms.

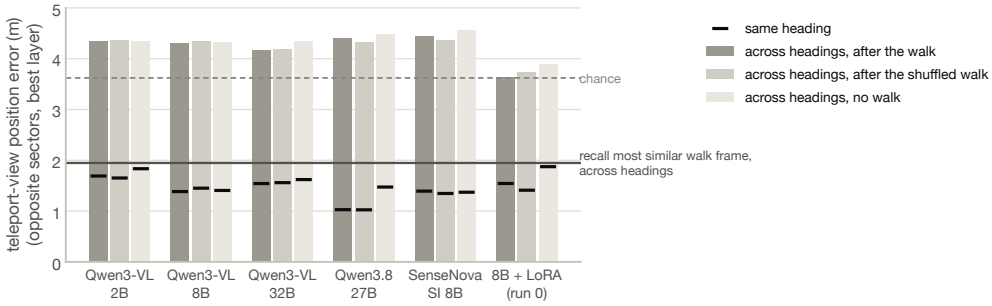


Figure F5. Run 0: position is not linearly decodable across headings from the view's token. (Best strict layer per condition; opposite 90° sectors; dark-gray line = recall baseline; dashed = chance 3.62 m; lower is better.) Bars show the query-view probe's across-sector error after the walk, a shuffled walk and no walk; ticks show within-sector error. This figure shows run 0 (two-epoch) results; the four-epoch units and population assay are in Section 4.

selected option reaches R^2 0.69 (residual 0.03–0.06). Cross-room position in room coordinates: best R^2 0.03 (error 3.38 m, chance 3.50 m), covering the base model and run 0.

Run 0 units. Under the mean-of-two heading test, run 0's heading-invariant share is 14% (walk) and 1% (single frame). The run 0 cross-sector query-view probe gives 3.63 m (chance 3.62 m; Figure F5).

Run 0 parity. The re-recorded run 0 matches on shares (within 0.4 points) and on the task readout, but its strict probe decoding differs by 0.104 m at one layer against a 0.10 m limit; units parity reads fails as registered.

Run 0 population assay. The strongest novel-view condition for run 0 (walk in context) gives HG 0.22 [0.17, 0.28]; without the walk, 0.06 [−0.02, 0.12]. Both read below partial; run 0's registered verdict, on walk frames, is positive.

G Lure-free accuracy

Appendix Table G1 gives each model's lure-free novel-trial accuracy in room A, with and without the walk.

Table G1. Lure-free novel-trial accuracy (%). $n = 102$, chance 50%, room A, 256-frame walk at 256 px unless noted. Claude models at high effort. "Best px" is the higher-scoring upsampled resolution; 512 and 1024 px are bicubic upsamplings of the 256 px renders (more tokens, no new image detail). Unparsed answers count as wrong.

Model	No walk	No walk (best px)	Walk
Qwen3-VL-8B	59	71 (512)	50
Qwen3.8-27B (direct)	56	64 (512)	58
Qwen3.8-27B (thinking)	61	63 (512)	65
Qwen3.6-35B-A3B	61	59 (512)	50
Gemma-4-31B	56	68 (1024)	53
Claude Haiku 5.5 (high)	67		56
Claude Sonnet 5.5 (high)	64		70
Claude Opus 5.5 (high)	63		75
Qwen3-VL-8B + LoRA, run 0 (about 64 frames)	53		92

H Claude effort sweep

We run Claude Haiku 5.5 (Anthropic, 2026a), Claude Sonnet 5.5 (Anthropic, 2026c) and Claude Opus 5.5 at each available effort level (none, low, medium, high), with and without the walk, on lure-free novel trials ($n = 102$). Claude Sonnet 5.5's "none" column uses the `between_tools` workaround, which still emits a median of 342 output tokens; it is not truly thinking-free. Claude Opus 5.5 does not support thinking off.

Appendix Table H1 and Figure H1 show the full effort sweep. Claude Haiku 5.5 does not gain from the walk: correctness averaged over four effort levels gives 27 better and 48 worse ($p = 0.020$, two-sided sign test). Claude Sonnet 5.5 gains at medium effort only (29 better, 12 worse, $p = 0.012$, exploratory: uncorrected for the four effort levels and trial-level in one room); at high effort the walk helps on 24 trials and hurts on 18 ($p = 0.44$).

Claude Sonnet 5.5 picks the first-listed option 44%–63% of the time across the none, low and medium effort levels, with and without the walk. Claude Opus 5.5's effort levels agree on 76% of novel trials. Costs: Claude Opus 5.5 $22.48, Claude Sonnet 5.5 $15.97, Claude Haiku 5.5 $0.99.

Table H1. Claude effort sweep: lure-free novel-trial accuracy (%). $n = 102$, chance 50%. "None" = thinking disabled for Claude Haiku 5.5 and the `between_tools` workaround for Claude Sonnet 5.5 (median 342 output tokens; not thinking-free). Claude Opus 5.5 does not support thinking off. Each cell is one run.

Model	Condition	None	Low	Medium	High
Claude Haiku 5.5	Walk	60	53	54	56
	No walk	68	59	62	67
Claude Sonnet 5.5	Walk	59	67	70	70
	No walk	60	62	53	64
Claude Opus 5.5	Walk	—	76	75	75
	No walk	—	61	66	63

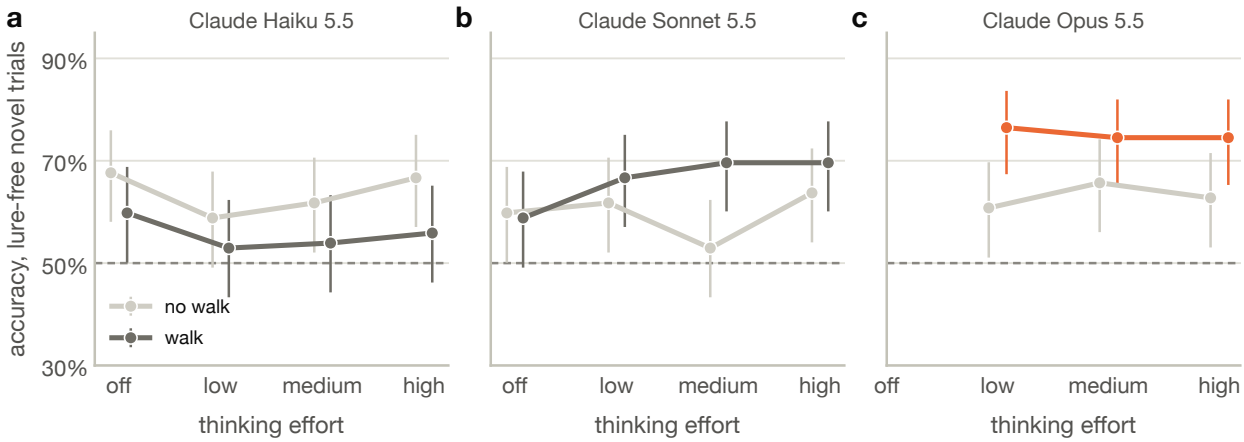


Figure H1. Claude Opus 5.5 gains from the walk, Claude Sonnet 5.5 only at medium effort, and Claude Haiku 5.5 not at all. Accuracy with 95% trial-level Wilson CIs; walk (dark gray, orange for Claude Opus 5.5) vs no walk (light gray), per model. $n = 102$ novel trials; dashed line marks 50% chance.

I Other providers

Appendix Table I1 shows novel-trial accuracy with and without the walk for non-Anthropic API models, using the same prompt and images as Claude Opus 5.5.

J Every Opus number in the main text

All Opus results are single runs of Claude Opus 5.5 with adaptive thinking. Appendix Table J1 lists each number the main text uses, with its effort level, walk length and trial set. Only the 256-frame comparison in the held-out rooms gives the fine-tuned model the same walk as Opus (Section 3.6).

K VGGT details

Subspace interchange patching. Appendix Table K1 shows the full patching results. The 10-d position PCA subspace carries nearly all the position and heading information the camera head reads: its share of the

Table I1. Other providers: lure-free novel-trial accuracy (%). (Room A, $n = 102$, chance 50%, 256-frame walk unless noted. Kimi K3 at reasoning low, temperature 1.0 (fixed by provider). DeepSeek V4.1 Flash: thinking on, reasoning low; 1 unparsed walk answer counted wrong. GPT-6 Astra: reasoning low. Qwen3.5-397B tested separately with a 160-frame walk (amendment 6). MiniMax-M3, MiMo-V2.6-Pro and Muse Spark Open tested via OpenRouter (amendment 5, $10.18). Five further open-weight VLMs could not be tested at the full walk: Nemotron 3.5 Super VL (context limit), Qwen3.5-397B (context limit; see the last row), Qwen3.8 Flash (image limit), Step-5-Preview (image limit), Inkling (request too large). Unparsed answers count as wrong for every model.)

Model	Walk	No walk	Better	Worse	p
Kimi K3	58	61	12	15	0.70
GPT-6 Astra	72	51	35	14	0.0038
DeepSeek V4.1 Flash	57	64	19	26	0.37
MiniMax-M3	55.9%	60.8%	23	28	0.58
MiMo-V2.6-Pro	51.0%	56.9%	19	25	0.45
Muse Spark Open	64.7%	58.8%	21	15	0.41
Qwen3.5-397B (160-fr.)	52.9	56.9	17	21	0.63

Table J1. Every Opus accuracy and gain in the main text. Accuracy with a 95% CI where one is reported. "Novel" and "lure-free" are defined in Section 2; the fine-tuned model sees 64-frame walks except in the 256-frame comparison.

Where	Condition	Effort	Trials (n)	Accuracy
Fig. 1d	room A, walk (256 frames)	high	lure-free novel (102)	75%
§3.1	room A, walk; lure trials	high	lure (50)	60%
§3.1	room A, walk; no-landmark trials	high	no landmark (29)	86%
Fig. 3a	room A, ordered walk	low	lure-free novel (102)	76%
Fig. 3a	room A, shuffled walk, real labels	low	lure-free novel (102)	80%
Fig. 3a	room A, shuffled and relabeled	low	lure-free novel (102)	79%
Fig. 3a	room A, 64-frame walk	low	lure-free novel (102)	77%
Fig. 3a	room A, no walk	low	lure-free novel (102)	61%
Fig. 1d, §3.2, Tab. 1	held-out rooms, walk (256 frames)	low	novel (1,299)	72.9% [70%, 75%]
§3.2	held-out rooms, no walk	low	novel (1,299)	63% [60%, 66%]
§3.2	held-out rooms, shuffled walk, real labels	low	novel (1,299)	78.8% [77.2%, 80.7%]
§3.2	held-out rooms, walk gain (points)	low	novel (1,299)	+9.7 [7.0, 12.1]
§3.2	held-out rooms, shuffled minus ordered (points)	low	novel (1,299)	+5.9
§3.5	held-out rooms, walk gain (points)	low	all	+11.5
§3.5	three Replica apartments, walk	low	all (503)	86.3% (gain +30.6)
§3.5	HSSD houses 1–8, walk	low	all (1,450)	89.8% (gain +28.1)
§3.5	11 photoreal scenes, walk	low	novel (424)	84.7%
§3.4	three Replica apartments, walk, scene mean	low	novel (58)	85.4%
§3.5	seven apartment arrangements, walk	low	novel (67)	91.0%
§3.5	9 new scenes, walk gain (points)	low	all	+29.3
§3.7	room A, the human page's trials, walk	high	novel (45)	33 correct

target–donor difference is 93% at block 21. The exact complement transfers −0.01 at block 21 (0.00 at 19); the complement of pose PCA 30 transfers 0.00; a random 10-d subspace transfers 0.00. The ridge probe's 2-d subspace holds at most 0.1% of the difference and transfers 0.00. Erasing position PCA 10 raises the camera-head error from 0.43 m to 3.38 m at block 21 (1.65 m at 19); erasing a random 10-d subspace gives 0.43 m (0.43 m at 19). Position is just as decodable at block 15 (probe R^2 0.98) but patching barely moves the output (0.03).

Table K1. Subspace interchange patching. Transfer for one target is the movement of the camera head's predicted position, projected onto the line from the target's clean prediction to the donor's clean prediction, as a fraction of that line's length: $(p - p_t)\cdot(p_d - p_t) / |p_d - p_t|^2$, with positions in room coordinates; reported as the mean over 128 targets (donor more than 3 m away, median 5.3 m; s.e.m. 0.018 for position PCA-10 at block 21). Share is the median, over target–donor pairs, of the fraction of the squared norm of their camera-token difference that lies in the subspace (%). Erase = camera-head error after setting the subspace's projection to its mean over fit frames (clean 0.43 m). "All but pose PCA 30" is the complement of the top 30 pose-PCA directions (a looser complement than position PCA 10's exact complement, which transfers −0.01).

	Share (%)		Transfer			Erase (m)
Subspace	Blk 19	Blk 21	Blk 15	Blk 19	Blk 21	Blk 21
Whole token	—	—	0.03	0.41	0.87	—
Position PCA 10-d	93	93	0.03	0.40	0.85	3.38
All but pose PCA 30-d	—	—	0.00	0.00	0.00	—
Heading PCA 6-d	58	65	0.00	0.07	0.32	—
Ridge probe 2-d	0.0	0.1	0.00	0.00	0.00	—
Random 10-d	—	—	0.00	0.00	0.00	—
Random 30-d	—	—	0.00	0.01	0.01	—

Patching controls (Experiment 13c). *Fixed alignment.* With the similarity transform fixed (not refitted per patched run), PCA 10 transfers 0.85 at block 21 (0.40 at 19) and the whole token 0.87 (0.41). Anchor predictions move only 0.4 cm, so the alignment is not driving the transfer. *Heading-matched donors.* With donors whose heading gap is 28° (instead of 91° in the original experiment), PCA 10 transfers 0.80 (0.47) and the whole token 0.82 (0.49). The transfer holds within a heading. *Heading carried by the subspace.* Patching PCA 10 turns the predicted heading by a median of 77° at block 21 (25° at 19) and by 23° with heading-matched donors (11° at 19).

λ-sweep. The λ-sweep ran on an RTX 4070 and reproduces the published run (clean error 0.43 m vs 0.43 m; block-21 PCA-10 transfer 0.85 vs 0.85). Two R^2 definitions apply: the shrunken predictor's own R^2 and the R^2 of a least-squares readout refit inside the same 2-d subspace. At the best λ (1), the shrunken predictor gives R^2 0.99 and the refit gives 0.99. At $\lambda = 10^6$, the shrunken predictor drops to 0.22 and the refit to 0.84, while transfer rises to 0.20. The Haufe pattern (Haufe et al., 2014), which recovers the source directions rather than the filter weights, transfers 0.20.

Mean-replacing neurons. Appendix Table K2 shows the effect of mean-replacing camera-token MLP neurons on the camera head's median error (768 test frames). Removing all 4,096 neurons per block in three blocks adds 6 cm to the camera error. Position is recovered from the residual stream and attention, not from these neurons alone.

Where the invariant units write (Experiment 23). This test was registered before the main recording; a smoke recording (384 fit frames, 8 pairs) had already started. Each of block 19's 4,096 MLP neurons has a *write vector*: its column of the second linear layer (fc2), scaled by the block's LayerScale. That vector is exactly what the neuron adds to the residual stream at the camera token, the same token and block where the patching test acts. We group the neurons by the mean-of-two heading test: 673 are invariant (one of their poles is an invariant place unit), 810 have a place pole but none invariant, and 2,613 have no place pole. For each write vector we measure the fraction of its squared norm inside position PCA 10 at block 19 (the

Table K2. Replacing camera-token MLP neurons with their mean: camera-head median error (m). Neurons mean-replaced over all 12,288 walk frames. Clean run: 0.42 m. Invariant = neurons with a place pole passing the older 180° halves test; heading-specific = place neurons with none passing; each set is a random draw of the smaller set's size per block (532 at block 19).

Neurons replaced	Block 19	Blocks 15+19+23
None (clean)	0.42	
Invariant (180° halves)	0.43	0.45
Heading-specific	0.42	0.43
Random (matched count)	0.42	0.43
All (4,096/block)	0.43	0.48

subspace whose patching transfers 0.40 there). A random direction puts 0.0098 of its squared norm in any 10-d subspace. The registered rule requires three conditions for "connect": the invariant neurons' median fraction exceeds twice random, and its differences from the other two groups' medians each have a 95% bootstrap interval above zero.

All three conditions are met. The invariant neurons' median is 0.057 [0.055, 0.061], or 5.9 times random, against 0.044 (4.5 ×) for non-invariant place neurons and 0.041 (4.2 ×) for non-place neurons. The differences are [0.010, 0.019] and [0.014, 0.021]; 96% of the invariant neurons load more than the 95th percentile of random directions. In walk mode (the anchors followed by 64 consecutive walk frames), the same weights also connect (6.1 times random). At block 21, where the subspace transfers 0.85, no group stands out: the invariant, non-invariant place and non-place medians are 0.030, 0.031 and 0.028 (all about 3.1 times random). On the read side at block 19 (each neuron's input direction against the MLP-input subspace), the invariant neurons score 0.018 against at most 0.007 for the other groups (1.8 times random, below the rule's 2× bar). Block-19 invariant writes projected into the block-21 subspace give 5.0 times random; the two subspaces overlap 0.69.

The invariant neurons' summed write inside the subspace is 2.1% of the position signal already in the residual stream (post hoc; all 4,096 neurons: 3.0%). The registered signed share is slightly against that signal, by 0.9% (all neurons: 1.4%), at the 18th percentile of equal-size random sets of place neurons. Replacing each set's activations with a donor frame's (interchange intervention) transfers at most 0.002 at either block, against 0.40 (block 19) and 0.85 (block 21) for the subspace. Mean replacement moves the predicted camera 1.1 cm–3.3 cm. The all-neuron ceiling is itself below the registered 0.1 bar, so no neuron set at this token could have passed. These results cover single mode at the camera token, blocks 19 and 21 only; patch tokens are untested.

Steering. We set neurons to their mean activity at a target square L (median 5.7 m away). Invariant place units (180° halves) transfer −0.001 at block 19 and −0.002 at block 21; all neurons transfer −0.003 across both blocks. A 4 × push gives −0.011. Neuron conditions move the camera at most 15 cm, whereas the whole residual token at L transfers 0.29 at block 19 and 0.64 at block 21 (Figure K1a).

Single unit. A single unit pushed to 4 × its peak moves the prediction 7 cm at block 21 (random non-place: 0.2 cm).

Joystick. Adding the "1 m east" direction inside the pose subspace at block 21 moves the prediction 0.18 m east per meter (Figure K1b; 0.16 at 19), in a straight line (R^2 1.00), with little sideways drift (0.01) and

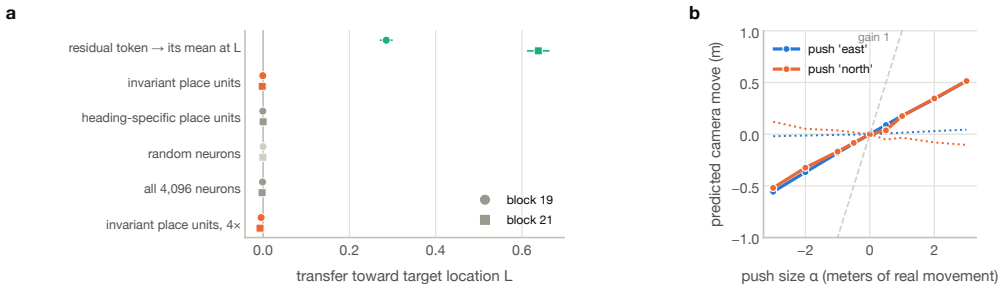


Figure K1. Place units read position but do not steer it. (Mean ± s.e.m.; circles = block 19, squares = block 21; invariant = older 180° halves test.) (a) Steering transfer toward L per condition; first row = whole-token reference. (b) Joystick at block 21: solid = along pushed axis, dotted = sideways, dashed = gain 1.

heading change at most 4°. Patch tokens and all-block interventions are untested.

Subspace definitions. Position PCA: the top principal directions of the mean residual token per 1 m position bin. Heading PCA: per 30° heading bin. Pose PCA: per 2 m × 90° joint bin. All fit on 2,048 frames not used for testing.

The ten directions in detail. The six pose features explain 84.5% [82.8%, 85.4%] of the token's variance at block 21, in pairs: directions 1–2 follow heading (62% of the variance), 3–4 position in the camera's axes (19%), 5–6 position in the room's axes (10%), and 7–10 are second order (5%). The camera-frame gradient turns with the camera; the room-frame gradient stays put. At block 21, heading alone closes 0.78 of the heading gap and moves translation 0.07; camera frame alone moves translation 0.57 and closes 0.01 of the heading gap. The second-order family, tested in the label-free arm only, is not read at either block. The strict invariant share falls from 34% at block 19 to 19% at block 21 (these two blocks only).

Per-part patching (Experiment 32). The setup is unchanged from Experiment 27: VGGT-1B in single mode (8 anchors and one walk frame, room A), the same 128 targets and donors, bf16 autocast. The patch point is the residual at the walk frame's camera token after global block g ($g = 21$, primary; $g = 19$, secondary). A condition writes $h_t + w$, where w is part of the difference $h_d - h_t$.

The label-free subspaces are Experiment 27's ordinary PCA (fit with no labels on the 2,048 fit frames). At each block the directions are grouped into families, fixed from the token PCA before the run, by which feature family explains each direction better: heading (directions 1–2 at both blocks), camera frame (3–4 at block 21; 3 and 6 at block 19), room frame (5–6 at block 21; 4 and 5 at block 19) and second order (7–10 at both blocks). 9 conditions at each block: the ten; each family alone; the ten without each family.

The feature-fit arm repeats the test along regression directions, because PCA directions mix features across pairs. The 2,048 fit frames' residuals are regressed on the six features ($\cos\theta$, $\sin\theta$; x, y; $x\cos\theta + y\sin\theta$, $x\sin\theta - y\cos\theta$; (x, y) from the room center, true pose). With B the coefficient matrix, B^+ its pseudo-inverse and $d = h_d - h_t$, a condition writes $w = B_{\text{keep}}(B^+ d)_{\text{keep}}$, an oblique projection that leaves the other families' codes as they were. 7 conditions at each block: all six features; each family alone; all six without each family.

Position transfer is the fraction of the target–donor line (room coordinates, mean over pairs). Heading gap closed is the fraction of the heading difference recovered. Translation transfer is the same fraction for the walk frame's camera-from-world translation (camera axes, scaled to meters by each run's anchor fit). Heading gap closed and translation transfer read the camera head's two outputs separately. All intervals are percentile bootstraps (10,000 resamples) over the 128 pairs, paired across conditions.

Table K3. Experiment 32's rule: heading and camera frame are needed at both blocks and second order is not read. Room frame is needed at block 19; at block 21 the two arms disagree, so its numbers carry no verdict. L = share of the transfer lost when the family is left out; A = share the family reaches alone; 95% intervals over the 128 pairs. The reference is the ten directions (label-free arm) or all six features (feature-fit arm). Second order has no feature-fit counterpart.

		Block 21			Block 19		
Family	Arm	L	A	Verdict	L	A	Verdict
Heading	label-free	0.48 [0.39, 0.57]	0.32 [0.24, 0.42]	needed	0.26 [0.15, 0.36]	0.07 [−0.03, 0.16]	needed
	feature-fit	0.47 [0.37, 0.56]	0.33 [0.24, 0.42]		0.27 [0.15, 0.38]	0.09 [−0.02, 0.19]	
Camera frame	label-free	0.45 [0.36, 0.54]	0.32 [0.24, 0.41]	needed	0.47 [0.38, 0.56]	0.28 [0.19, 0.36]	needed
	feature-fit	0.34 [0.25, 0.43]	0.27 [0.19, 0.34]		0.46 [0.35, 0.56]	0.29 [0.20, 0.38]	
Room frame	label-free	0.19 [0.16, 0.22]	0.17 [0.15, 0.20]	no verdict	0.48 [0.41, 0.55]	0.41 [0.33, 0.50]	needed
	feature-fit	0.32 [0.28, 0.36]	0.24 [0.20, 0.28]		0.48 [0.41, 0.57]	0.42 [0.36, 0.50]	
Second order	label-free	0.04 [0.01, 0.07]	0.04 [0.02, 0.06]	not read	0.06 [0.00, 0.10]	0.06 [0.02, 0.10]	not read

The registered decision rule defines L = share of the transfer lost when a family is left out, and A = share the family reaches alone, both relative to the reference (the ten in the label-free arm, all six features in the feature-fit arm). Needed: $L \geq 0.25$ with CI above 0. Not read: $L < 0.10$ and $A < 0.12$, both interval ends below 0.25. Read but redundant: $L < 0.10$ and $A \geq 0.25$ with CI above 0. Otherwise unclear, and the numbers are reported with no verdict. The weaker arm governs; second order has no feature-fit counterpart.

The ten reproduce Experiment 27 within 0.01 at each block, with the largest single-pair difference 0.003 (block 21) and 0.007 (block 19). The rerun clean residuals equal the stored ones exactly at both blocks.

Unregistered. The two arms disagree at block 21 because the PCA pair is not the whole room code. At block 21, 71% of the fitted room directions' energy lies in PCA pair 5–6 and 25% in pair 3–4, the camera-frame family; at block 19, 75% and 20%. So leaving out the room pair leaves about a quarter of the room code in place, which accounts for the gap between the arms. Room frame alone moves the prediction 0.81 m in all, 0.29 m off the position line, with a turn of 2.9°. The parts do not add: alone they sum to 0.74 of 0.86.

Token PCA (unregistered). The token PCA uses Experiment 27's recordings: the walk frame's camera token after global blocks 19 and 21 in single mode, room A. Directions are Experiment 27's ordinary PCA, fit with no labels on the 2,048 fit frames. Six features of the true pose define three families: heading ($\cos\theta$, $\sin\theta$); room-frame position (x, y); and camera-frame position ($(x,y)\cdot f$ and $(x,y)\cdot r$, the form of the translation VGGT's camera head outputs; correlation with the head's t_z and t_x: -0.993). A mirror control (the camera-frame products with the rotation reversed) and second-order features ($\cos 2\theta$, $\sin 2\theta$, x^2+y^2) are scored the same way. Each family's share is the variance it explains in 6-fold cross-validation (grouped by walk, 24 walks) over the 4,096 other even-walk frames, relative to the token's total held-out variance. On the odd walks, the six features explain 89.4% of the top ten directions' variance at block 21, against 89.0% on the even walks. Raw pixels are scored under their own PCA on the fit frames.

As a consistency check, not independent evidence: the median share of a neuron's held-out variance in the room pair at block 21 is 0.149 for invariant place neurons (mean of two) against 0.063 for the other place neurons (a unit that reads room position is heading-invariant by construction).

Maps across rooms. We place 8 anchor views at the same 8 poses in rooms A and B. Because the anchors sit at the same poses, they define the same coordinate frame in both rooms.

Block-19 place units keep their maps across rooms (Figure K2a,b; $r = 0.95$, within A 0.95, random pairs 0.17).

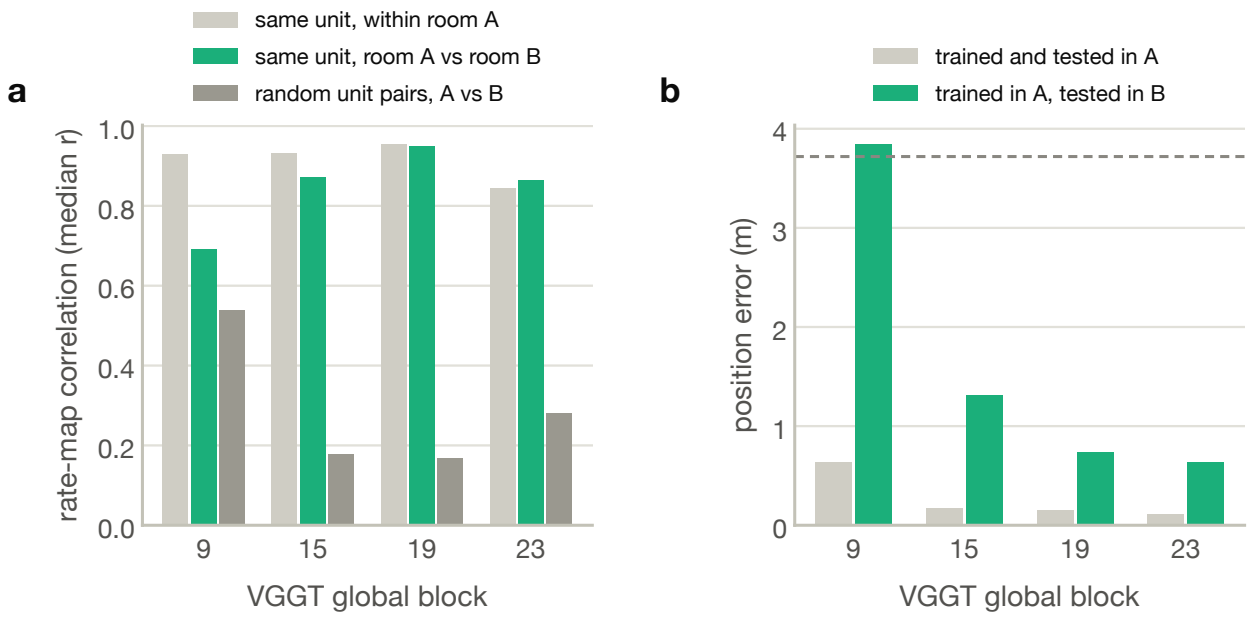


Figure K2. VGGT's maps keep across rooms when the anchors share the same poses. (a) Rate-map correlation per block: within room A (even vs odd walks), room A vs room B, and random pairs. (b) Decoder trained in room A, tested in room A or room B, per block.

A decoder trained in room A works in room B (0.74 m vs chance 3.72 m). Block 9 is more room-specific (3.84 m). The camera head is less accurate in room B (0.85 m vs 0.42 m).

Cross-room patching (Experiment 14d). Subspaces fit in room A are patched into room-B runs (128 targets; clean error 0.85 m in room B vs 0.42 m in room A). At block 21, position PCA 10 transfers 0.87 (A-fit) vs 0.87 (B-fit); PCA 2 transfers 0.19 vs 0.22; the ridge probe at $\lambda = 10^4$ transfers 0.15 vs 0.16. At block 19, the same pattern holds: 0.45 (A-fit) vs 0.42 (B-fit). All paired A-fit minus B-fit differences fall between -0.04 and 0.03, within the registered tolerance of 0.1. Subspace overlaps range from 0.70 to 0.92. The 10-d causal subspace is not a room-A coordinate basis: it carries over to a room with different walls, posters and pillars.

Anchor controls (Experiment 14e). All results are at block 19 in room A. The camera head works with every anchor set: 0.42 m (standard), 0.47 m (reordered), 0.42 m (rotated). Split-half stability of the standard maps is 0.95; the standard maps correlated against themselves turned by 45° give 0.66, confirming that the test can distinguish a turned map from a kept one.

All anchor poses rotated 45° about the room center. Rate maps correlate 0.95 with the turned standard map and 0.68 with the unturned map (-0.01 turned the other way); 96% of 1,505 place units are closer to the turned map. Among the 957 place units that pass the older 180° halves test: 0.97 and 99% closer.

The same anchors reordered (only the first camera changes, a 45° change in its heading): 0.85 against the turned map, 0.03 the other way; 71% closer to the turned map. The reordered condition fits the turned map less well than the full rotation (0.85 vs 0.95), as expected if the first camera's position also matters, but only rotation was registered so we do not claim it.

Position decoders do not transfer between anchor sets: 2.66 m–3.80 m across sets, against 0.13 m–0.15 m within a set (chance 3.72 m).

The maps move with the anchors, so VGGT's place units code position in the frame the anchors define, consistent with the first camera's frame, not the room's own frame. The rotation parallels the finding that hippocampal place fields turn with a rotated cue card (Muller and Kubie, 1987).

L Full statistics for the main text

Table L1: **Full statistics behind the main text's numbers.** The main text gives one number per result; this table gives the ranges over the fine-tuned model's five runs, 95% CIs and the shares under the mean of the N–S and E–W correlations ("mean of two", in brackets). Rows follow the main text; a few give appendix-only detail for the section named.

Sec.	Result	Full statistics
3.1	Base 8B walk gain (64 frames) in room A, 95% CI	−8.8 [−21.0, 2.6]
3.1	Irrelevant walk minus no walk (base 8B), 95% CI	−5.9 [−15.2, 3.7]
3.1	Larger models (8B, 32B, 27B): share of nearest-frame picks on novel views that face within 45° of the query (range)	69%–80%
3.1	Opus's dependence on DINOv2 appearance similarity, held-out rooms, low effort, novel trials, with and without the walk	with walk: 1.6, $p = 0.53$; without: 7.8, $p = 0.0039$
3.1	Opus's lure-edit comparison	Opus: 53% vs 66%
3.2	Opus's held-out-room walk gain, 95% CI	9.7 [7.0, 12.1]
3.2	Opus room-A walk gain averaged over efforts, 95% CI	12.1 [2.9, 21.2]
3.2	Opus room-A shuffled minus ordered, low effort, 95% CI	3.9 [−4.5, 12.5]
3.2	Opus held-out shuffled minus ordered, 95% CI	5.9 [3.9, 8.2]
3.2	Opus held-out shuffled minus no walk, 95% CI	15.6 [12.3, 18.9]
3.2	Opus room-A ordered minus relabeled, low effort, 95% CI	−2.9 [−10.6, 4.8]
3.2	Opus held-out gain	9.7
3.2	Fine-tuned lead over Opus	24.2
3.3	Renumbering cost, runs 5 and 7, 95% CIs	−0.2 [−0.8, 0.4] and 0.0 [−0.7, 0.9]
3.3	8-frame block share of order effect, runs 5 and 7, 95% CIs	0.80 [0.76, 0.84] and 0.74 [0.68, 0.79]
3.3	8-frame block share on log-odds scale, runs 5 and 7, 95% CIs	0.55 [0.49, 0.62] and 0.48 [0.39, 0.57]
3.3	Reversal share, runs 5 and 7, 95% CIs	0.48 [0.42, 0.54] and 0.47 [0.41, 0.54]
3.3	Reversal share on log-odds scale, runs 5 and 7	0.70 and 0.70
3.3	Fine-tuned model, shuffled walk, held-out rooms: range over the five runs	54.4%–82.1%
3.3	Fine-tuned model, shuffled walk, room A: mean (range over runs)	71% (59%–81%)
3.4	Fine-tuned model's lead over retrieval baseline in procedural rooms, 95% CI	29.4 [24.9, 34.2]
3.4	Fine-tuned model accuracy when retrieval baseline is right vs wrong	97.7% on 879 trials right vs 95.8% on 420 wrong
3.4	Opus accuracy when retrieval baseline is right vs wrong	76.7% vs 66.0%
3.4	Opus lead over retrieval baseline in procedural rooms, 95% CI	+5.2 [−0.3, 10.9]
3.4	Fine-tuned model's lead over retrieval baseline in photoreal scenes, 95% CI	13.2 [5.6, 19.4]
3.4	Per-bin leads in photoreal scenes where retrieval baseline does best	9.6 (heading gap 90–120°) to 15.0 (options in different rooms, new view in the near option's room, houses only)
3.4	Retrieval baseline score in apartments	82.2% on 58 novel trials
3.4	Fine-tuned model's lead in houses only, 95% CI	17.5 [13.1, 22.0]
3.4	Opus lead over retrieval baseline in photoreal scenes, 95% CI	10.1 [2.8, 16.9]

Sec.	Result	Full statistics
3.4	Sequence matcher accuracy, 95% CI	73.3% [70.2%, 76.0%]
3.4	Matcher lead over retrieval baseline, 95% CI	5.6 [1.1, 9.6]
3.4	Matcher relative to Opus, 95% CI	0.4 [−3.0, 4.0]
3.4	Fine-tuned model lead over matcher, 95% CI	23.8 [21.0, 26.8]
3.4	Fine-tuned model lead over matcher per room	16.1–31.8 in each of 10 rooms
3.4	Matcher order gain, 95% CI	−1.0 [−2.2, 0.2]
3.4	Matcher permuting frame numbers	0.0
3.4	Matcher validation scores (order-free vs ordered)	77.6% vs 77.0%
3.4	Matcher in photoreal, 95% CI	47.8% [43.7%, 51.9%]
3.4	Matcher in new houses, fine-tuned model's lead	52.2%; lead 38.5 [33.8, 42.5]
3.4	Pixel variant narrows gap, 95% CI	17.1 [15.3, 19.1]
3.4	Held-out procedural rooms in the retrieval comparison	10
3.4	Photoreal novel trials, scene means: retrieval baseline and fine-tuned model	74.9% and 88.1%
3.4	New houses, novel trials: retrieval baseline (scene mean)	71.8%
3.4	Sequence matcher in the new houses: novel trials	947
3.4	Control walks for the walk-shift test: mean step when moving; share of frames revisited after 64 or more steps	0.22 m; 30%
3.5	Fine-tuned model in the 11 photoreal scenes of Figure 7b (Replica apartments and HSSD houses 1–8), all trials with the walk: range over runs; trial counts	93.0%–94.0%; 1,953 trials, 424 novel
3.5	Fine-tuned model minus Opus, all trials, range over runs: the three Replica apartments and HSSD houses 1–8	apartments by 4.6–6.0, houses by 3.9–5.0
3.5	The three Replica apartments, novel trials: Opus against the fine-tuned model (range over runs), p range	89.7% vs 84.5%–87.9%; $p = 0.51$–1.0
3.5	Opus walk gain, the three Replica apartments, all trials, 95% CI	30.6 [25.4, 35.7]
3.5	Opus, the three Replica apartments, novel trials	90%
3.5	Opus's walk gain, all trials: the three Replica apartments against the procedural rooms	30.6 against 11.5
3.5	Opus houses walk gain, 95% CI	28.1 [26.2, 30.5]
3.5	Opus shuffled vs ordered walk in houses, 95% CI	0.9 [−0.5, 2.3], 5/2 houses
3.5	Opus shuffled minus ordered walk, all trials, the 11 scenes (three apartments, HSSD houses 1–8), 95% CI	1.7 [0.2, 3.4], 7/11 scenes up
3.5	HSSD houses 1–8, novel trials: fine-tuned model (range over runs) against Opus	88.7%–90.5% vs Opus 84.8%
3.5	Base model house walk gain, p-values	8.8 points, $p = 3 \times 10^{-8}$; novel $p = 0.077$
3.5	Same-room house trials lead, 95% CI	19.8 [9.9, 26.7] on 176 novel trials
3.5	VGGT novel accuracy in houses	74.9%
3.5	New houses criteria	house met, FRL met, Opus walk-gain met
3.5	New houses shuffled walk margin	smallest margin over shuffled walk 16.5
3.5	Opus walk gain in new scenes detail	382 trials right only with walk vs 40 right only without; walk helps in 9, hurts in 0
3.5	FRL apartment detail	four-epoch 97.7%–98.7%, Opus 99.3%, base 76.6%
3.5	Fine-tuned model vs Opus in new houses	+4.4 to +6.4 all trials, +7.2 to +7.9 novel
3.5	Fine-tuned model vs Opus over all houses	+4.7 to +5.4
3.5	Fine-tuned model vs Opus in apartments	+2.4 to +3.4 all trials; novel: 91.0% vs 86.6%–89.6% (not significant)
3.5	New houses retrieval baseline lead, 95% CI	18.9 [14.4, 23.0]
3.5	New houses per-bin retrieval detail	distance bin 4 m or more: retrieval baseline 83.4%, lead 11.7 [8.8, 14.1]; HSSD1–8: 11.6

Sec.	Result	Full statistics
3.5	Opus in new houses vs retrieval baseline	83.9% novel; +10.4 vs 73.6% there
3.5	The three Replica apartments: lure-free trials (familiar)	503 (445)
3.5	HSSD houses: lure-free trials (novel)	1,450 (366)
3.5	FRL apartment arrangements: lure-free trials (novel)	399 (10)
3.5	New houses: lure-free trials (novel)	3,067 (947)
3.5	Fine-tuned model in the new houses, all trials with the walk (scene mean)	93.7%
3.5	New houses with Opus answers	5
3.5	Opus's walk gain, the three Replica apartments, novel trials (points)	41.4
3.5	New houses novel trial count	947 novel trials
3.5	New houses bin where retrieval baseline does best	distance bin 4 m or more: lead 11.7
3.6	GPT-6 Astra walk gain, 95% CI	20.6 [7.5, 35.1]; 9 walks up, 1 down
3.6	Astra shuffled minus ordered, 95% CI	5.9 [−2.5, 15.4]
3.6	Astra novel scores	72% vs 51%, $p = 0.0038$
3.6	Qwen3.5-397B novel scores	52.9% vs 56.9%, $p = 0.63$
3.6	VSI-Bench fine-tuned model change, 95% CI	−2.1 [−3.1, −1.1]
3.6	VSI-Bench metric task losses	counting −5.4, absolute distance −6.4
3.6	VSI-Bench view-relating, 95% CI	−0.9 [−2.7, 1.0]
3.6	MindCube-Tiny gain, 95% CI	4.1 [1.0, 7.3]; 1,050 questions, chance 33.0
3.6	Gemma detail: base model; held-out (validation-room) accuracy at the end of training; update at which the loss left its plateau	base 51.9%; training accuracy 54.7%–81.3%; plateau 160–485; 1 run never exited; bar not met
3.6	Gemma best run leads Opus by	15.7 in held-out rooms
3.6	Gemma in new houses	47.6%–84.2%
3.6	Matched-Opus lead, 95% CI	+18.6 [16.6, 20.8]
3.6	Matched-Opus fine-tuned model scores, 95% CI	91.4% [90.3%, 92.6%], runs 90.6%–92.7%
3.6	Matched-Opus Opus scores, 95% CI	72.9% [70.4%, 75.1%]
3.6	Matched-Opus per-run and per-room leads	single runs +17.7 to +19.8; all 10 rooms, +13.4 to +25.8
3.6	Matched-Opus longer walk cost, 95% CI	−5.7 [−6.6, −4.7]; down in all 10 rooms
3.6	Matched-Opus 64-frame lead	+24.2
3.6	Opus room A 64 vs 256 frames	77% with 64, 76% with 256
3.6	Fine-tuned Gemma model, held-out rooms, novel trials with the walk (mean of its runs)	69.2%
3.6	256-frame comparison: runs that meet its criterion on their own (of five)	5
3.7	Participant 1 study times	21–56 s per walk
3.7	Participant 1 p-value and lure	$p = 2 \times 10^{-5}$; 17 of 23 lure
3.7	Participant 2 study times and run duration	3 s and 3 s on walks 11 and 15; whole run 6.4 min
3.7	Participant 2 detail	19 of 45, $p = 0.37$; 6 of 23 lure
3.7	Base model lure performance	4 of 23 lure
4	Pixel place units	72%
4	VLM strict single-frame max	0.3%
4	Strict test threshold	0.5
4.2	Invariant synthetic units, mean-of-two pass rate	100%
4.2	Conjunctive synthetic units, mean-of-two pass rate	0.9%
4.2	View-dependent synthetic units, mean-of-two pass rate	0%
4.2	Gain-modulated synthetic units ($R = 5$), mean-of-two pass rate	97.5%
4.2	Gain-modulated units, rate ratio at half pass (mean of two)	$R \approx 18$

Sec.	Result	Full statistics
4.2	Invariant synthetic units (stability 0.5–0.6), mean-of-two pass rate	49%
4.2	Invariant synthetic units (stability 0.7–0.8), mean-of-two pass rate	98%
4.2	RNN heading-invariant share, mean of two	98%–100%
4.3	Single-frame strictly invariant share, mean of two: VLMs; any model (VGGT without anchors)	1%; 6%
4.3	Strictly invariant share of place units with the walk in context, Qwen-family VLMs including the two-epoch fine-tune (range over models)	2%–29% [7%–51%]
4.3	The same share for a moving average of single-frame activations, Qwen-family models (range over models)	82%–88% [90%–94%]
4.3	Qwen-2B walk strict share, mean of two	47%
4.3	Qwen-8B walk strict share, mean of two	7%
4.3	Qwen-2B EMA strict share, mean of two	91%
4.3	Pixel EMA strict share, mean of two	97%
4.3	VLM single-frame strict share range, mean of two	0%–1%
4.3	VGGT single-frame share, mean of two	45%
4.3	VGGT walk share, mean of two	63%
4.4	Fine-tuned walk share, mean of two	18.2%–31.4%
4.4	Fine-tuned single-frame share, mean of two	0.6%–1.3%
4.4	Fine-tuned EMA share, mean of two	90.0%–91.3%
4.4	Invariant synthetic pass rate at layer 35's stability, mean of two	85%
4.4	Median place-unit stability at layer 35	0.62–0.67
4.4	Invariant synthetic pass rate at layer 35's stability bin	59%
4.5	Base 8B walk HG, 95% CI	0.73 [0.72, 0.75]
4.5	Fine-tuned EMA HG	0.99
4.5	Base model teleport-view HG, 95% CI	0.00 [−0.06, 0.05]
4.5	Base model teleport-view HG without walk, 95% CI	−0.04 [−0.09, 0.01]
4.5	Teleport views: HG after the ordered walk minus after the shuffled walk, range over runs	0.03–0.11
4.5	Teleport-view unseen heading error	2.57–2.84 m
4.5	Teleport-view seen heading error	1.38–1.58 m
4.5	Teleport-view chance error	3.67 m
4.5	Threshold comparison, base model: HG for the teleport views without the walk against single frames	−0.04 vs −0.28
4.5	Threshold comparison, fine-tuned model: HG for the teleport views without the walk against single frames (ranges over runs)	0.20–0.30 vs −0.13 to −0.03
5	VGGT anchors single-frame strict	34%
5.1	PCA-10 transfer at block 19	0.40
5.1	PCA-10 complement transfer at block 21	−0.01
5.1	Random 10-d subspace transfer at block 21	0.00
5.1	Camera-head error, clean	0.43 m
5.1	Camera-head error, PCA-10 erased at block 21	3.38 m
5.1	Camera-head error, random 10-d erased at block 21	0.43 m
5.1	Heading PCA-6 transfer at block 21	0.32
5.1	Position minus ordinary PCA transfer at block 21, 95% CI	−0.003 [−0.010, 0.004]
5.1	Position minus ordinary PCA transfer at block 19	−0.006
5.1	Difference-matched control transfer at block 21	0.86
5.1	Random 10-d subspace transfer at block 21 (subspace comparison)	0.00
5.1	Position PCA-10 share of fit frames' variance	0.94

Sec.	Result	Full statistics
5.1	Ordinary PCA-10 share of fit frames' variance	0.95
5.1	Top-2 ordinary PCA heading gap closed	0.78
5.1	Position PCA-2 heading gap closed	0.05
5.1	Six pose features, share of token variance at block 21, 95% CI	84.5% [82.8%, 85.4%]
5.2	Note: the invariant units in this test are those that pass the mean-of-two test; the six-pair set was not tested	—
5.2	Chance alignment of writes with PCA-10	0.0098
7	Fine-tuned model's per-bin margin over every retrieval rule in the procedural rooms	at least 26.0 points in every bin
7	Opus's shuffled-over-ordered margin in the apartments, with CI	3.6 [1.2, 6.0]
7	GPT-6 Astra's shuffled-vs-ordered p-value	$p = 0.26$
7	Fine-tuned model's lead over Opus in the held-out rooms: range over runs (each the mean over rooms)	23.4–24.6 points (Opus 72.9%)
7	Matched-frame (256) lead of the fine-tuned model over Opus, 95% CI	+18.6 [16.6, 20.8] over rooms (+17.7 to +19.8 across runs)
7	Fine-tuned model vs Opus in photoreal houses	13 houses by +4.7 to +5.4
7	Fine-tuned model vs Opus in apartments	apartments by +2.4 to +3.4; CI over 7 apartments does not stay above zero
7	All seven apartment arrangements (67 novel trials): Opus against the fine-tuned model (range over runs)	Claude Opus 5.5 91.0% vs 86.6%–89.6% (not significant)
7	Single-frame strictly invariant share, highest VLM	at most 0.3%
7	Fine-tuned model's walk strict-test pass rate	at most 17%
7	Fine-tuned model's walk HG range	0.77–0.85
7	Fine-tuned model's teleport-view HG range	0.39–0.49
7	VGGT strictly invariant share on single frames with anchors	34%
7	VGGT pose subspace transfer at block 21	0.85
7	VGGT 2-d PCA transfer	0.18
7	VGGT 2-d ridge transfer	0.20
7	VGGT steering gain: mean of the east and north gains at blocks 19 and 21; east gain at block 21	about $1/6$; 0.18 m per m
7	VGGT 10-d subspace R-squared in x,y	11%
7	VGGT second-order family: transfer lost when it is left out (L)	0.04
7	VGGT MLP neuron swap transfer	swapping all 4,096 neurons transfers at most 0.002, against 0.40 for the subspace
7	Fine-tuned model's order effect: reversal and local shares in two runs	runs 5 and 7: reversal removes 0.48 and 0.47; local share 0.80 and 0.74 in points
7	VGGT block-19 units under mean of two	most counted units are rate-modulated; rate maps turn with the anchors
7	Retrieval margin in the houses, overall and per-bin	overall 17.5 and 18.9; bins down to 11.6 and 11.7
7	Ordered-over-shuffled margins, range across runs	15.1–42.9 in the held-out rooms
7	Shuffled-walk accuracy range	54.4% to 82.1%
7	Training recipe update count	568 updates
7	Novel trial counts, room B	106 novel
7	RNN validation HG values	HG 0.98 and 0.98
7	Older cross-sector decoder on base RNN	7.57 m at 98% [99%] invariant units
F.3	Fine-tuned model, best median errors in walk's own frame	3.33 m–3.51 m (chance 4.01 m)
F.3	Base model, walk-own-frame cell count	288 cells
E	RNN strict invariant share	97%–100%